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# AI-POWERED DECISION SUPPORT SYSTEM FOR WILDFIRE DISASTER RESPONSE AND MANAGEMENT (DISASTERAWARE)

A THESIS SUBMITTED TO
THE GRADUATE SCHOOL OF NATURAL AND APPLIED SCIENCES
OF
MIDDLE EAST TECHNICAL UNIVERSITY

BY

ÇAĞLAR AKMAN

IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR
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**AI-POWERED DECISION SUPPORT SYSTEM FOR WILDFIRE DISASTER RESPONSE AND MANAGEMENT (DISASTERAWARE)**

submitted by **ÇAĞLAR AKMAN** in partial fulfillment of the requirements for the degree of **Doctor of Philosophy** i**n Electrical and Electronic Engineering, Middle East Technical University** by,

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## ABSTRACT

### AI-POWERED DECISION SUPPORT SYSTEM FOR WILDFIRE DISASTER RESPONSE AND MANAGEMENT (DISASTERAWARE)

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Operational wildfire decision support systems choose from closed catalogues of predefined alternatives, cannot revise their rule bases during an incident, and rarely expose the reasoning behind their recommendations. This thesis closes these gaps with DisasterAware, a distributed, concept-driven fuzzy decision support framework built on two strictly separated cores: a physics-based simulation core and a decision core that reads it only through observations. Local agents turn ten confidence-gated observation features into five decision concepts, a sparse Takagi-Sugeno rule base maps them to six intervention families, and a non-inferential coordination layer ranks regions and shares capacity. The central contribution is the open decision space. A staged adaptation loop tunes rule parameters, instantiates rules on uncovered situations, and, when the base has no answer, prompts a pretrained language model as a schema-constrained proposer. Every modification must clear deterministic verification gates and a satisficing forecast test before it acts, so the doctrine grows from five seed rules only as the incident demands; a graduated fail-safe and a no-harm rule protect life-safety orders, and every decision remains traceable to its concepts, rules, and confidences. The contributions are tested end to end. The simulation core reproduces the front propagation of four historical fires with 86 to 94 percent agreement. On this validated ground the full open configuration reduces mean burned area by 82 percent against no action, nearly halves the loss of the static rule base, decides in under one percent of the cycle, and carries sixteen times more agents at under 1.3 times the cost.

**Keywords:** Wildfire decision support system, Distributed multi-agent reasoning, Concept-driven Takagi-Sugeno fuzzy inference, Open decision space with generative rule proposal and verification, Explainable and uncertainty-aware decision support system

# ÖZ

## ORMAN YANGINLARINDA AFET MÜDAHALESİ VE YÖNETİMİ İÇİN YAPAY ZEKA DESTEKLİ KARAR DESTEK SİSTEMİ (DISASTERAWARE)

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İşletimdeki orman yangını karar destek sistemleri önceden tanımlı seçeneklerden oluşan kapalı kataloglardan seçim yapar, olay sürerken kendi kural tabanlarını değiştiremez ve önerilerinin gerekçesini çoğunlukla göstermez. Bu tez, bu boşlukları DisasterAware ile kapatır: fizik tabanlı bir benzetim çekirdeği ile bu çekirdeği yalnızca gözlemler üzerinden okuyan bir karar çekirdeğinden oluşan, kesin ayrımlı iki çekirdek üzerine kurulu, dağıtık ve kavram-güdümlü bulanık bir karar destek çerçevesi. Yerel ajanlar güven-kapılı on gözlem özniteliğini beş karar kavramına dönüştürür, seyrek bir Takagi-Sugeno kural tabanı kavramları altı müdahale ailesine eşler, çıkarım yapmayan bir üst koordinasyon katmanı bölgeleri sıralayıp ortak kapasiteyi paylaştırır. Tezin ana katkısı açık karar uzayıdır. Aşamalı bir uyarlama döngüsü kural parametrelerini ayarlar, kapsanmayan durumlara yeni kurallar yazar ve tabanın yanıtı kalmadığında önceden eğitilmiş bir dil modelini şema-kısıtlı öneri üreticisi olarak çağırır. Her değişiklik uygulanmadan önce belirlenimci doğrulama kapılarını ve tahmine dayalı bir yeterleme (satisficing) testini geçmek zorundadır; böylece doktrin beş çekirdek kuraldan yalnızca olayın gerektirdiği kadar büyür, kademeli emniyet mekanizması ile zararsızlık kuralı can güvenliği emirlerini korur ve her karar kavramlarına, kurallarına ve güven değerlerine kadar izlenebilir kalır. Katkılar uçtan uca sınanmıştır. Benzetim çekirdeği dört tarihsel yangının cephe ilerleyişini yüzde 86-94 uyumla yeniden üretir. Bu doğrulanmış zeminde tam açık yapılandırma ortalama yanan alanı eylemsizliğe kıyasla yüzde 82 azaltır, sabit kural tabanının kaybını yaklaşık yarıya indirir, kararını karar çevriminin yüzde birinden kısa sürede verir ve ajan sayısının on altı katını 1,3 kattan az maliyetle taşır.

**Anahtar Kelimeler:** Orman yangını karar destek sistemi, Dağıtık çok-ajanlı akıl yürütme, Kavram-güdümlü Takagi-Sugeno bulanık çıkarım, Üretken kural önerisi ve doğrulama ile açık karar uzayı, Açıklanabilir ve belirsizliğe duyarlı karar destek sistemi

To my beloved wife Emine and our sons Kuzey, Rüzgar, and Atlas for their endless love, patience, and encouragement. You are my calm after every storm.

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To my wife, Emine, my constant companion on every road and the meaning of my life. She stood behind everything, carrying countless sacrifices quietly and creating the space this work needed. She organized, managed, and coordinated every aspect of our family life so that I could devote myself fully to this doctoral journey, and she held everything together when the demands of research were overwhelming. This degree is as much hers as it is mine.

To my sons, Kuzey, Rüzgar, and Atlas, I hope this dissertation opens a path for you and shows that perseverance and discipline can turn the most challenging dreams into reality. May you walk this path further than I ever could. You are my greatest inspiration.

Thank you all — this accomplishment belongs as much to you as it does to me.

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| | | |
|---|---|---|
| **AI** | : | Artificial intelligence |
| **DEM** | : | Digital elevation model |
| **DSS** | : | Decision support system |
| **evFIS** | : | Evolving fuzzy inference system |
| **GIS** | : | Geographic information system |
| **HIL** | : | Human-in-the-loop |
| **LIME** | : | Local interpretable model-agnostic explanations |
| **LLM** | : | Large language model |
| **LoS** | : | Line of sight |
| **MAE** | : | Mean absolute error |
| **MF** | : | Membership function |
| **ML** | : | Machine learning |
| **POD** | : | Probability of detection |
| **ROS** | : | Rate of spread |
| **SHAP** | : | Shapley additive explanations |
| **UI** | : | User interface |
| **VIIRS** | : | Visible infrared imaging radiometer suite |
| **WFDSS** | : | Wildfire decision support system |
| **WUI** | : | Wildland-urban interface |

## LIST OF SYMBOLS

The following list defines all symbols used in this thesis, organized by functional category. Four notational conventions apply throughout: Subscript k denotes the discrete time step; arguments (x, y) denote the grid cell; superscript $i$ identifies the operational region; a hat accent ( $\widehat{\ }$ ) indicates an estimated quantity, observed or forecast; and an overbar ( $\overline{\ }$ ) indicates a confidence-gated or spatially averaged quantity.

### 1. Fire State and Propagation Variables

| | |
|---|---|
| $B_k(x,y)$ | Burning status indicator |
| $B_{k+1}^{\mathrm{pers}}(x,y)$ | Burning persistence term |
| $B_{k+1}^{\mathrm{prop}}(x,y)$ | Burning propagation term |
| $F_{\mathrm{burn},k}(x,y)$ | Fractional burn rate |
| $I_k(x,y)$ | Local fire intensity proxy |
| $I_{\mathrm{Ign},k}(x,y)$ | External ignition injection indicator |
| $R_{\mathrm{spread},k}(x,y)$ | Rate of spread at time step $k$ |
| $\tau_k(x,y)$ | Time since ignition |
| $\Psi_k(x,y)$ | Propagation influence |
| $\Theta_{\mathrm{ign}}$ | Ignition threshold |
| $\Phi$ | State transition operator |

### 2. Fuel Variables

| | |
|---|---|
| $F_{load,k}(x,y)$ | Available combustible fuel mass |
| $F_{moist,k}(x,y)$ | Surface fuel moisture |
| $F_{red,k}(x,y)$ | Suppression-induced fuel reduction |
| $F_{type}(x,y)$ | Fuel type classification |
| $U_{Fuel,k}(x,y)$ | Fuel-related input |
| $\varepsilon_{fuelload}$ | Fuel extinction threshold |

### 3. Terrain / Geographic Variables

| | |
|---|---|
| $G_{access}(x,y)$ | Terrain accessibility index |

| | |
|---|---|
| $G_{aspect}(x,y)$ | Terrain orientation (aspect) |
| $G_{elev}(x,y)$ | Terrain elevation |
| $G_{slope}(x,y)$ | Terrain slope |
| $U_{Geo}(x,y)$ | Terrain-related input |

**4. Meteorological Variables**

| | |
|---|---|
| $W_{gust,k}(x,y)$ | Wind gust field |
| $W_{prec,k}$ | Precipitation (domain-uniform scalar, no spatial variation) |
| $W_{rh,k}(x,y)$ | Relative humidity field |
| $W_{temp,k}(x,y)$ | Air temperature field |
| $W_{wd,k}(x,y)$ | Wind direction field |
| $W_{ws,k}(x,y)$ | Wind speed field |
| $U_{Meteo,k}(x,y)$ | Meteorological input |
| $\delta_{Meteo}$ | Meteorological perturbation applied by operator |

**5. Value / Asset Variables**

| | |
|---|---|
| $V_{bld}(x,y)$ | Building footprint indicator |
| $V_{crit}(x,y)$ | Critical facility index |
| $V_{evac}^{norm}(x,y)$ | Accessibility to evacuation routes |
| $V_{pop}^{norm}(x,y)$ | Population density at grid cell (person/km$^2$) |
| $V_{prio}(x,y)$ | Asset priority index |
| $U_{Val,k}(x,y)$ | Value-related contextual input |
| $\sigma_{Val}$ | Value priority scaling factor in $\Theta_{UI}$ |

**6. Resource / Suppression Variables**

| | |
|---|---|
| $R_{avail,k}(x,y)$ | Operational availability of suppression resources |
| $R_{cap,k}(x,y)$ | Available suppression capacity |
| $R_{eff,k}(x,y)$ | Nominal suppression effectiveness |
| $R_{time,k}(x,y)$ | Estimated travel time for suppression resources |
| $U_{Res,k}(x,y)$ | Resource-related contextual input |
| $u_{m,k}^{i}(x,y)$ | Candidate intervention degree for family m in region $i$ |

| | |
|---|---|
| $\delta_{Res}(x,y)$ | Resource availability perturbation applied by operator |
| $\eta_{cap,k}$ | Normalized suppression capacity factor |
| $\eta_{avail,k}$ | Normalized resource availability factor |
| $\eta_{reach,k}$ | Reachability (travel-time) factor |
| $\eta_{eff,k}$ | Suppression effectiveness factor |

**7. Observation / Sensing Variables**

| | |
|---|---|
| $O_k^i(x,y)$ | Region-specific observable information at time step k |
| $O_k(x,y)$ | Observable information at time step $k$ |
| $h^i(\bullet)$ | Region-specific observation function for $\Omega_i$ |
| $\gamma_k^i$ | Observation confidence for region $i$ |
| $\varepsilon_k^i(x,y)$ | Observation uncertainty |

**8. Spatial Domain / Grid Variables**

| | |
|---|---|
| $\mathcal{G}$ | Spatial grid domain |
| $N$ | Total number of operational regions / local agents |
| $N_x$ | Number of grid cells along the horizontal axis |
| $N_y$ | Number of grid cells along the vertical axis |
| $\mathcal{N}^8(x,y)$ | 8-connected neighborhood |
| $S_k(x,y)$ | Wildfire system state over spatial grid at time step k |

**9. Decision Support System (DSS) Variables**

| | |
|---|---|
| $\mathcal{F}_{DSS,k}$ | Contextual input set |
| $\mathcal{F}_{DSS,k}^*$ | UI-modified decision-context set at step k |
| $\mathcal{F}_{in,k}$ | External input set |
| $\mathcal{F}_{in,k}^*$ | UI-modified external input set at step k |
| $U_{DSS,k}(x,y)$ | DSS-generated intervention input parameters |
| $U_{DSS,k}^i(x,y)$ | Regional intervention output from agent $i$ |
| $U_{DSS,safe}$ | Conservative baseline intervention under fail-safe policy |
| $\mathcal{U}_{DSS}$ | Feasible intervention space |
| $U_{Ign,k}(x,y)$ | Ignition input |

| | |
|---|---|
| $\pi_{DSS}^{i}$ | Regional decision-support mapping |
| $A_{DSS}$ | Coordination operator aggregating the regional decisions |
| $\sigma_i$ | Attention share assigned to region i by the coordinator |
| $\delta_{Ign}(x, y)$ | Ignition perturbation applied by operator |
| $\lambda_k$ | Fail-safe blending factor between the DSS decision and the conservative baseline |
| $\Theta_{UI}$ | User interaction operator |

**10. Fuzzy Logic Variables**

| | |
|---|---|
| $\mathcal{N}_{concept}$ | Number of decision concepts |
| $N_{feature}$ | Number of input features |
| $\mu_{A_{r,n}}$ | Membership of the antecedent term of rule r over concept n |
| $v_{r,m}$ | Singleton consequent value of rule $r$ for intervention family m |
| $\alpha_r$ | Activation (firing) strength of rule $r$, the minimum of its antecedent memberships |
| $\alpha_{min}$ | Coverage floor of the seeded rule base (0.05) |
| $w_j$ | Non-negative priority weight of cost term j |

**11. Functions / Mappings**

| | |
|---|---|
| $f_{intensity}$ | Fire intensity approximation function |
| $f_{ros}$ | Rate-of-spread mapping function |
| $f_{supp}$ | Suppression-to-mitigation mapping |
| $g_{dir,k}$ | Directional spread weighting factor |

**12. Feature Variables and Parameters**

| | |
|---|---|
| $z_{k,l}^{i}(x, y)$ | $l^{th}$ feature at $(x, y)$, region $i$, step $k$; $z \in [0,1]$, $l = 1 \dots 10$ |
| $\beta_t$ | Travel-time decay coefficient |
| $\Delta t$ | Simulation time step |
| $\delta$ | Admission margin of the generative verification gates ($\delta \geq 0$) |

**13. Decision Cost and Quality Variables**

| | |
|---|---|
| $J$ | Total normalized decision cost (weight sum of cost terms) |
| $J_{phys}$ | Physical decision cost (burned area, asset loss, population) |

| | |
|---|---|
| $J_{burn}$ | Burned-area cost term |
| $J_{asset}$ | Asset-loss cost term |
| $J_{pop}$ | Population-exposure cost term |
| $J_{resp}$ | Response cost term |
| $J_{delay}$ | Response-delay cost term |
| $Q_k$ | Decision quality at step $k$ |
| $J_{TH}$ | Fixed satisficing ceiling on the normalized decision cost (0.35) |
| $\delta_{gain}$ | Required relative gain over the no-action forecast (0.05) |
| $\delta_v$ | Vocabulary margin of the package gate G5 (0.0001) |
| $N_{term}$ | Number of linguistic terms per fuzzy variable |
| $N_{input}$ | Largest number of features feeding a single concept |
| $a_k^{eff}$ | Gated concept activation at step $k$ |
| $a_k^{obs}$ | Observation-based concept activation at step $k$ |
| $a_k^{per}$ | Persistence-prior concept activation at step $k$ |
| $\rho$ | Persistence decay factor of the effective concept activation |
| $e_k$ | Contextual-bandit state (deficit band and coverage-gap flag) |
| $q_k$ | Selected adaptation stage (contextual-bandit action) |
| $\eta$ | Quality gate for the graduated fail-safe (accept if $Q_k \geq \eta$) |
| $\varphi_{(i,j)}$ | Bearing angle from cell $(i,j)$ toward $(x,y)$ in directional spread |

# CHAPTER 1

# INTRODUCTION

Decision makers in disaster response and other critical domains must combine many streams of information under uncertainty, time pressure, and resource limits, and a poor decision can have irreversible consequences. Decision Support Systems (DSS) are designed to extend the range of information an operator can weigh under such pressure (Sprague & Carlson, 1982; Power, 2002; Turban, Sharda, & Delen, 2011; Sharda, Delen, & Turban, 2018). Despite decades of development alongside data analytics and computational intelligence (Power & Heavin, 2017; Sharda, Delen, & Turban, 2018; Arnott & Pervan, 2005), classical DSS still assume full observability, stationary dynamics, and well-defined objectives (Keeney & Raiffa, 1976), assumptions that rarely hold in real operations, where ***epistemic uncertainty*** persists (French, Maule, & Papamichail, 2019; Walker, et al., 2003).

DSS research has therefore shifted toward hybrid architectures that combine simulation, expert knowledge, and uncertainty-aware reasoning (Marakas & O'Brien, 2013; Arnott & Pervan, 2005). At the same time, artificial intelligence (AI) methods have raised predictive accuracy but often operate as opaque systems whose ***reasoning is difficult to expose at the decision level*** (Ribeiro, Singh, & Guestrin, 2016; Abdul, Vermeulen, Wang, Lim, & Kankanhalli, 2018; Sanneman & Shah, 2020). In critical domains such as disaster management, effective human-machine teaming depends on decision outputs that operators can interpret, trace, and align with their own reasoning (McNeese, Schelble, Canonico, & Demir, 2021); hence, next-generation ***DSS must reason with incomplete and unreliable observations, preserve interpretability, and adapt as conditions evolve*** (French, Maule, & Papamichail, 2019; Thompson & Calkin, 2011; Comfort, Boin, & Demchak, 2010; Walker, et al., 2003).

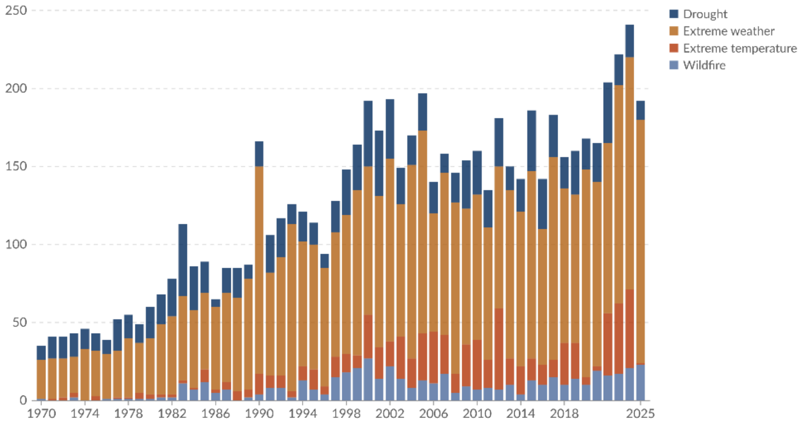


Figure 1.1 Global trend in reported natural disasters

Figure 1.1 shows the steep rise in reported natural disasters since 1970, with weather- and climate-related events, including wildfires, taking a growing share (CRED/UCLouvain, 2026). The EM-DAT documentation attributes much of the rise before 2000 to improvements in reporting and coverage rather than to hazard frequency, and it advises against reading the earlier period as a trend in hazard activity. Wildfire frequency, spatial extent, and intensity have all risen under climate change, land-use transformation, and the expansion of the wildland-urban interface (Bowman, et al., 2020; UNEP, 2022), and the resulting ecological, economic, and human impacts increasingly outpace conventional response capabilities (Thompson & Calkin, 2011; Pacheco, et al., 2015; Finney, An overview of FlamMap fire modeling capabilities, 2006).

Wildfire management is a demanding decision domain because fire behavior evolves in space and time under the joint influence of weather, terrain, fuel, and suppression. Wind and humidity set ignition probability and spread rate, slope and aspect steer propagation, fuel structure determines intensity and burn duration, and suppression actions feed back into the fire dynamics (Rothermel, 1972; Finney, An overview of FlamMap fire modeling capabilities, 2006; Sullivan, 2009a; Sullivan, 2009b; Sullivan, 2009c). Operational wildfire response relies not only on numerical indicators, but also on incomplete and linguistically expressed situational assessments such as “high spread potential,” “limited suppression feasibility,” or “uncertain fire boundary.” Fuzzy reasoning is well suited to this setting (Bellman &

Zadeh, 1970), since it supports approximate inference over partial and imprecise evidence without probability distributions, which are rarely available in real time (Thompson & Calkin, 2011; Pacheco, et al., 2015).

Many DSS approaches exist for wildfire management, from optimization-based resource allocation and simulation-based systems to geospatial risk assessment and fuzzy reasoning (Thompson & Calkin, 2011; Pacheco, et al., 2015; Soualah, 2024; Vásquez, 2021; Finney, An overview of FlamMap fire modeling capabilities, 2006). Mature deployed systems such as WFDSS and FlamMap, together with recent ML-based hybrids, provide substantial operational value, yet each addresses only part of these requirements. The remaining challenge is therefore architectural. **There is a need for a framework that simultaneously addresses adaptability under dynamic conditions, explainability of decision outputs, and support for distributed multi-level coordination within one coherent design** (Thompson & Calkin, 2011; Vásquez, 2021; McNeese, Schelble, Canonico, & Demir, 2021). The DisasterAware framework is motivated not by a single methodological deficiency but by a set of structural limitations that recur across existing DSS. The next section examines these limitations, and the thesis contributions are then mapped to them.

## 1.1 Gap Analysis

Despite advances in DSS research, the literature still exhibits several unresolved limitations that constrain operational effectiveness. Existing systems have delivered important capabilities in fire spread prediction, geospatial risk analysis, resource planning, and situation monitoring. However, many of these contributions are built on assumptions that are difficult to sustain under dynamic conditions: **Predefined decision alternatives**, **centralized reasoning**, **limited treatment of uncertain observations**, **weak explainability**, and **constraint handling** that presumes complete information (Arnott & Pervan, 2005; Turban, Sharda, & Delen, 2011). These limitations consolidate into three structural gaps, summarized in Table 1.1, each reflecting a structural rather than incidental shortcoming of current architectures.

Table 1.1 DSS gaps identified in wildfire decision support literature

| Gaps of Existing Wildfire DSS Architectures |
| --- |
| **G1:** Many of the reviewed DSS frameworks rely on **predefined decision alternatives** evaluated within **centralized, level-undifferentiated architectures**, preventing adaptation to dynamic, unforeseen conditions and limiting scalability, local responsiveness, and level-appropriate coordination. Field studies report that practitioners use these systems both for decision tasks and to confirm and document decisions already made. |
| **G2:** Many of the reviewed systems represent **epistemic uncertainty** only through prior probability distributions or calibrated belief intervals and expose the reasoning behind a recommendation only through **post-hoc attribution**, leaving decision sensitivity unquantified and eroding operator trust. Fuzzy approaches that avoid distributional assumptions remain outside mainstream wildfire DSS architectures. |
| **G3:** In many of the reviewed systems, **simulation dynamics and decision reasoning remain tightly coupled, and operational constraints are managed** through offline optimization requiring complete state information, so no end-to-end workflow built on a strictly separated simulation core demonstrates real-time, cell-level decision support under operational conditions. |

### G1 Closed Decision Spaces in Centralized Architectures

Many classical DSS frameworks organize decision support around an explicit set of candidate alternatives defined before evaluation begins (Power, 2002; Pacheco, et al., 2015; Sprague & Carlson, 1982). Discrete multi-criteria frameworks similarly evaluate a fixed alternative set against predefined criteria (Malczewski, 1999; Keeney & Raiffa, 1976). Wildfire management violates this assumption. Fire behavior evolves continuously with wind shifts, fuel variability, and terrain interactions, producing operational conditions that no fixed strategy set can represent

(Pacheco, et al., 2015; Vásquez, 2021; Calkin, O'Connor, Thompson, & Stratton, 2021). Field studies confirm the mismatch. Practitioners often use existing DSS to validate and document decisions already made from experience (Fillmore & Paveglio, 2023; Rapp, Rabung, Wilson, & Toman, 2020). A recent review reports the same confirmatory pattern and the absence of effectiveness evaluation frameworks (Rapp, et al., 2026). The closed decision space is not an incidental limitation but a structural barrier to adaptive operational use. Centralized DSS architecture concentrates observation aggregation, decision evaluation, and response generation in a single analytical core. Every observation and every candidate decision must pass through this single core, responsiveness degrades when the operational state changes rapidly or the spatial domain expands. Recent reviews of wildfire DSS call for tools that deliver responses within operationally useful timeframes (Vásquez, 2021). Meeting these timeframes while coordinating across regions argues for distributed rather than centralized reasoning.

A limitation is the absence of explicit hierarchical differentiation between decision levels (Anthony, 1965). Wildfire operations span local intervention, regional coordination, and strategic prioritization (Turoff, Chumer, Van de Walle, & Yao, 2004; Comfort, Boin, & Demchak, 2010), but most DSS functions as a single-layer process, conflating inferential responsibilities with coordinative ones. Assessments of the U.S. Risk Management Assistance program likewise identify the timely delivery of decision products as a critical need, so coordination support must keep pace with the incident it serves (Calkin, O'Connor, Thompson, & Stratton, 2021).

**G2 Unrepresented Epistemic Uncertainty and Limited Explainability**

Wildfire decisions are shaped as much by epistemic uncertainty, arising from incomplete or unreliable information, as by stochastic variability in fire dynamics (Morgan & Henrion, 1990; Walker, et al., 2003; Thompson & Calkin, 2011). Existing uncertainty-aware DSS approaches (Comes, Hiete, Wijngaards, & Schultmann, 2011) typically require either prior probability distributions (Bayesian) or pre-calibrated belief intervals (Dempster-Shafer), both of which are rarely available at cell-level resolution in real-time operational settings (Shafer, 1976; Pearl, 1988). Fuzzy logic has been proposed as a principled alternative for graded representation of operational evidence without distributional assumptions (Mendel, 2017; Zadeh, Fuzzy sets, 1965), yet its systematic integration into wildfire DSS architecture remains limited. Field studies report that managers question the reliability of model outputs and do not let DSS dictate their decisions (Rapp, Rabung, Wilson, & Toman, 2020). A tool hiding the observational uncertainty behind a recommendation gives the operator no basis for judging its reliability.

AI-based prediction models, including those entering wildfire management, increasingly operate as black boxes whose internal reasoning is difficult to interpret (Ribeiro, Singh, & Guestrin, 2016; Guidotti, et al., 2018). Emergency response requires traceable recommendations that operators can justify to multiple stakeholders (Fillmore & Paveglio, 2023). Post-hoc attribution methods such as SHAP and LIME provide statistical approximations of model behavior rather than causal rule-level traceability (Lundberg & Lee, 2017; Ribeiro, Singh, & Guestrin, 2016). Empirical studies (Schultz, Miller, Greiner, & Kooistra, 2021) report that managers use decision support outputs mainly to confirm decisions already made, and that the value of a tool lies in its ability to communicate the rationale behind a decision clearly and transparently (Fillmore & Paveglio, 2023; Rapp, Rabung, Wilson, & Toman, 2020). Recent hybrid predictive-fuzzy frameworks demonstrate that accuracy and interpretability are not mutually exclusive (Dubey & Dubey, 2025), but such an architecture remains absent from mainstream DSS.

**G3 Sim-Decision Coupling without an End-to-End Operational Workflow**

Many DSS tightly integrate fire-spread simulation with decision reasoning. Modular architecture is essential for traceability and independent validation of components (Sugumaran & Degroote, 2010). Tight coupling prevents either component from being updated or verified in isolation (Pacheco, et al., 2015; Vásquez, 2021). As an example, WFDSS’s fire-behavior systems and decision records are architecturally merged within a single platform (Noonan-Wright, et al., 2011). In the architectural perspective of this thesis, this integration leaves either component difficult to replace independently. Recent hybrid architectures reinforce the same coupling. The transformer-fuzzy decision model of Dubey & Dubey (2025) couples prediction and response reasoning within a single integrated pipeline. As wildfire simulation science advances separately from DSS research, this coupling increasingly blocks the incorporation of new modeling capabilities into operational decision frameworks. Optimization-based DSS represent operational constraints such as resources, travel time, and suppression effectiveness through formal cost functions (Keeney & Raiffa, 1976; Pais, Carrasco, Moudio, & Shen, 2021). Recent operational toolkits follow the same formulation (Orphanoudakis, Betzelos, & Leligou, 2025). Many deterministic formulations are evaluated under fully specified state inputs, which are rarely available during active wildfire (Martell, 2007; Finney, 2006).

The gap is not lack of awareness but the absence of mechanisms integrating constraints into real-time, cell-level reasoning under incomplete and evolving observations. Distributed wildfire response further breaks the completeness assumption. Resources are dispersed, availability changes continuously during an incident, and local accessibility depends on conditions that can only be estimated from partial observations. Published DSS rarely provide operational mechanisms for constraint-sensitive reasoning at this granularity. These constraints compound the coupling problem. Without separable components and constraint-aware reasoning, complete workflows that connect observation, reasoning, and operator interaction in a single demonstrable pipeline remain rare, sustaining the distance between published capability and field adoption.

## 1.2 Research Question and Hypothesis

Formally, this thesis addresses the following research question: ***How can a wildfire decision support system generate adaptive intervention guidance over an open decision space, expose the reasoning behind every recommendation it derives from incomplete observations together with the sensitivity of the closed-loop outputs,***

***and demonstrate the resulting pipeline end to end against a strictly separated simulation core, under distributed, decision-cycle-compatible latency constraints, with analytically stated and empirically evaluated pipeline properties?***

This question gives rise to the following main hypothesis: ***A hierarchically distributed DSS architecture built on concept-driven fuzzy reasoning, equipped with adaptive and generative rule evolution behind a satisficing verification gate, intrinsic rule-level traceability with quantified sensitivity, and a strictly separated simulation core, addresses the three identified structural gaps within a single coherent framework and states its pipeline properties analytically while evaluating them empirically, a combination rarely available in existing operational DSS.*** This hypothesis is decomposed into three testable sub-claims:

***H1 (Scalability and Adaptability):*** A hierarchically distributed architecture with concept-abstracted local fuzzy agents and non-inferential global coordination generates adaptive interventions over an open decision space under evolving wildfire conditions; the concept hierarchy reduces the antecedent combination space and the distributed implementation keeps decision latency compatible with the decision cycle as the number of local agents grows by an order of magnitude.

***H2 (Auditable Traceability and Parameter Sensitivity):*** The combination of concept-driven fuzzy reasoning, intrinsic rule-level traceability, confidence-gated uncertainty encoding, and systematic sensitivity analysis produces intrinsically traceable, structurally inspectable decision outputs and auditable decision records that do not require post-hoc attribution methods such as SHAP or LIME.

***H3 (Simulated Operational Feasibility Under Uncertainty):*** An end-to-end workflow that couples the strictly separated simulation core and decision core, together with satisficing-based decision acceptance and cost-aware constraint integration, produces feasible interventions across the tested incompletely observed, resource-constrained simulated scenarios, where every fielded intervention passed the applicable pre-execution satisficing, quality, resource, and no-harm gates.

## 1.3 Main Contributions of the Thesis

The thesis makes three architectural and methodological contributions that collectively address the structural gaps identified in Section 1.1. The contributions have distinct primary objectives, each addressing one gap as summarized in Table 1.2. They operate as interdependent parts of a single framework and are realized across the background in Chapter 2, the DisasterAware framework and its simulation core in Chapter 3, the decision support system design and its analytical properties in Chapter 4, and the empirical evaluation in Chapter 5.

Table 1.2 Mapping of thesis contributions to identified gaps

| Addressed Gap | Contribution |
|---|---|
| **G1: Closed decision spaces in centralized architectures** | C1: Open decision space through concept-driven abstraction, adaptive rule generation, and hierarchically distributed reasoning with non-inferential global coordination Evaluated dimensions: Decision quality across the configuration ladder, adaptation activity and vocabulary growth, and decision latency at scale. |
| **G2: Unrepresented epistemic uncertainty and limited explainability** | C2: Sensitivity analysis built on confidence-gated uncertainty propagation and intrinsic rule-level traceability Evaluated dimensions: Decision trace records and the sensitivity of closed-loop outputs to operational parameters. |
| **G3: Simulation-decision coupling without an end-to-end operational workflow** | C3: End-to-end use-case demonstration over strictly separated simulation and decision cores with satisficing-based acceptance Evaluated dimensions: Simulation-core fidelity, campaign feasibility under partial observation and resource limits, and constraint compliance. |

### C1 Open Decision Space through Adaptive, Concept-Driven Reasoning

To address G1, this thesis replaces the fixed menu of predefined alternatives with an open, concept-driven decision field. Ten observation features are hierarchically aggregated into five domain-meaningful decision concepts, so that rules are written in operational language rather than over raw signals. The field is then reshaped online. An evolving Takagi-Sugeno rule base adapts membership functions (MFs) and term resolution (Takagi & Sugeno, 1985; Angelov & Filev, 2004), a generative model proposes genuinely new candidate rules when the contextual stage selector escalates to the generative stage, and every candidate passes a satisficing verification gate before it may act (Simon, 1956). Reasoning itself is confined to local fuzzy agents, while a non-inferential global layer ranks regional priorities, assigns resources and attention shares, and adjusts the acceptance gates of monitored regions, which keeps the open field scalable. The reshaping acts on three nested levels, the membership boundaries and consequents of existing rules, the resolution of their term sets, and the vocabulary itself. At the vocabulary level, the framework may introduce intermediate concepts, blend existing interventions, and compose verified effects, such as wetting, clearing, and coating, into new interventions. It cannot create new physical effects. Nothing enters the base without passing the gate chain, which ends in a reseeded forecast comparison on the simulator.

### C2 Sensitivity Analysis with Traceability and Confidence-Gated Uncertainty

To address G2, the reasoning pipeline is built so that explanation and uncertainty handling are structural rather than retrofitted. Epistemic uncertainty is propagated without prior distributions by blending each concept activation with a persistence-based estimate in proportion to observation confidence. Every decision remains traceable to the concepts that fired it, the rules involved, and the confidence behind them, and a sensitivity analysis quantifies how the operational parameters of the closed loop, including the decision cycle, the no-harm horizon, the observation coverage, the satisficing bound, and the quality gate, affect selected closed-loop outputs, in contrast to post-hoc attribution methods such as SHAP or LIME.

### C3 End-to-End Use-Case Demonstration

To address G3, the thesis enforces a strict separation between the simulation core and the decision core, which grounds analytically stated pipeline properties under incomplete observation and allows either side to be validated independently. On this foundation the framework is exercised in an end-to-end simulated use case that couples the spread simulation, the decision pipeline, and an operator dashboard, with satisficing acceptance and a graduated fail-safe screening every intervention before it is fielded. The use case addresses the research-to-practice distance reported in field studies (Fillmore & Paveglio, 2023; Rapp, Rabung, Wilson, & Toman, 2020). In this use case, the paired with-DSS and without-DSS runs share the same initial state, exogenous drivers, and random-number stream, so subsequent differences arise from the issued orders alone.

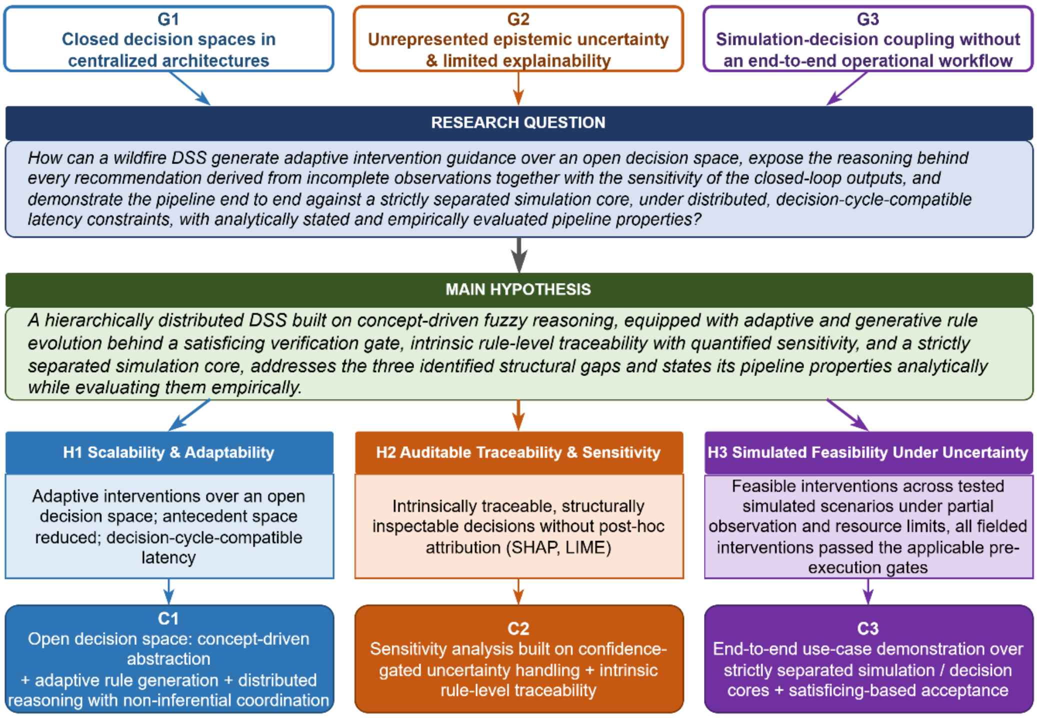

Figure 1.2 Research question, hypothesis, gaps, and contributions

Collectively, these contributions transform wildfire DSS from a predefined, centralized toolkit into an open, distributed, and uncertainty-aware reasoning system whose operational properties are analytically stated and empirically evaluated. The main hypothesis posits that a hierarchically distributed, concept-driven, confidence-gated, and satisficing-based architecture meets the requirements of the research question, and it is decomposed into three testable sub-claims: H1 scalability and adaptability, H2 auditable traceability and parameter sensitivity, H3 simulated operational feasibility under uncertainty. Each sub-claim is realized by one contribution that addresses one structural gap, so every gap maps to a targeted architectural contribution. The argument runs from the literature gaps through the research question and the hypotheses to the contributions, as illustrated in Figure 1.2.

## 1.4 Organization of the Thesis

The remainder of the thesis follows the path a decision travels, from background to evidence. Chapter 2 includes the background material, covering wildfire dynamics and modeling, the foundations of DSS, the input space that turns heterogeneous sensing into observation features, concept-driven fuzzy reasoning with the rule-explosion argument, and the decision space with its interventions, evaluation measures, and adaptive-generative loops. Chapter 3 introduces the DisasterAware framework as the computational foundation. It presents the system architecture, the external input layers for meteorology, topography, fuel, ignition, and decisional data, and the simulation core whose state transition and hybrid spread model couple to the decision support core, with testability rather than maximal fidelity as its design criterion. Chapter 4 includes the DSS itself, moving from the feature space and fuzzy variables to the five decision concepts, the intervention rule base, and the layered architecture of observation, concept mapping, local DSS, and global coordination, and closing with the epistemic encoding of incomplete observations and the explainability overlay. Chapter 5 provides empirical evidence. It validates the simulation core against historical fires, locates the operating point through sensitivity analysis, reports the performance of the decision layer from the improvement ladder to the open decision space, and closes with a criterion-by-criterion compliance map. Chapter 6 summarizes the contributions and outlines future research.

# CHAPTER 2

# BACKGROUND INFORMATION

This chapter presents the background on which the remainder of the thesis builds. Its purpose is twofold: To introduce the concepts on which the design chapters rely, and to justify, on the basis of the literature, the choices that shape DisasterAware. The chapter is organized as follows. Section 2.1 reviews wildfire dynamics and spread modeling. Section 2.2 introduces the foundations of decision support, including closed versus open decision spaces and centralized versus distributed architecture. Section 2.3 presents the input space, from heterogeneous sensing sources to the ten observation features. Section 2.4 develops concept-driven fuzzy reasoning. Section 2.5 defines the decision space: The admissible interventions, the cost, acceptance, and uncertainty machinery by which candidate decisions are judged, and the adaptive and generative loops that keep the decision space open.

## 2.1 Wildfire Dynamics and Spread Modeling

This section reviews the mechanisms of wildfire spread and the principal approaches to modeling it. The treatment is confined to the material required by the subsequent chapters: The physical drivers of fire behavior, the families of spread models, and the properties a decision layer requires from the model to which it is coupled.

### 2.1.1 Drivers of Fire Behavior

Fire behavior at any point of a landscape is governed by three groups of drivers, namely **weather**, **terrain**, and **fuel**. Weather contributes the most rapidly varying quantities: Air temperature, relative humidity, wind speed and direction, wind gusts, and precipitation. Wind pushes the flame front toward unburned fuel, while humidity and precipitation set the moisture content of fine fuels and, with it, the ease of ignition (Rothermel, 1972). Terrain contributes elevation, slope, aspect, and accessibility. A front burning upslope preheats the fuel above it and accelerates, aspect determines how much solar radiation dries the fuel over the day, and accessibility bounds where crews can act. Fuel contributes the most slowly varying quantities. The vegetation type, the available fuel load, and the fuel moisture determine how intense the combustion becomes and how long a cell keeps burning (Scott & Burgan, 2005).

These drivers are summarized operationally by a small set of quantities: The rate of spread (ROS) of the front, its direction, and the fire line intensity. The Rothermel model expresses ROS as a function of fuel properties, fuel moisture, wind, and slope, and it still underlies most operational tools (Rothermel, 1972; Andrews, 2018); the detailed parameterization used in this thesis is given in Appendix A. The significance of the model for this thesis lies not in its exact functional form but in the causal chain it encodes. Weather, terrain, and fuel jointly determine local spread, and any intervention that modifies one of these inputs alters the behavior of the fire.

### 2.1.2 Families of Spread Models

The literature groups spread models into three families. **Physical and quasi-physical models** resolve combustion chemistry and heat transfer; they are the most detailed and the most expensive to run (Sullivan, Wildland surface fire spread modelling 1: Physical and quasi-physical models, 2009a). **Empirical and quasi-empirical models** fit observed spread rates to environmental variables; they are fast and robust within the conditions for which they are calibrated (Sullivan, Wildland surface fire spread modelling 2: Empirical and quasi-empirical models, 2009b). **Simulation models** propagate a fire front across a landscape by applying one of the former families as a local kernel; FlamMap is the best-known operational example (Sullivan, Wildland surface fire spread modelling 3: Simulation and mathematical analogue models, 2009c; Finney, An overview of FlamMap fire modeling capabilities, 2006).

Grid-based simulation discretizes the landscape into cells, each holding fuel, moisture, and burn state. A cell ignites with a probability which grows with the number of burning neighbors, their alignment with the wind, and the local fuel condition, and it stops burning when its fuel is consumed or suppressed. This representation matches the resolution at which incident decisions are made, so it is preferred when the model is coupled to a decision layer. **DisasterAware follows this pattern with a grid-based simulation model whose local spread kernel belongs to the Rothermel-type quasi-empirical family.**

### 2.1.3 Requirements of the Decision Layer on the Spread Model

A wildfire DSS does not require the highest-fidelity fire simulator available; it requires a testable environment. Testability comprises four properties: **Scenarios can be reproduced exactly**, the **per-cell state can be observed and logged**, the **inputs through which interventions act can be controlled**, and **the outcome of a decision can be measured against the outcome of not deciding**.

Any spread model that offers these four properties, at any level of fidelity, is sufficient to run and evaluate the decision layer; higher fidelity refines the numerical results but alters neither the architecture nor the claims of this thesis. Beyond testability, the coupled model must run at an update rate compatible with the decision cycle.

DisasterAware therefore adopts a hybrid grid model, formalized in Chapter 3, that combines Rothermel-type local spread with explicit per-cell state tracking, in which every grid cell carries its remaining fuel, its moisture, and its burn status, is driven by external weather and terrain layers, and accepts interventions only at the input level, so the decision core never manipulates the physical state directly.

Figure 2.1 summarizes this coupling: Weather, terrain, and fuel drive fire behavior; fire behavior updates the cell states; the DSS observes cell states and issues interventions; and interventions feed back into the model inputs.

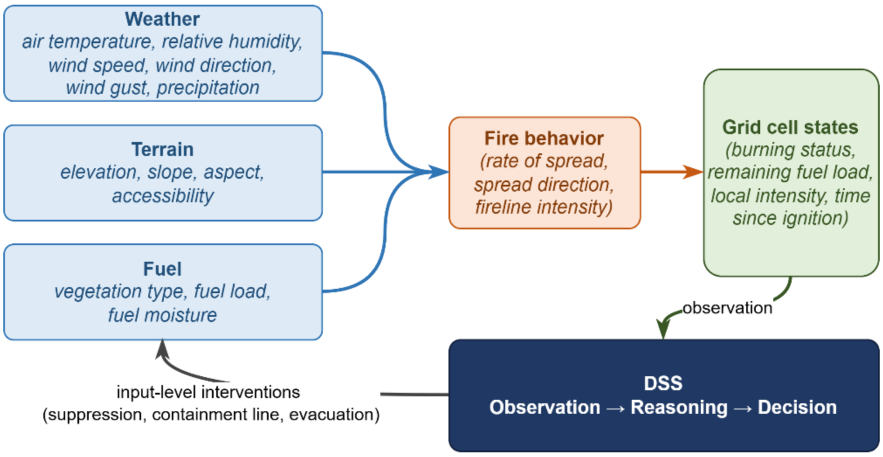


Figure 2.1 Coupling between fire behavior, the grid state, and the decision layer

## 2.2 Decision Support Foundations

This section establishes DSS foundations on which the remainder of the thesis rests. It first reviews the core elements that any operational DSS must supply and, in doing so, introduces the two spaces the framework mediates between, namely the input space of observable quantities and the decision space of orderable actions. It then examines the distinction between closed and open decision spaces, which motivates the central contribution of this thesis, and closes with a comparison of centralized and distributed architectures, which determines where reasoning takes place. Together these discussions supply the vocabulary in which the input space, the reasoning machinery, and the decision space of the subsequent sections are defined.

### 2.2.1 Core Elements of Decision Support Systems

A DSS mediates between two spaces, namely the **input space** and **decision space**. The input space comprises everything the system can observe or estimate: Sensor fields, forecasts, terrain and fuel maps, resource states, and the quantities derived from them. The decision space comprises everything the system is authorized to decide, namely the set of actions, with their intensities, locations, and timings, which can be ordered at a given moment.

A DSS is a software system that assists an operator in choosing among actions whose consequences are uncertain (Sprague & Carlson, 1982; Power, Decision support systems: Concepts and resources for managers, 2002). Decision support occupies a deliberate middle ground between full automation and unaided judgment (Power, A Brief History of Decision Support Systems, 2007). In the classical framework of Gorry and Scott Morton it targets semi-structured decisions, those for which parts of the reasoning can be modeled while the remainder requires human judgment (Gorry & Scott Morton, 1971; Simon, The New Science of Management Decision, 1960).

Wildfire response is a canonical instance of this class. The physics of spread can be modeled, and candidate responses can be scored, yet accountability, values, and situational judgment remain with the incident commander; the system therefore recommends, while the decision authority remains with the operator. Independent of domain, a working DSS supplies the same core elements (Turban, Sharda, & Delen, 2011; Sharda, Delen, & Turban, 2018):

**Input interface,** which transforms raw observations into usable, bounded variables; **Knowledge base,** which encodes the behavior of the domain, represented in this thesis in the form of fuzzy rules; **Reasoning engine,** which produces candidate decisions from the current inputs; **Decision evaluation,** which scores every candidate before it is allowed to act; **Presentation and oversight,** which enable the operator to inspect, question, and override the outcome.

In time-critical domains two further elements become essential: Explicit uncertainty handling, since inputs are incomplete and unreliable, and timeliness, since a correct recommendation that arrives late has no operational value (French, Maule, & Papamichail, 2019).

### 2.2.2 Closed versus Open Decision Spaces

In a **closed decision space,** the **alternatives are enumerated** before reasoning begins, and the system can only rank them. This structure characterizes classical multi-criteria decision making (Chen & Hwang, 1992; Zavadskas & Turskis, 2011), in which a fixed alternative set is scored against weighted criteria (Hwang & Yoon, 1981; Triantaphyllou, 2000; Malczewski, 1999). The approach fails in wildfire response for a structural reason. Fire behavior evolves continuously, and an incident routinely produces conditions that no predefined alternative covers. Field studies document the consequence. Commanders use existing tools to justify decisions they have already made, not to generate new options (Fillmore & Paveglio, 2023; Rapp, Rabung, Wilson, & Toman, 2020).

An **open decision space** requires three capabilities: **(i) modification of existing alternatives during operation**, **(ii) construction of genuinely new alternatives**, and **(iii) verification of every new alternative before it acts**, since an unchecked generated action constitutes an operational hazard (Seshia, Sadigh, & Sastry, 2022).

Naturalistic decision research established early that experienced commanders do not select from menus at all. In the recognition-primed model, a decision maker constructs a single plausible course of action from recognized patterns, evaluates it by mental simulation, and repairs or replaces it when the simulation fails, so that option generation rather than option ranking is the expert behavior a support system should serve (Klein, 2017; Ross, Shafer, & Klein, 2006). The same expert behavior has a direct computational counterpart. The evolving fuzzy systems literature made rule structure itself adaptive. Evolving Takagi-Sugeno models create a new rule online whenever incoming data are poorly represented by the existing base, and subsequent variants supplied principled mechanisms for splitting, merging, and pruning rules on streaming data (Angelov & Filev, 2004; Lughofer, 2011). An online value-based selector contributes to the timing of openness, learning from the outcomes of past trials when to invoke the structure-changing stages rather than discovering new actions itself (Sutton & Barto, 2018). Most recently, generative models, including large-language-model agents applied to wildfire analysis and response, have been used to draft candidate rules or actions that lie outside the current structure altogether (Xie, et al., 2024; Chen, et al., 2025; Chandra, Kumar, Patra, & Agarwal, 2025).

DisasterAware combines the **three mechanisms**: An **evolving Takagi-Sugeno** (1985) rule base for modification, a **generative proposer** for construction, and a satisficing acceptance gate for verification (Simon, Rational choice and the structure of the environment, 1956), so that the decision space remains open while every realized action remains admissible. The generative stage composes a fixed set of verified physical effects into new interventions, from re-weighted macros to clause actuators with their own targeting geometry, and only where the map affords them. The same stage may define a new intermediate concept as a normalized weighted combination of existing features and concepts, so the reasoning vocabulary can grow alongside the intervention vocabulary.

### 2.2.3 Centralized and Distributed Decision Support Architectures

In centralized architecture every observation flows to a single reasoning core and every decision flows back out of it. This pattern offers simplicity and global consistency, but the reasoning core constitutes a bottleneck. Communication grows with the size of the domain, latency grows with communication, and a failure of the core silences the entire system (Shim, et al., 2002; Arnott & Pervan, 2005).

In a distributed architecture reasoning moves to local nodes and information flows in both directions. Local nodes send compact summaries upward when regional balance is needed, and the upper layer returns allocations and priorities downward. Multi-agent systems research provides the coordination mechanisms (Sycara, 1998; Wooldridge, 2009), and wildfire DSS reviews explicitly call for such designs (Vásquez, 2021).

DisasterAware follows the **distributed pattern** with one restriction that later chapters exploit. Local DSS agents perform all inference on region-level observations, while the global layer performs no inference at all; it only assigns resources and normalizes priorities across regions. **Upward flow**, from each Local DSS agent to the global coordination layer, carries three compact quantities, namely the **region priority** derived from the top-level operational priority concept, the **resource request** implied by the proposed interventions, and a **confidence summary** of the observations behind them. **Downward flow**, from the global layer back to every Local DSS agent, returns the **resource allocation**, the **normalized priority weights**, and the **acceptance thresholds** in force. Neither direction carries raw sensor fields or rule activations, which keeps the channel narrow, the coordination auditable, and the local decision path short. Figure 2.2 contrasts the two patterns and marks the restriction that keeps the global layer non-inferential, which preserves scalability.

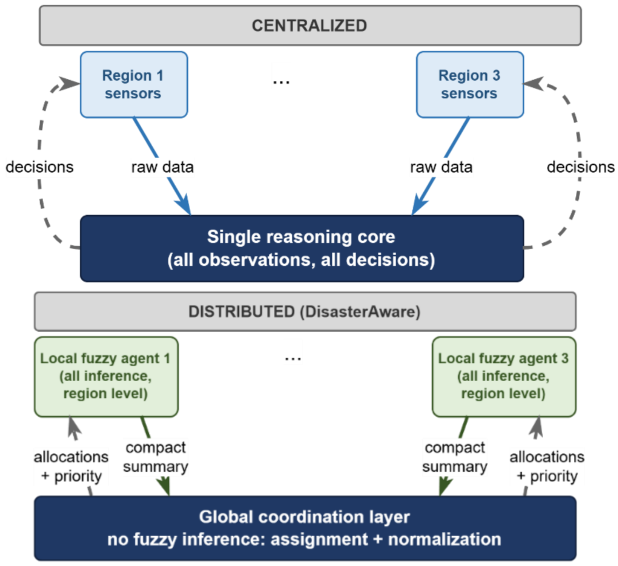


Figure 2.2 Information flow in centralized and distributed DSS architectures

## 2.3 The Input Space: Heterogeneous Sensing to Observation Features

A wildfire DSS draws on heterogeneous sensing sources: Weather stations and forecast services, satellite and aerial imagery, in-situ sensors and unmanned aerial platforms tasked over the incident, geographic information layers for terrain, fuel, assets, and roads, and the resource tracking and field reports of the response organization itself. None of these sources observe the fire state directly; each delivers a partial view with its own refresh cycle. For this reason, the features are defined as bounded, observation-derived quantities rather than raw signals.

Concretely, five families of sources feed the framework. Meteorological networks and numerical forecasts deliver gridded temperature, humidity, wind, and precipitation at a fixed cadence. Spaceborne and airborne imagery provides active-fire detections, burned-area mapping, and periodic vegetation and fuel updates, with wide coverage but coarse revisit intervals (Jain, et al., 2020). In-situ sensing, including ground stations and unmanned aerial platforms tasked over the incident, offers high-rate local observations with narrow footprints. Geographic information layers contribute to the quasi-static context, namely terrain, fuel models, road networks, assets, and population (Scott & Burgan, 2005) (Scott & Burgan, 2005). Finally, the response organization itself constitutes an information source. Resource tracking, dispatch logs, and field reports carry the evolving state of crews, equipment, and access (Martell, Forest fire management, 2007).

These sources are heterogeneous in every operationally relevant respect. They differ in modality, spatial coverage, refresh rate, and reliability, and they produce information of various kinds, from continuous physical fields to categorical status reports. No single raw signal can drive decision rules directly. The heterogeneous stream is therefore reduced by feature extraction to a small set of bounded, observation-derived quantities, and it is exactly this set that constitutes the input space of the framework.

The ten features and the reasoning behind each are compiled in Table 2.1; their formal derivation from the simulation state, together with the full literature grounding, is given in the design specification. The set spans threat, exposure, feasibility, and timing, so no operational question of Section 2.4 is left without a supporting feature and none of the ten is redundant.

Table 2.1 The ten observation features, their sources, and their rationale

| Feature | Primary source | Rationale |
|---|---|---|
| **Fire intensity $z_1$** | Fire state estimation from imagery and in-situ sensing | Fireline intensity in the tradition of Byram and the Rothermel model is the standard descriptor of how intensely a front is burning |
| **Spread potential $z_2$** | Fire state and weather | Rate of spread from fuel, wind, and slope in the Rothermel tradition remains the operational basis of spread assessment |
| **Weather severity $z_3$** | Weather stations and forecasts | Fire-weather index systems condense a small set of weather observations into spread and intensity potential |
| **Ignition proximity $z_4$** | GIS road and infrastructure layers | Human-caused ignitions cluster near roads and infrastructure, so distance to them drives ignition probability |
| **Fuel load $z_5$** | Fuel model layers | Standard fuel models parameterize combustible load and drive every surface-fire prediction |
| **Asset exposure $z_6$** | GIS asset and urban interface layers | Values-at-risk mapping quantifies what a fire threatens and supports risk-based response |
| **Resource accessibility $z_7$** | Terrain, fuel, and access layers | Suppression difficulty combines the terrain, fuel, and access conditions of a site |
| **Access and road status $z_8$** | GIS road status and field reports | Access governs both suppression logistics and evacuation egress |
| **Suppression availability $z_9$** | Resource tracking | Fireline production rates and available capacity determine whether suppression can be mounted |
| **Temporal urgency $z_{10}$** | Ignition timing and incident clock | Incident command is time-critical; recognition-primed decision research formalizes acting under urgency |

Heterogeneity also imposes admission criteria on the input space. A quantity is admitted to the feature set only if it satisfies four conditions.

- **Observability.** It must be computable from what sensors and reports actually deliver, never from hidden simulator state.
- **Boundedness.** It must admit normalization to a fixed interval, so that membership functions remain stable across scenarios.
- **Operational meaning.** An expert must be able to read it and phrase rules over it without translation.
- **Non-redundancy.** It must contribute discriminative information that the remaining features do not already carry.

Sufficiency of the resulting set is argued on three grounds and then evaluated. By construction, the ten features span the threat, exposure, feasibility, and timing dimensions of the operational questions, so no question lacks a supporting feature; each feature carries established wildfire-science grounding rather than convenience, as the table records; and none duplicates information carried by the others. The design chapters show that the concept mappings require no further input, and the evaluation chapter exercises the closed loop under biased operational parameters and degraded observation; a per-feature ablation is left to future work.

## 2.4 Concept-Driven Fuzzy Reasoning

This section assembles the reasoning machinery of DisasterAware. Fuzzy reasoning is chosen for three reasons established in the preceding sections. The evidence reaching a wildfire DSS is graded and linguistic rather than crisp; the knowledge that binds observations to interventions exists chiefly as expert phrasing, which fuzzy rules encode without translation; and the resulting inference chain remains inspectable, which the traceability requirement of this thesis makes mandatory. The section first recalls how fuzzy inference turns linguistic expertise into computable rules, then confronts the combinatorial obstacle that arises when such inference is attempted over the full observation vector and finally introduces the conceptual middle layer that resolves the obstacle while preserving interpretability. The section thereby introduces the two design questions addressed in the design chapters: **Which features are admitted to the input space, and over which concepts the decision rules are written.**

### 2.4.1 Linguistic Uncertainty and Fuzzy Inference

Operational assessments in wildfire response arrive as graded, linguistic judgments. A spread potential is high, a suppression opportunity is marginal, a boundary estimate is uncertain. Fuzzy set theory supplies the formal counterpart of such judgments by admitting partial membership (Zadeh, Fuzzy sets, 1965; Mendel, 2017). A fuzzy set over a universe of discourse is characterized by a membership function that assigns each element a degree of membership, as formalized in ( 1 ); under this semantics a single wind reading may simultaneously hold membership 0.7 in strong and 0.3 in moderate. In ( 1 ), $Z$ denotes the universe of discourse of an input, $\mathcal{A}$ a linguistic term over it, and $\mu_{\mathcal{A}}(z)$ the degree to which a reading $z$ belongs to $\mathcal{A}$. This graded semantics removes the brittle thresholds of crisp classification and preserves expert phrasing under formalization (Pedrycz, 1993). Domain knowledge enters through linguistic rules of the canonical conjunctive form ( 2 ), whose activation strength is the conjunction of its antecedent memberships ( 3 ).

$$\mu_{\mathcal{A}}: Z \to [0,1], \ \ \mathcal{L} = \{(z, \mu_{\mathcal{A}}(z)) : z \in Z\} \tag{1}$$

$$R_r\text{: IF } z_1 \text{ is } \mathcal{A}_{r,1} \text{ AND } \dots \text{ AND } z_n \text{ is } \mathcal{A}_{r,n} \text{ THEN } u \text{ is } \mathcal{B}_r \tag{2}$$

$$\alpha_r = \min_i \ \mu_{\mathcal{A}_{r,i}}(z_i) \tag{3}$$

In ( 2 ) and ( 3 ), $z_1, \dots, z_n$ denote the input variables, $\mathcal{A}_{r,i}$ the linguistic terms selected by rule $R_r$, $\mathcal{B}_r$ the consequent term over the output $u$, and $\alpha_r$ the firing strength of rule $r$, with the minimum operator realizing the conjunction of the antecedents. Mamdani inference aggregates the consequents of the active rules and defuzzifies the aggregate into a crisp output (Mamdani, 1974), with the centroid form of ( 4 ) as the standard choice.

$$u^* = \frac{\int u \cdot \mu_{agg}(u)\mathrm{du}}{\int \mu_{agg}(u)\mathrm{du}}, \quad \text{where } \mu_{agg}(u) = \max_r \min(\alpha_r, \mu_{\mathcal{B}_r}(u)) \tag{4}$$

In ( 4 ), $\mu_{agg}$ is the aggregated output membership, obtained by clipping each active consequent at its firing strength and taking the pointwise maximum, and $u^*$ is the resulting crisp output. The Takagi-Sugeno variant replaces the fuzzy consequent with a functional one and combines outputs as an activation-weighted average ( 5 ).

$$u = \frac{\sum_r \alpha_r \cdot g_r(z)}{\sum_r \alpha_r} \tag{5}$$

In ( 5 ), $g_r(z)$ is the functional consequent of rule $r$, typically affine in the inputs. The identification theory behind this form is presented in (Takagi & Sugeno, 1985). DisasterAware adopts the zero-order Takagi-Sugeno form throughout the decision layer. Each rule consequent is a singleton intensity, and the candidate intervention is the activation-weighted average of the singletons of the rules that fire, so the inference stays interpretable while the consequents remain exposed as numeric parameters that the adaptation loop can tune.

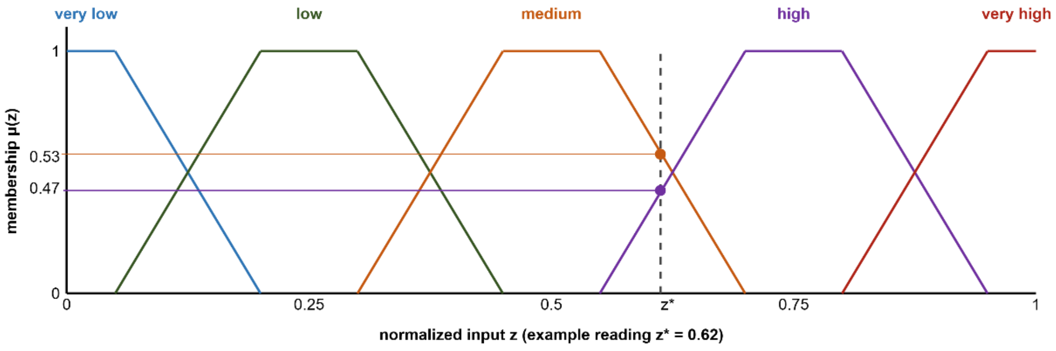


Figure 2.3 Five-term trapezoidal membership partition with an example reading

Figure 2.3 illustrates the five-term trapezoidal partition of a normalized input used throughout the design chapters; the five-term resolution is a design choice of this thesis, fixed for every feature and concept in Chapter 4. Adjacent terms overlap so that every reading activates at most two terms. Suppose the normalized spread potential of a cell reads $z_2 = 0.62$. Fuzzification against the partition of the figure yields membership 0.53 in MEDIUM and 0.47 in HIGH, and every other term stays at zero. Two rules fire, IF spread potential is MEDIUM THEN suppression effort is moderate with strength 0.53, and IF spread potential is HIGH THEN suppression effort is HIGH with strength 0.47. The zero-order Takagi-Sugeno aggregation weights each singleton consequent by the strength of the rule that carries it, so the recommendation is 0.53 times 0.5 plus 0.47 times 0.8, which yields a suppression effort of 0.64 on the normalized intensity scale, and every intermediate quantity remains visible to the operator.

### 2.4.2 Mamdani and Takagi-Sugeno Inference: Mechanism and Choice

Two classical mechanisms called Mamdani (1974) and Takagi-Sugeno (1985) turn the activations of ( 3 ) into one crisp action. The difference between them determines what an online learner can touch, what a verification gate can compare, and what a forecast loop can afford. In both mechanisms, crisp inputs are fuzzified against the term partitions, and each rule fires with the strength of its weakest antecedent. The methods differ in the form of the consequent.

**Mamdani inference** keeps the consequent linguistic. Each active rule clips or scales a consequent membership set at its firing strength, the clipped sets are aggregated pointwise, and the aggregate is defuzzified into a crisp value, typically by the centroid of ( 4 ). The strengths of this construction are real: The knowledge remains linguistic end to end, the output set makes the ambiguity of a decision visible, and the scheme reads naturally for expert elicitation. Its costs are equally real. Every evaluation carries a numeric integration over the output universe; the output surface is piecewise and develops plateaus between rule cores, so nearby situations can read identically; and adaptation can act only on membership geometry, because the consequents are sets rather than parameters.

**Takagi-Sugeno inference** replaces the consequent set with a function of the inputs, as in ( 5 ), and obtains the crisp output directly as the firing-strength weighted average of the rule consequents. This thesis uses the zero-order special case, in which every consequent is a constant intensity:

$$u = \frac{\sum_{r \in R} \alpha_r \cdot v_r}{max(\sum_{r \in R} \alpha_r, 1)}, \quad v_r \in [0,1] \tag{6}$$

In ( 6 ), $R$ is the set of active rules, $\alpha_r$ is the firing strength of rule $r$ from ( 3 ), and $v_r$ is its constant consequent, read directly as an intervention intensity on the unit interval. The lower bound of one in the denominator implements the coverage discipline of the framework. When the whole base fires weakly, the output is attenuated rather than renormalized upward, so a barely covered situation cannot masquerade as a confident command. The output is crisp by construction, no defuzzification step exists, and the value interpolates smoothly between the actions of the rules that share the situation. The design chapters instantiate this form once per intervention family and region, with consequents $v_{r,m}$ and outputs $u^i_{m,k}$ indexed by the family m and the region $i$.

Table 2.2 Mamdani and zero-order Takagi-Sugeno inference compared

| Aspect | Mamdani | Takagi-Sugeno (zero-order) |
|---|---|---|
| **Consequent form** | Fuzzy set over the output universe | Crisp value in [0,1] per rule |
| **Output computation** | Clip, aggregate, then defuzzify by centroid | Firing-strength weighted average, closed form |
| **Cost per evaluation** | Numeric integration over the output universe | One multiply-add per active rule |
| **Output surface** | Piecewise, plateaus between rule cores | Smooth interpolation between rule actions |
| **Adaptation surface** | Membership geometry only | Memberships and consequent values; consequents are bounded numbers a learner can nudge |
| **Verification of a generated rule** | Requires defuzzification semantics per proposal | Consequents are auditable numbers; simulation gates compare scalars |
| **Interpretability** | Linguistic input and output | Linguistic antecedents; numeric output read as an intensity command |

Figure 2.4 contrasts the two pipelines step by step, and Table 2.2 condenses the comparison. The front of the pipeline is identical; everything that differs follows from the consequent form.

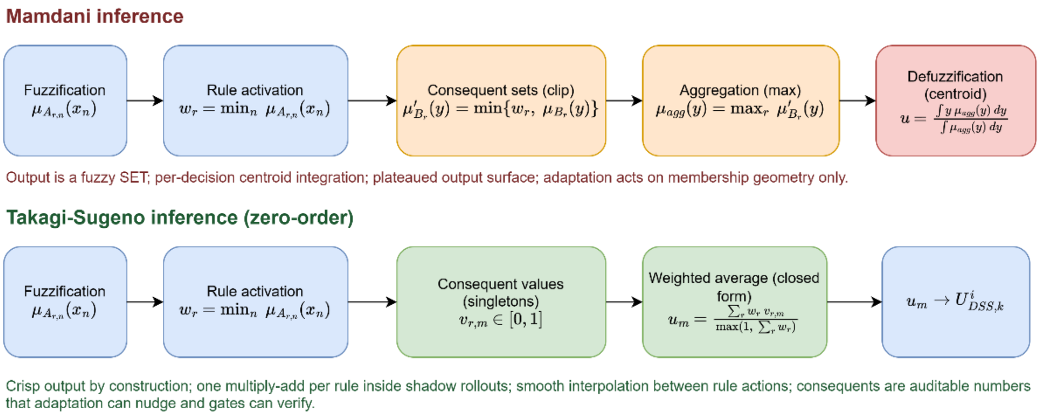


Figure 2.4 Mamdani and zero-order Takagi-Sugeno inference pipelines

**The choice of Takagi-Sugeno** in DisasterAware follows from four requirements of the architecture:

1. **Evaluation economics**: Every candidate decision and every adaptation trial is judged inside shadow forecasts, so the inference closed form of ( 6 ) is executed thousands of times per incident; a per-evaluation centroid integration would price the forecast loop out of its decision cycle.

2. **Actuation semantics**: The outputs of the decision layer are intensity commands on the interventions, and a zero-order consequent is exactly such a command; Mamdani would compute a linguistic output only to defuzzify it into the number the actuator needed all along.

3. **Adaptivity**: The evolving-fuzzy stage nudges consequents by bounded steps and inserts terms into partitions, and the generative stage proposes rules whose consequents must be audited by simulation gates; both demand consequents that are numbers with predictable effect on the output, which is precisely what the weighted average provides and what the identification theory behind Takagi-Sugeno models is built for.

4. **Interpretability** is retained where it matters. The antecedents remain linguistic over the concept vocabulary, so every decision still reads as a sentence, while the numeric consequent is the part a commander audits as a dosage rather than a word.

Mamdani inference remains the right tool where the output itself must stay linguistic, for advisory displays, elicitation sessions, and human-facing summaries, and nothing in the architecture forbids attaching such a presentation layer downstream. The decision path, by contrast, is Takagi-Sugeno throughout. It takes crisp inputs and produces crisp outputs, it is differentiable in its parameters, and it is inexpensive enough to run inside the shadow forecasts that assess a candidate before it is trusted.

### 2.4.3 The Rule Explosion Problem

The obstacle to applying fuzzy inference directly over a rich observation vector is combinatorial. A complete rule base over the full feature set, with each feature partitioned into a fixed number of linguistic terms, contains the number of rules given in ( 7 ), where $N_{\text{feature}}$ denotes the number of features and $N_{\text{term}}$ the number of linguistic terms per feature. As an example, 10 features with 5 linguistic terms each would demand $5^{10} = 9{,}765{,}625$ rules, close to 10M, and even a modest 3 terms per feature already yields $3^{10} = 59{,}049$ rules. A base of this size cannot be elicited from experts, maintained by engineers, or audited by operators (Mendel, 2017).

$$|R| = {N_{\text{term}}}^{N_{\text{feature}}} \tag{7}$$

### 2.4.4 Concepts as an Intermediate Abstraction

The decisive reduction is achieved through a conceptual middle layer. Rather than writing rules over raw features, DisasterAware first composes the features into a compact, multi-level concept hierarchy. Six base concepts summarize physically related features; three mid-level and two higher-level decision concepts built on them answer, each exactly once, the recurring operational questions; and a single top concept, operational priority, serves cross-region coordination. These counts follow from the construction rather than from free choice. Each base concept summarizes one group of physically related features, and the grouping of the ten features yields exactly six such groups, stated group by group in the closing paragraph of this section and, with their aggregation weights, in Table D.1; the five decision concepts answer the five operational questions of Table 2.3 one for one; and cross-region coordination needs a single quantity. The construction that fixes the hierarchy, with its aggregation weights, is given in Chapter 4. The questions, the decision concepts, and what constructs each are listed in Table 2.3.

Table 2.3 Operational questions and the decision concepts derived from them

| Operational question | Decision concept |
| --- | --- |
| **How severe is the fire threat in this cell?** | Fire Threat Level |
| **Can an intervention feasibly be mounted here?** | Suppression Feasibility |
| **How exposed are the assets here?** | Asset Exposure Risk |
| **How urgent is action here?** | Intervention Urgency |
| **Is evacuation pressure building here?** | Evacuation Pressure |

The derivation follows two principles. Each question must be one that incident commanders demonstrably ask during response operations (Thompson & Calkin, 2011; Klein, 2017), and **each concept must answer its question completely while answering no other, which keeps the concept set complementary and free of redundancy**.

Rules are then written over these concepts, while features only feed the concept mappings. With $N_{\text{feature}}$ features grouped into $N_{\text{concept}}$ concepts drawing on at most $N_{input}$ features each, the rule burden collapses from a product to a sum, as summarized in ( 8 ). The active rule base itself stays compact: The evolving mechanism instantiates only the concept combinations that observed situations activate, and representative rules are enumerated in Table E.1, and the complete concept construction mappings in Table D.1, so every rule that can act in the field is stated explicitly.

$$N_{\text{term}}{}^{N_{\text{feature}}} \dashrightarrow N_{\text{concept}} \cdot N_{\text{term}}{}^{N_{input}} + N_{\text{term}}{}^{N_{\text{concept}}} \qquad (8)$$

Here $N_{\text{concept}}$ is the number of decision concepts at the top of the hierarchy and $N_{\text{input}}$ the largest number of features feeding a single concept; a reduction whose exact magnitude for the configuration of this thesis is established in Chapter 4. In DisasterAware the concept mappings are realized as weighted aggregations rather than as rule bases, so only the decision rules over the concepts materialize as a rule base. The intermediate layer provides more than combinatorial economy. Concepts constitute the traceability vocabulary of the entire framework. Every recommendation can be read back as a statement over fire threat, suppression feasibility, asset exposure, intervention urgency, and evacuation pressure, and it is precisely this granularity that the evaluation machinery of Section 2.5 and the operator-facing explanations exploit. The magnitude of the reduction is substantial. Directly over $N_{\text{feature}} = 10$ features with $N_{\text{term}} = 5$ linguistic terms, a flat rule base would span $5^{10} \approx 9.8 \cdot 10^6$ antecedent combinations. Over the $N_{\text{concept}} = 5$ decision concepts at the top of the hierarchy the antecedent space shrinks to $5^5 = 3125$ combinations, and the evolving rule base instantiates only the small active subset that observed situations require.

The base layer itself draws directly on the features. Fire severity combines fire intensity $z_1$ and spread potential $z_2$; spread hazard combines spread potential, weather severity, and ignition proximity; fuel hazard combines weather severity and fuel load; asset value derives from asset exposure; crew reachability combines resource accessibility with access and road status; and logistics support derives from suppression availability. One derivation chain then runs from end to end.

Fire severity, spread hazard, and fuel hazard combine into fire threat level, the quantity that answers the first operational question and fires decision rules. Base concepts thus function as interpretable intermediate hazards, while the five decision concepts at the top of the hierarchy carry the operational meaning. The hierarchy is a seed rather than a ceiling. The generative stage of the design chapters may add an intermediate concept as a weighted combination of existing features and concepts, and an admitted concept enters the hierarchy on the same footing as a seeded one.

## 2.5 The Decision Space: Actions and Their Evaluation

The concept hierarchy of the previous section supplies the vocabulary over which decision rules are written; the output of those rules is an assignment of intensities to concrete actions. This section therefore fixes the decision space in two steps: First the vocabulary of interventions that can be ordered at all, then the cost model, the acceptance rule, and the uncertainty treatment by which candidate assignments over that vocabulary are judged.

### 2.5.1 Actions and Interventions in Wildfire Decision Support

A wildfire DSS is operationally useful only if its outputs correspond to orders that an incident commander can issue. The decision space of DisasterAware is therefore built from a fixed vocabulary of intervention families, each applied with an intensity over the grid, so that every rule developed in the later chapters binds to a concrete, executable action (Thompson & Calkin, 2011; Pacheco, et al., 2015). Six intervention families cover the levers reported in the wildfire response literature.

Table 2.4 lists each family, the concrete orders it represents, the input-level effect through which it acts on the simulation inputs, and the literature that grounds it. The list is short and closed at the level of families, which keeps the action vocabulary interpretable and allows every recommendation to be read as an executable order; **openness enters not through the invention of new physical effects but through the generation of new rules, through interventions composed from the verified effects and the existing families, and through intermediate concepts defined over the existing features and concepts.**

Table 2.4 Intervention families, representative orders, and input-level effects

| Intervention | Representative orders | Input-level effect |
|---|---|---|
| **Suppression effort (Martell, Forest fire management, 2007)** | Direct attack on the active front; indirect attack with backburns and aerial retardant drops | Reduces local fire intensity and the ignition probability of neighboring cells |
| **Resource deployment (Pais, Carrasco, Moudio, & Shen, 2021)** | Dispatch, staging, and repositioning of crews, and aircraft across regions | Changes the suppression capacity available to each region |
| **Containment line (Ager, Vaillant, & Finney, 2010)** | Firebreak construction and control-line placement along selected boundaries | Removes fuel along the line and blocks propagation paths |
| **Asset protection (Calkin, O'Connor, Thompson, & Stratton, 2021; Noonan-Wright, et al., 2011)** | Point protection of structures and critical infrastructure in the fire path | Lowers the loss suffered by a protected asset if the fire reaches it |
| **Evacuation (Turoff, Chumer, Van de Walle, & Yao, 2004)** | Evacuation orders, shelter-in-place, staging of the exposed population | Reduces population exposure in threatened cells |
| **Public warning (Comfort, Boin, & Demchak, 2010)** | Alerts and information updates issued to the affected public | Raises preparedness and shortens evacuation lead time |

Beneath the six doctrine intervention families the simulator verifies a short action list of physical effects: Wet, clear, ignite, coat, draft, evacuate, and prime. Each entry is a field the engine already writes and the cost model already prices, and the design chapters formalize the list in Table 4.9. Three further base channels built on these effects, water drafting, aerial retardant drop, and tactical burn, are validated in the engine although no seed rule orders them, and they are announced to the generative stage together with their map availability, so a water-dependent channel is offered only where the map carries a water body. A composed intervention may draw on the full action list and on every base channel, and it enters the rule base only through the verification gates of Section 4.5. The generative stage never defines a new physical effect; it composes and deploys the effects the simulator already provides.

Inaction is not modeled as a seventh type. No-intervention is the zero-intensity point of every type rather than a separate alternative, which keeps the decision space continuous and lets the cost model of this section price inaction the same way it prices action. The design has two practical consequences. First, deliberate restraint becomes an explicit, costed recommendation rather than a silent default, and the operator can verify that action is evaluated and that restraint carried the lower expected cost. Second, the adaptation loop can move continuously between restraint and engagement, since raising an intensity from zero requires no structural change to the decision but only a different point in the same intervention field.

A decision layer requires an explicit evaluation criterion before it can adapt, since every candidate intervention must be reduced to a scalar measure of expected cost. This section grounds the cost model of DisasterAware; the acceptance mechanism, the uncertainty treatment, and the sensitivity analysis that build on it are developed in the design and evaluation chapters.

### 2.5.2 Decision Cost and Its Components

A candidate intervention is evaluated through its expected cost. Wildfire economics quantifies this through the cost-plus-loss principle. The total cost of a response comprises the expenditure on suppression and the net value lost to the fire (Martell, Forest fire management, 2007; Thompson & Calkin, 2011). DisasterAware adopts the same additive form, so the cost of a decision is the weighted sum of the response effort it spends and the losses it fails to prevent, in the form given in ( 9 ).

$$J(U_{DSS,k}) = \sum_j w_j \cdot J_j(U_{DSS,k}), \sum_j w_j = 1, \ J_j \in [0,1] \quad (9)$$

In ( 9 ), $U_{DSS,k}$ denotes the intervention field at decision step $k$, realized region by region in the design chapters as $U^i_{DSS,k}$, $J_j$ the individual loss and response terms, each normalized to [0,1], and $w_j$ their non-negative priority weights. The loss terms are the recognized values at risk in wildfire response. The components $J_j(U_{DSS,k})$, their scope, and the rationale for their inclusion are listed in Table 2.5.

Table 2.5 Decision cost components, their scope, and their rationale

| Cost | What it prices | Normalization |
|---|---|---|
| **Burned area $J_{burn}$** | Land and ecological loss over the affected cells. (Martell, Forest fire management, 2007; Thompson & Calkin, 2011) | Burned cells divided by the burnable cells of the scenario |
| **Asset loss $J_{\mathrm{asset}}$** | Structures in the wildland-urban interface together with power, transport, and facilities. (Calkin, O'Connor, Thompson, & Stratton, 2021; Pais, Carrasco, Moudio, & Shen, 2021) | Value lost divided by the total exposed value |
| **Population exposure $J_{\mathrm{pop}}$** | Life safety of the exposed population. (Turoff, Chumer, Van de Walle, & Yao, 2004; Comfort, Boin, & Demchak, 2010) | Person-steps exposed divided by the population at risk times the horizon |
| **Response cost $J_{\mathrm{resp}}$** | Every ordered action, priced by applied intensity, including deployment, evacuation logistics, and warning. | Committed capacity divided by the total available capacity |
| **Response delay $J_{delay}$** | Mean number of decision steps between threat onset in a cell and the first intervention reaching it. (French, Maule, & Papamichail, 2019) | Delay steps divided by the scenario horizon |

Burned area captures land and ecological loss; asset loss captures, in a single term, both the property of the wildland-urban interface and the power, roads and facilities whose failure cascades beyond the fire (Pais, Carrasco, Moudio, & Shen, 2021); population exposure captures life safety; a response cost captures the resources committed by every ordered action, not only suppression; deployment, evacuation logistics, and public warning all contribute in proportion to the intensity at which they are applied, and deployed capacity is priced regardless of its eventual effectiveness, so that protective actions carry a nonzero cost for the adaptation loop; and a response-delay term penalizes slow action, measured as the average number of decision steps between a cell first exceeding the threat threshold and the first intervention reaching that cell.

The weights $w_j$ are not free parameters. Their initial ordering follows the priority-setting literature (Dunn, Calkin, & Thompson, 2017), using pairwise comparison in the tradition of the analytic hierarchy process over the doctrine of wildfire risk management (Saaty, 1980; Thompson & Calkin, 2011); the vector is then normalized to sum to one, as ( 9 ) requires. The weights used in this thesis are $w_{burn} = w_{asset} = w_{pop} = 1.0 \; and \; w_{resp} = w_{delay} = 0.2$ before normalization by their total, which places the physical losses of life, assets, and land jointly first and prices response effort and delay as secondary, in line with published incident-management priority orderings; the cross-run comparisons of the evaluation chapter are read on the physical cost, the burned-area, asset, and population terms with their weights renormalized, so no conclusion of this thesis hinges on the response and delay weighting.

The terms are expressed in different units, such as area, monetary value, persons, and time; each term is therefore normalized to [0,1] with respect to a scenario reference scale before weighing. Without this normalization the weighted sum, and any sensitivity analysis over the weights, would lose meaning; the sensitivity analysis reported in the evaluation chapter relies on this property. The burned area and asset-loss terms are disjoint by construction. Burned area prices land and ecological loss only, while structure and infrastructure values are counted exclusively in their own terms, so no value is counted twice. The reference scales are stated per component in the last column of the cost table, so the terms become dimensionless and mutually summable before the weights apply.

These reference scales also make the model transferable to a live incident. In the field the static scenario totals are replaced by a rolling reference envelope, namely the burned area, the exposed value, and the exposed population that the spread forecast predicts for the current operational period if no intervention is ordered, which is the operational counterpart of probabilistic spread products already in use. Two safeguards keep the construction sound. All denominators are frozen at the start of each operational period, so the adaptation loop cannot improve its apparent score by shrinking its own reference, and every denominator is bounded below, so the ratios stay defined when little or no spread is forecast. Life safety is subject to one further safeguard. A candidate whose predicted population exposure exceeds a hard ceiling is rejected at the acceptance gate outright, which aligns the weighted model with the doctrine that treats life safety as an inviolable constraint.

The component set is deliberately compact. Fewer terms would conceal trade-offs that commanders must weigh, while additional terms are unnecessary at operational granularity, since finer losses aggregate within the structure and infrastructure terms and are separated by the weights rather than by new components. A compact, interpretable sum allows the decision layer to evaluate candidates online and enables the operator to see why one intervention outranks another.

### 2.5.3 Satisficing Acceptance and the Graduated Fail-Safe

The cost model of the previous subsection serves as an acceptance criterion rather than as an objective for global optimization. Global minimization over a spatial decision field is computationally intractable under real-time constraints (Papadimitriou & Tsitsiklis, 1987) and presupposes information that incomplete observation cannot supply. DisasterAware follows the satisficing principle of bounded rationality (Simon, Rational choice and the structure of the environment, 1956; Gigerenzer & Selten, 2001; Simon, The Sciences of the Artificial, 1996). A candidate intervention is accepted when its expected cost clears an adaptive threshold, as given in ( 10 ).

$$\text{accept}(U_{DSS,k}) \Leftrightarrow \hat{J}_k(U_{DSS,k}) \leq \min\left(J_{TH}, (1-\delta_{gain}) \cdot \hat{J}_k(0)\right) \quad (10)$$

In ( 10 ), $J_{TH}$ is the fixed ceiling on the normalized cost and $\delta_{gain}$ the required relative gain over the no-action forecast $\hat{J}_k(0)$, with $J_{TH} = 0.35$ and $\delta_{gain} = 0.05$ in this thesis. The hat marks a cost evaluated on a shadow forecast rather than on the realized state. The bound is the tighter of the two conditions, so a situation in which inaction is already cheap demands a stronger candidate; the exact form is given in the design chapters. When no candidate clears the threshold, the framework does not force a binary choice between acting and abstaining. A graduated fail-safe attenuates the best available intervention toward a conservative baseline, ( 11 ).

$$U_{DSS,k}{}' = \lambda_k \cdot U_{DSS,k} + (1-\lambda_k) \cdot U_{DSS,safe}\,, \ \ \lambda_k \in [0,1] \quad (11)$$

In ( 11 ), $U_{DSS,safe}$ is the conservative fallback intervention and $\lambda_k$ the attenuation weight set by decision quality. The design chapters realize this weight as $\lambda_k = min(\frac{Q_k}{\eta}, 1)$, where $Q_k$ is the measured decision quality and $\eta$ the quality gate, with $\eta = 0.60$ in this thesis, so attenuation deepens in proportion to the quality deficit. A value of 1 realizes the intervention in full, intermediate values yield a proportionally reduced commitment, and 0 corresponds to complete fallback to the safe posture. The transition from full intervention through partial intervention to fallback is therefore continuous, auditable, and reversible as observation quality recovers.

The fallback $U_{DSS,safe}$ is not the absence of a decision but a pre-authorized conservative posture. Under it, monitoring and public warning remain active, deployed units hold their current safe positions, and no new offensive commitment is made. In the intervention vocabulary of Table 2.4, $U_{DSS,safe}$ holds suppression effort, resource deployment, containment line, and asset protection at zero and carries the current evacuation and public warning intensities forward. For a candidate with suppression effort 0.8, containment line 0.6, evacuation 0.7, and public warning 0.5, full fallback therefore keeps the evacuation at 0.7 and the warning at 0.5 and zeroes the offensive components. When, for instance, an indirect attack fails the acceptance test because confidence has collapsed over the target sector, attenuation drives the offensive component toward zero while the evacuation and warning components are retained; the original intensity returns once fresh observations restore confidence.

### 2.5.4 Epistemic Uncertainty, Traceability, and Sensitivity

The uncertainty that dominates an active incident is epistemic. The state is under-observed, not merely noisy (Morgan & Henrion, 1990; Walker, et al., 2003). Bayesian updating and Dempster-Shafer belief calibration both presuppose distributional knowledge that cell-level, real-time sensing does not supply (Pearl, 1988; Shafer, 1976; Zadeh, 1986). DisasterAware instead employs confidence gating. Every concept activation is blended with a persistence-based estimate in proportion to the confidence of the observations that produced it, following ( 12 ). $a_k^{obs}$ is the concept activation computed from current observations, $a_{k-1}^{per}$ its persistence-based estimate, carried as the decayed effective activation of the previous step, and $\gamma_k$ the observation confidence; a low-confidence input is thus attenuated toward the most recent trustworthy estimate rather than trusted outright. The design chapters carry each activation as a five-term membership vector and apply this blend to every term; a scalar reading is used only where a single value per concept is needed.

$$a_k{}^{eff} = \gamma_k \cdot a_k{}^{obs} + (1 - \gamma_k) \cdot a_{k-1}{}^{per},\ \ \gamma_k \in [0,1] \qquad (\ 12\ )$$

Inference runs over explicit concepts and rules, so every decision remains traceable to the concept activations that produced it, the rules they fired, and the confidence behind them. Explanation is thus a structural property of the pipeline, in contrast to post-hoc attribution methods (Ribeiro, Singh, & Guestrin, 2016; Guidotti, et al., 2018). On this chain, a sensitivity analysis quantifies how a perturbation of an operational parameter of the closed loop propagates to the decision, as in ( 13 ) where $Sens_{i,k}$ measures the change of the decision cost $J_k$ induced by a change of the varied quantity $p_{k,i}$, with the evaluation chapter reading the physical part of the cost as the observed output. Table 2.6 summarizes what the pipeline traces for every decision, what is actually sensed, and which quantities the sensitivity analysis varies and observes.

$$Sens_{i,k} = (\Delta J_k)/(\Delta p_{k,i}) \qquad (13)$$

Table 2.6 Scope of traceability and sensitivity analysis

| Scope | Content |
|---|---|
| **Traced for every decision** | Concept activations, the rules they fired with their firing strengths, and the observation confidence behind each input |
| **Sensed from the field** | Feature values $z_f$ with per-cell confidence $\gamma_k$; nothing hidden from observation enters the decision path |
| **Sensitivity inputs (what is varied)** | Closed-loop operational parameters: The decision cycle, the no-harm horizon, the observation coverage, the satisficing bound, and the quality gate, together with the fire load, the resource level, the regional partitioning, and the attention threshold |
| **Sensitivity outputs (what is observed)** | Decision cost $J_k$, read across runs as its physical part, intervention intensities $U_{DSS,k}$, and the acceptance outcome |

### 2.5.5 Generative Models and Large Language Models

Generative models produce new content rather than only classifying or scoring existing content. The form most relevant to decision support is the large language model (LLM). An LLM is a neural network, usually a transformer, trained on large text corpora to predict the next token in a sequence (Vaswani, et al., 2017; Brown, et al., 2020). Through this pre-training it acquires broad linguistic competence and a wide store of world knowledge, which it can recombine into text that follows the structure and the vocabulary asked of it.

A pre-trained model can be applied to a new task without further training, through in-context learning. The task, the relevant context, and often a few examples are placed in the prompt, and the model answers from its fixed weights (Brown, et al., 2020). Domain knowledge can be supplied at query time instead of being written into the weights, and the model is steered by instructions and examples rather than retrained. This is what lets a single general model function as a task-specific component.

In DSS, generative models are used as proposers. They draft candidate actions, rules, or plans that a separate mechanism then evaluates. Their strength is the ability to suggest options that a fixed, hand-written repertoire does not contain. Their weakness is that they can produce fluent output that is factually wrong or unsafe (Amodei, et al., 2016), a failure often called hallucination (Ji, et al., 2023), and their internal reasoning is not directly auditable. A generated artifact is treated as untrusted by construction and is admitted only after explicit verification. In this thesis the model is a replaceable proposal mechanism with a fixed contract. It receives a deterministic summary of the situation together with the closed vocabulary, it must answer in the canonical rule form, and nothing it produces carries authority of its own, since admission rests entirely on the verification gates, which contain no model. Recent work in wildfire analysis has begun to deploy LLM agents as assistants whose outputs are drafted for expert review (Xie, et al., 2024).

### 2.5.6 Adaptive and Generative Decision Loops

The preceding sections defined the admissible interventions, the reasoning that produces candidate decisions, and the criteria by which candidates are judged. This section presents the mechanism that keeps the decision space open without compromising safety. DisasterAware organizes this as a staged adaptation loop that escalates only when cheaper corrections have demonstrably failed. The adaptation stages ① and ② follow the evolving-fuzzy-systems literature (Angelov & Filev, 2004; Lughofer, 2011), the third extends it with generation (Xie, et al., 2024; Chen, et al., 2025; Chandra, Kumar, Patra, & Agarwal, 2025).

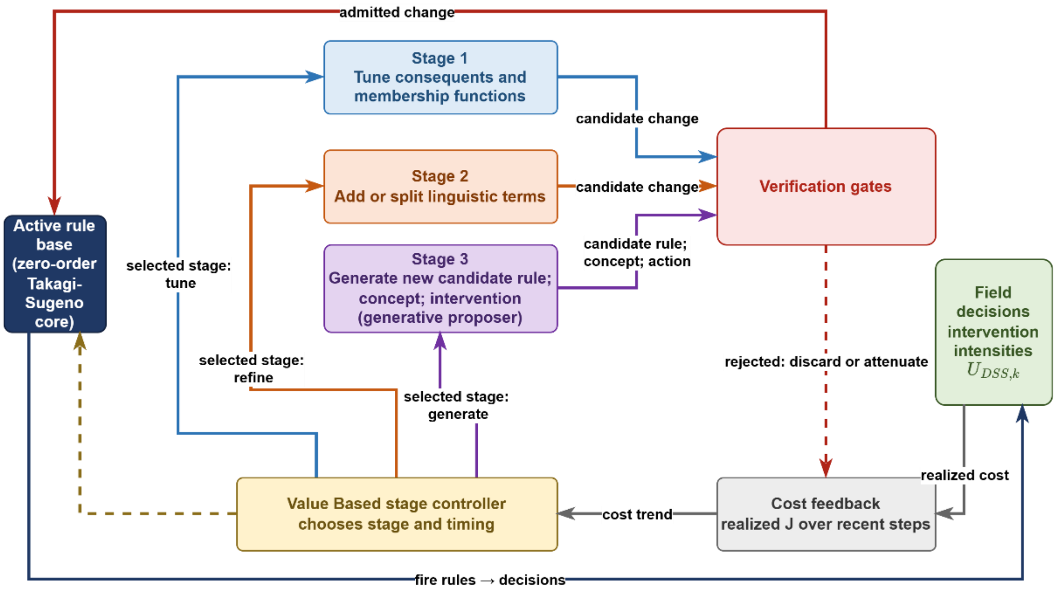


Figure 2.5 Staged adaptation loop with generative proposal and verification

Figure 2.5 depicts the complete loop. The rule base issues decisions, the realized cost is fed back, the value-based stage controller selects one of the three operations forming a ladder of increasing invasiveness. **Parameter tuning** is the least invasive: The evolving-fuzzy stage adjusts the membership-function parameters of existing rules. **Resolution increase** goes one step further, refining the term set by adding a linguistic term or splitting the region of the input space where the residual cost concentrates. **Generative proposal** is the last resort: When neither of the earlier operations removes a persistent cost, a generative model drafts a candidate rule outside the current rule structure, alone or as a package that brings one new vocabulary object: An intermediate concept combined from existing features and concepts, a macro intervention that blends existing interventions, or a clause actuator composed from the verified physical effects. The proposer supplies candidates only;

the stages and gates that admit, field, and retain them are deterministic and model-free. Large-language-model agents have recently begun to function as such proposers in wildfire analysis and response.

The loop is disciplined by two safeguards. The value-based stage controller decides when escalation is warranted, driven by the same cost function that evaluates decisions (Sutton & Barto, 2018). Every candidate, regardless of its origin, must pass the satisficing acceptance gate before it may act, according to the admission condition in ( 14 ).

$$r' \text{ admitted } \Leftrightarrow J_{R\cup\{r'\}} \leq J_R - \delta,\ \ \delta \geq 0 \qquad (14)$$

In ( 14 ), $J_R$ denotes the cost achieved under rule base $R$ and $\delta$ a non-negative improvement margin, set to zero for a plain rule and to the vocabulary margin $\delta_v$ for a package that grows the vocabulary, with the design chapters reading the physical part of the cost for this comparison; an unverified generated rule therefore never reaches the field (Seshia, Sadigh, & Sastry, 2022). The action vocabulary of Section 2.5 remains fixed and interpretable, the rule structure over it evolves and is occasionally extended by generation, and everything that acts has been verified. The design chapters formalize each stage of the loop; the evaluation chapter measures its effect.

The adaptation loop is governed by a value-based, contextual-bandit formulation defined over the following elements. The state vector comprises the recent trajectory of the decision cost, aggregate statistics of the rule base such as the rule count and the mean firing strength, and a summary measure of observation confidence. The action set consists of the three adaptation stages: Tuning the consequents and the membership parameters of existing rules, growing the rule base, and requesting a generative proposal. The reward is the reduction of the forecast physical cost measured on the shadow run at the moment the modification is accepted, diminished by a penalty proportional to structural growth, so that improvements confirmed by the forecast are favored while superfluous structural complexity is discouraged.

The three adaptation stages also admit a direct interpretation in decision-space terms, and this interpretation is adopted throughout the subsequent chapters. Parameter tuning reshapes the boundaries of decision regions that already exist, so the same rules fire over slightly shifted conditions. Resolution increases partition a region more finely, so situations that previously received one response can now be distinguished. Generative proposal opens a region that no existing rule could reach, extending the reachable part of the decision space itself, and the verification gate decides whether that new region becomes part of the admissible space. When the proposal arrives as a package, the enlargement reaches the vocabulary itself: A new intermediate concept adds an antecedent axis to the space, and a new composed intervention adds a consequent that later rules may order. Openness therefore does not imply arbitrariness. Every enlargement of the space is priced by the cost model and admitted only through the verification gate.

# CHAPTER 3

# DISASTERAWARE SIMULATION FRAMEWORK

This chapter defines the structural, computational, and physical foundations of the DisasterAware framework, which serves as the physical backbone of the entire architecture. Its objective is to formalize how wildfire dynamics are represented and computed before the cognitive reasoning mechanisms of the Decision Support System are introduced in the next chapter.

Wildfire evolution is modeled as a spatio-temporal process over a discretized spatial domain partitioned into grid cells. Each cell represents a localized fire condition described by structured state variables: Whether combustion is active, the amount of remaining combustible fuel, local fire intensity, and combustion duration. This state-based representation treats fire propagation as a constrained dynamical system rather than a purely geometric spread.

System evolution is driven by several categories of external forcing. Meteorological drivers such as wind speed and humidity affect spread acceleration and combustion sustainability; terrain properties such as slope and aspect introduce directional bias; fuel characteristics regulate ignition thresholds and burn duration; and operational intervention parameters alter local fuel availability through suppression actions. These influences are kept architecturally separate from the internal wildfire state to preserve clarity and computational stability.

A key design principle is the strict separation between physical state evolution and decision logic. The simulation layer models wildfire behavior under environmental and operational inputs without performing reasoning, optimization, or policy selection. Human interaction and scenario manipulation are permitted only through controlled input-level modifications, never through direct state overwriting.

The Decision Support System is introduced here only to clarify its structural interface with the simulation environment; its internal fuzzy reasoning, layered decision synthesis, and optimization mechanisms are deferred to the next chapter. The remainder of this chapter is organized as follows:

Section 3.1 presents the layered architectural structure and inter-layer data flow of the DisasterAware framework.

Section 3.2 formalizes the external input structure: The simulation input set together with its meteorological, topographic, fuel, ignition, and decisional data layers, and the contextual inputs used by the DSS.

Section 3.3 formalizes the simulation core: The discrete-time state representation and its state variables, the hybrid fire spread transition mechanism, and the interaction and scenario control operator.

Section 3.4 introduces the observation interface that exposes the state to the Decision Support System, states the structural positioning of the DSS Core within the framework, and defers its design specification to the next chapter.

The chapter thereby supplies the physical and computational ground on which the decision layers of the later chapters operate.

## 3.1 Overall System Block Diagram

Figure 3.1 presents the architectural structure of the DisasterAware framework. The system is designed as a discrete-time, state-driven, layered DSS architecture with a clear separation between input acquisition, physical state evolution, decision evaluation, and human interaction. The architectural objective is to preserve physical consistency within wildfire simulation while enabling adaptive and interpretable decision-making.

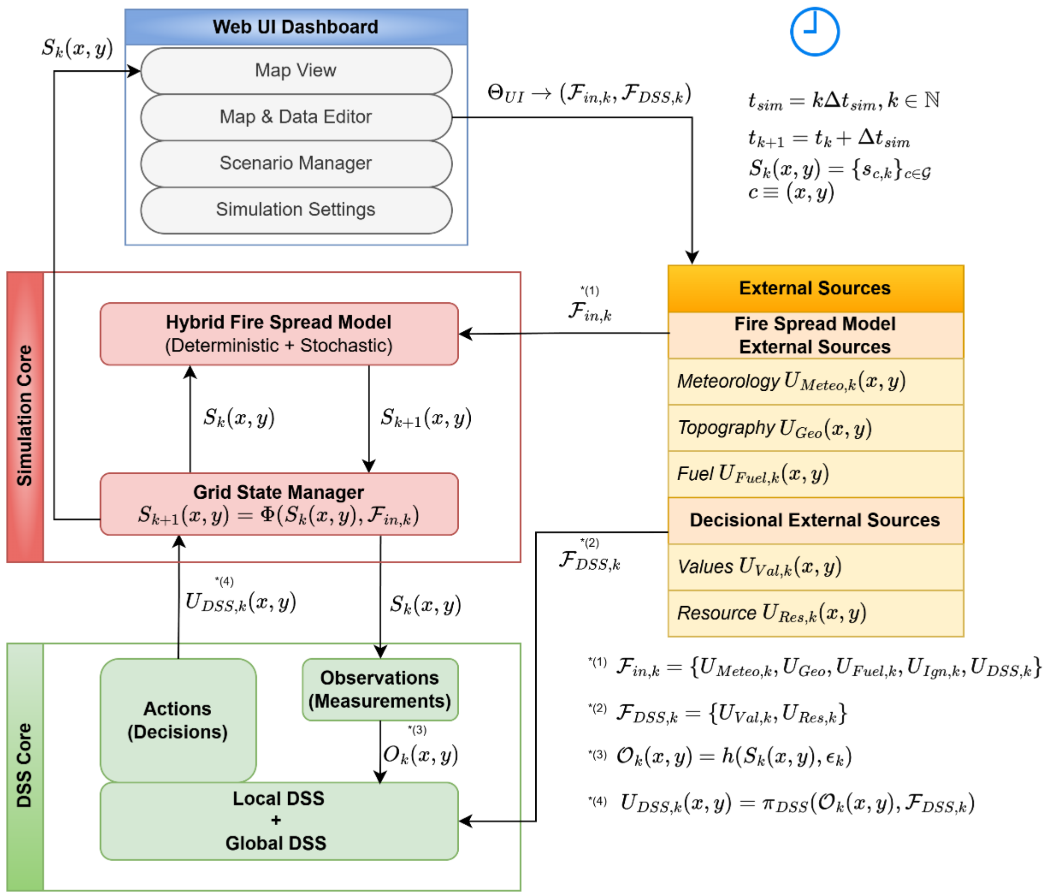


Figure 3.1 DisasterAware system block diagram (simulation focused)

The architecture is organized into four principal layers: External Data Sources, Simulation Core, Decision Support System (DSS) Core, and Web UI Dashboard. Each layer performs a strictly defined function and communicates through structured interfaces rather than direct state manipulation. This separation keeps the physical state consistent and every interface inspectable.

The **External Data Sources** layer provides structured exogenous inputs including meteorological, terrain, fuel, and decisional parameters. These inputs function as boundary conditions for the simulation. Environmental inputs are independent of the system state, while decision-driven inputs generated by the DSS depend on observable quantities derived from the system state.

At the center of the framework lies the **Simulation Core**, which serves as the authoritative computational engine. Wildfire propagation is modeled over a spatial grid where each cell contains a structured representation of local fire conditions. The complete wildfire state includes burning status, available combustible mass, local fire intensity proxy, and time since ignition. This state is maintained and updated at every simulation time step. The Grid State Manager enforces physical constraints such as non-negativity of fuel load and bounded combustion behavior. No external module, including the DSS or the UI, is permitted to directly overwrite internal state variables. All modifications must occur through controlled parameter influence, preserving simulation integrity.

The **Decision Support System Core** operates as a cognitive layer built on top of the Simulation Core. The DSS operates on observable quantities $O_k(x, y)$ derived from the system state through the observation process. This observation layer extracts structured indicators from the raw state variables, such as fire front behavior, acceleration zones, exposure metrics, and operational accessibility conditions.

The DSS re-evaluates the situation once per decision cycle, an interval set in the evaluation chapter so that orders are revised before they age. Within a cycle, threshold symptoms such as a coverage gap additionally engage the adaptation mechanisms of the next chapter. When a decision is generated, it is translated into operational parameters that influence subsequent state transitions indirectly through the simulation engine.

The **Web UI Dashboard** forms the operational boundary between human operators and the computational layers. Unlike the Simulation Core, which is time-synchronous, the UI layer operates asynchronously. The operator can pause the simulation; resume execution; roll back to a stored checkpoint; restart from initial conditions; inject new ignition scenarios; modify operational resource availability; and cancel an active simulation.

The UI never directly modifies internal state variables. Instead, it adjusts permissible configuration parameters that affect future simulation steps. This ensures that human intervention remains controlled, traceable, and reversible. The data flow between layers follows a strict directional structure. External inputs feed the Simulation Core at each time step. The Simulation Core updates the authoritative wildfire state. The DSS observes the current state through observable information $O_k(x, y)$, uses contextual decision inputs $\mathcal{F}_{DSS,k}$, evaluates risk, and generates structured intervention recommendations when required. The UI visualizes both physical state evolution and DSS outputs while allowing controlled scenario manipulation.

As shown in Figure 3.1, this layered organization enforces a clear separation between wildfire physics, decision cognition, and human supervision. The architecture is designed to ensure: (i) Physical consistency of fire propagation; (ii) Modular extensibility; (iii) Horizontal scalability; (iv) Traceable intervention logic; (v) Robust human-in-the-loop (HIL) control.

By maintaining strict boundaries between simulation dynamics and decision reasoning, the framework establishes a stable computational foundation upon which distributed fuzzy reasoning and adaptive coordination mechanisms can be constructed in subsequent chapters.

## 3.2 Simulation Input and External Sources

The simulation input set $\mathcal{F}_{in,k}$ collects every parameter that influences wildfire evolution at discrete time step $k$. These inputs include both exogenous environmental factors and decision-driven intervention parameters generated by the DSS. These inputs function as boundary conditions and forcing mechanisms for the state transition process. At each discrete simulation step $k$, the external input set is updated and provided to the Simulation Core.

External inputs can be categorized into three distinct groups: (i) Environmental input: Including meteorological, terrain, and fuel-related factors that govern physical wildfire dynamics; (ii) Exogenous event inputs: Such as ignition events, which trigger fire initiation independently of system evolution; (iii) Decision-driven inputs:

Intervention parameters generated by the DSS, representing operational actions influencing wildfire evolution.

$\mathcal{F}_{in,k}$ ( 15 ) represents all external parameters that influence wildfire dynamics, namely meteorology $U_{Meteo,k}$ ( 16 ), topography $U_{Geo}$ ( 17 ), fuel $U_{Fuel,k}$ ( 18 ), ignition $U_{Ign,k}$ ( 19 ) and inputs generated by DSS $U_{DSS,k}$ *( 20 )*. This structured separation ensures that environmental forcing, ignition triggering, and operational intervention remain conceptually and functionally distinct.

$$\mathcal{F}_{in,k} = \{U_{Meteo,k}, U_{Geo}, U_{Fuel,\mathrm{k}}, U_{Ign,k}, U_{DSS,k}\} \qquad (15)$$

In addition to the simulation input set $\mathcal{F}_{in,k}$, the DSS process relies on a distinct contextual input set defined as $\mathcal{F}_{DSS,k} = \{U_{Val,k}, U_{Res,k}\}$. This set represents value-related and resource-related information used by the DSS for decision evaluation. Unlike $\mathcal{F}_{in,k}$, which directly influences the state transition operator, $\mathcal{F}_{DSS,k}$ does not affect wildfire dynamics directly. Instead, it provides contextual information that guides decision-generation within the DSS layer.

### 3.2.1 Meteorology Data Layer

Meteorological inputs $U_{Meteo,k}$ ( 16 ) are spatio-temporal environmental drivers that influence ignition sustainability, rate of spread, fuel drying, and fire intensity.

$$U_{Meteo,k}(x,y) = \begin{bmatrix} W_{temp,k}(x,y) \\ W_{rh,k}(x,y) \\ W_{ws,k}(x,y) \\ W_{wd,k}(x,y) \\ W_{gust,k}(x,y) \\ W_{prec,k} \end{bmatrix} \qquad (16)$$

These fields are updated at every discrete simulation step so that wildfire behavior follows the evolving weather. These variables listed in Table 3.1 are primarily **time-dependent** and may optionally be **spatially dependent** when high-resolution gridded weather fields are available.

Table 3.1 Meteorology data layer parameters

| Parameter | Type | Unit | Purpose in Simulation |
|---|---|---|---|
| **Air Temperature** $\boldsymbol{W}_{\mathbf{temp,k}}(\boldsymbol{x},\boldsymbol{y})$ | Spatio-temporal | °C | Represents ambient air temperature, influencing fuel drying rate, ignition thresholds, and fire intensity potential. |
| **Relative Humidity** $\boldsymbol{W}_{\mathbf{rh,k}}(\boldsymbol{x},\boldsymbol{y})$ | Spatio-temporal | % | Represents atmospheric moisture content and directly regulates live and dead fuel moisture levels. |
| **Wind Speed** $\boldsymbol{W}_{\mathbf{ws,k}}(\boldsymbol{x},\boldsymbol{y})$ | Spatio-temporal | m/s | Serves as the primary driver of fire spread rate and significantly increases the rate of spread under high-speed conditions. |
| **Wind Direction** $\boldsymbol{W}_{\mathbf{wd,k}}(\boldsymbol{x},\boldsymbol{y})$ | Spatio-temporal | rad | Determines the directional propagation of the fire front and controls anisotropic spread behavior. |
| **Wind Gust Speed** $\boldsymbol{W}_{\mathbf{gust,k}}(\boldsymbol{x},\boldsymbol{y})$ | Spatio-temporal | m/s | Represents short-term wind fluctuations and stochastic acceleration in fire propagation. |
| **Precipitation** $\boldsymbol{W}_{\mathbf{prec,k}}$ | Temporal | mm/h | Increases fuel moisture and can suppress, delay, or weaken active combustion processes. |

### 3.2.2 Topography Data Layer

Terrain parameters $U_{Geo}$ ( 17 ) define static spatial constraints over which wildfire propagation occurs. Unlike meteorological variables, these parameters remain constant during short-horizon simulation but modulate directional spread, acceleration, and accessibility. The geometry parameters are listed in Table 3.2.

$$U_{Geo}(x,y) = \begin{bmatrix} G_{elev}(x,y) \\ G_{slope}(x,y) \\ G_{aspect}(x,y) \\ G_{access}(x,y) \end{bmatrix} \tag{17}$$

Table 3.2 Geometry (topography) data layer parameters

| Parameter | Type | Unit | Purpose in Simulation |
|---|---|---|---|
| **Elevation** $\boldsymbol{G}_{\mathbf{elev}}(\boldsymbol{x},\boldsymbol{y})$ | Spatial | m | Represents the altitude of each grid cell and influences local wind behavior, vegetation type distribution, and environment gradients. |
| **Slope** $\boldsymbol{G}_{\mathbf{slope}}(\boldsymbol{x},\boldsymbol{y})$ | Spatial | rad | Represents terrain inclination and increases fire spread rate in the uphill direction due to flame heat transfer effects. |
| **Aspect** $\boldsymbol{G}_{\mathbf{aspect}}(\boldsymbol{x},\boldsymbol{y})$ | Spatial | rad | Represents slope orientation and influences solar radiation exposure, fuel drying patterns, and directional fire propagation. |
| **Accessibility Index** $\boldsymbol{G}_{\mathbf{access}}(\boldsymbol{x},\boldsymbol{y})$ | Spatial | [0,1] | Represents terrain traversability and influences responder mobility, intervention timing, and operational feasibility. |

### 3.2.3 Fuel Data Layer

Fuel-related inputs $U_{Fuel,k}$ ( 18 ) describe the combustible material available at each grid cell. $F_{type}(x,y)$ is the vegetation class and $F_{load,0}(x,y)$ is the available combustible mass at the initial time. Fuel moisture $F_{moist,k}(x,y)$ is initialized from meteorological drivers or scenario configurations and thereafter evolves inside the transition operator under precipitation and the optional equilibrium moisture mode. The fuel parameters are listed in Table 3.3.

$$U_{Fuel,k}(x,y) = \begin{bmatrix} F_{type}(x,y) \\ F_{load,0}(x,y) \\ F_{moist,k}(x,y) \end{bmatrix} \tag{18}$$

Table 3.3 Fuel data layer parameters

| Parameter | Type | Unit | Purpose in Simulation |
|---|---|---|---|
| **Fuel Type** $\boldsymbol{F}_{\textbf{type}}(\boldsymbol{x},\boldsymbol{y})$ | Spatial | – | Defines the vegetation within each grid cell and determines combustion characteristics such as burn rate and heat release behavior. |
| **Fuel Load** $\boldsymbol{F}_{\textbf{load,0}}(\boldsymbol{x},\boldsymbol{y})$ | Spatial | $[0,1]\,\frac{kg}{m^2}$ (norm.) | Represents the available combustible mass in each grid cell and directly controls fire intensity, burn duration, and energy release. |
| **Fuel Moisture** $\boldsymbol{F}_{\textbf{moist,k}}(\boldsymbol{x},\boldsymbol{y})$ | Spatio-temporal | – (mass fraction) | Represents moisture of surface fuels and determines ignition probability and combustion sustainability. |

### 3.2.4 Ignition Layer

Ignition is separate from fire intensity. It is the activation of combustion at a specific grid cell and time step. Ignition may originate from predefined initial conditions, external events, or spotting effects. It functions as a triggering mechanism within the transition dynamics and does not constitute a persistent state variable. The ignition input is defined in ( 19 ) where $I_{Ign,k}\ (x,y)$ is a binary ignition injection indicator.

$$U_{Ign,k}(x,y) = [I_{Ign,k}(x,y)] \tag{19}$$

### 3.2.5 Decisional Data Layers

The Decisional External Sources ($\mathcal{F}_{DSS,k}$) layer represents the non-physical inputs that shape decision making within the distributed DSS. Unlike the meteorological, topographic, and fuel layers that govern fire dynamics, this layer carries the contextual information needed to evaluate operational priorities and constraints, and it therefore determines how simulation outcomes are interpreted and acted upon.

The DSS generates decision-driven intervention parameters $U_{DSS,k}$ *( 20 )* based on the observed quantities $O_k(x, y)$ and $\mathcal{F}_{DSS,k}$. $R_{cap,k}(x, y)$ is the suppression capacity allocated to the cell, $R_{avail,k}(x, y)$ is operational availability status of resources, $R_{eff,k}(x, y)$ is suppression effectiveness, and $R_{time,k}(x, y)$ is estimated travel time to reach grid cell $(x, y)$. These parameters do not represent physical measurements but structured representations of operational actions, which are incorporated into the external input set to influence wildfire dynamics. The design chapter derives these fields from the intervention intensities over the six families of Table 2.4: Each local agent turns its intensity vector into the slice of $U_{DSS,k}$ for its region, so *( 20 )* is input-level realization of the decision, distinct from the resource pool $U_{Res,k}$ of Table 3.5.

$$U_{DSS,k}(x, y) = \begin{bmatrix} R_{cap,k}(x, y) \\ R_{avail,k}(x, y) \\ R_{eff,k}(x, y) \\ R_{time,k}(x, y) \end{bmatrix} \qquad (20)$$

$\mathcal{F}_{DSS,k}$ has two components: The Value Data and the Resource Data layers. The Value layer defines spatial exposure and asset importance, which supports risk-based prioritization. The Resource layer defines the available suppression capabilities and logistical constraints, which support feasible intervention planning. Together they turn the physical fire propagation output of the Simulation Core into actionable decision variables, keeping fire behavior modeling and decision reasoning separate.

#### 3.2.5.1 Value Data Layer

The Values-at-Risk Data layer represents exposed assets, infrastructure, and socio-economic elements that may be affected by wildfire propagation. The value data layer parameters are listed in Table 3.4. All variables in this layer are spatially dependent and static during short-term operational simulations, although their priority weights may be dynamically adjusted based on operational objectives.

Table 3.4 Value data layer parameters

| Parameter | Type | Unit | Purpose in Simulation |
| --- | --- | --- | --- |
| **Building Footprint** $\boldsymbol{V}_{\mathbf{bld}}(\boldsymbol{x},\boldsymbol{y})$ | Spatial | {0,1} | Represents the presence of residential or commercial structures and enables identification of wildland-urban zones. |
| **Critical Facility Index** $\boldsymbol{V}_{\mathbf{crit}}(\boldsymbol{x},\boldsymbol{y})$ | Spatial | {0,1} | Identifies high-priority assets such as hospitals, power plants, fuel depots, and water infrastructure. |
| **Population Density** $\boldsymbol{V}_{\mathbf{pop}}(\boldsymbol{x},\boldsymbol{y})$ | Spatial | person/km² | Represents exposed population per grid cell and supports evacuation and safety prioritization. |
| **Accessibility to Evacuation Routes** $\boldsymbol{V}_{\mathbf{evac}}(\boldsymbol{x},\boldsymbol{y})$ | Spatial | m | Indicates proximity to evacuation infrastructure and supports emergency planning decisions. |
| **Protection Priority Score** $\boldsymbol{V}_{\mathbf{prio}}(\boldsymbol{x},\boldsymbol{y})$ | Spatial | – | Aggregated weighting factor combining multiple exposure indicators to rank cells by importance. |

The protection priority score $V_{prio}(x,y)$ aggregates the four value indicators into a single scalar index that ranks each grid cell by its overall protection importance. The aggregation is as a normalized weighted sum ( 21 ) where $V_{pop}^{norm} = \frac{V_{pop}}{max\, V_{pop}}$ is the max-normalized population density, and $V_{evac}^{norm} = \frac{V_{evac}}{max\, V_{evac}}$ is the max-normalized evacuation distance (higher value indicates poorer access to evacuation routes, hence earlier protective intervention).

$$V_{prio}(x,y) = w_{bld} \cdot V_{bld}(x,y) + w_{crit} \cdot V_{crit}(x,y) + w_{pop} \cdot V_{pop}^{norm}(x,y) + w_{evac} \cdot V_{evac}^{norm}(x,y) \quad (21)$$

The weights satisfy $w_{bld} + w_{crit} + w_{pop} + w_{evac} = 1$. The weight assignment reflects the operational priority. Critical facilities ($w_{crit} = 0.40$) receive the highest weight because their loss has cascading consequences (e.g., hospitals, power infrastructure, water supply). Population density ($w_{pop} = 0.25$) ranks second because human life exposure scales with density. Building footprint ($w_{bld} = 0.20$) captures structural asset density. Evacuation accessibility ($w_{evac} = 0.15$) receives the lowest weight because it represents a logistical constraint rather than a direct measure of asset value; however, it remains non-zero because locations with poor evacuation access require protective intervention earlier in the decision timeline.

#### 3.2.5.2 Resource Data Layer

The Resource Data layer represents the operational capabilities available for wildfire intervention. While the fire behavior model estimates how fire evolves under environmental conditions, this layer defines the constraints and effectiveness of suppression actions. The resource data layer parameters are listed in Table 3.5.

Table 3.5 Resource data layer parameters

| Parameter | Type | Unit | Purpose in Simulation |
|---|---|---|---|
| **Suppression Capacity** $\overline{\boldsymbol{R}}_{\mathbf{cap,k}}(\boldsymbol{x},\boldsymbol{y})$ | Spatio-temporal | $m^3$ / step | Represents available suppression capability (i.e., water volume or manpower at grid cell at time step k). |
| **Travel Time** $\overline{\boldsymbol{R}}_{\mathbf{time,k}}(\boldsymbol{x},\boldsymbol{y})$ | Spatio-temporal | min or h | Represents estimated time required to reach a grid cell from current position. |
| **Suppression Efficiency** $\overline{\boldsymbol{R}}_{\mathbf{eff,k}}(\boldsymbol{x},\boldsymbol{y})$ | Spatio-temporal | [0,1] | Quantifies how effectively a resource reduces fuel load or fire intensity. |
| **Resource Availability** $\overline{\boldsymbol{R}}_{\mathbf{avail,k}}(\boldsymbol{x},\boldsymbol{y})$ | Spatio-temporal | [0,1] | Degree to which the resource is currently deployable. |

## 3.3 Simulation Core

The Simulation Core is the computational foundation of the framework. It represents the wildfire as a state defined over a spatial grid and advances it in discrete time through a single transition operator built on a hybrid fire spread model. Each cell carries its burning status, fuel load, fire intensity, and time since ignition, and these evolve together under the meteorological, topographic, and fuel inputs that govern fire dynamics. The operator is deterministic; stochastic effects such as wind gusts and ember spotting enter only through the external input fields and never write to the state directly. The Simulation Core estimates only how the fire behaves over space and time and leaves the interpretation of that behavior to the DSS, which keeps fire modeling and decision making clearly separated. Its design criterion is testability and reproducibility rather than maximal physical fidelity.

### 3.3.1 Spatio-Temporal State Transition

Each grid cell in $\mathcal{G}$ corresponds to a spatial coordinate pair $(x, y)$ and represents a localized fire condition ( 22 ). $N_x$ is the number of grid cells along the horizontal spatial axis, and $N_y$ is the number of grid cells along the vertical spatial axis.

$$\mathcal{G} = \{ (x, y) \mid x = 1, \dots, N_x, y = 1, \dots, N_y \} \qquad (22)$$

At each simulation step, the wildfire state over the grid is represented by $S_k$. The system state is defined as the collection of local cell states over the domain ( 23 ). $B_k(x, y)$ is the burning status of cell $(x, y)$, $F_{load,k}(x, y)$ is available combustible mass, $I_k(x, y)$ is local fire intensity proxy, and $\tau_k(x, y)$ is time since ignition.

$$S_k = \{ s_k(x, y) | (x, y) \in \mathcal{G} \}$$
$$s_k = \begin{bmatrix} B_k(x, y) \\ F_{load,k}(x, y) \\ I_k(x, y) \\ \tau_k(x, y) \end{bmatrix} \qquad (23)$$

The state vector includes only internal wildfire variables. Environmental factors and operational interventions are not part of the state and are represented as external inputs influencing the state transition. Grid domain and the cell state vector representation is illustrated in Figure 3.2.

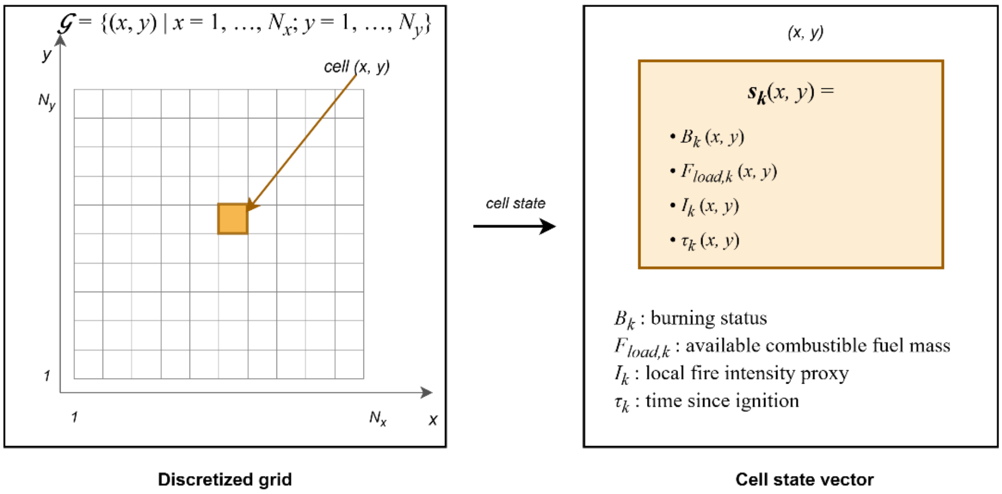


Figure 3.2 Grid domain representation and per-cell state structure

### 3.3.2 Hybrid Fire Spread Model

The hybrid model integrates deterministic combustion dynamics with stochastic variability to represent realistic wildfire behavior. The global state evolution ( 24 ) is governed by the transition operator where $\mathcal{F}_{in,k}$ ( 15 ) includes meteorological forcing, topography, fuel conditions, ignition injection, and DSS-generated intervention parameters. Φ denotes the structured state transition operator composed of burning status evolution, fuel mass evolution, and fire intensity update as defined in the subsequent subsections. The transition operator itself is deterministic. Stochastic variability may arise only through exogenous input fields (e.g., wind gust fluctuations), which are treated as externally provided spatio-temporal inputs.

$$S_{k+1}(x, y) = \Phi\left(S_k(x, y), \mathcal{F}_{in,k}\right)$$
$$\mathcal{F}_{in,k} = \{U_{Meteo,k}, U_{Geo}, U_{Fuel,k}, U_{Ign,k}, U_{DSS,k}\} \tag{24}$$

The overall state transition and hybrid fire spread model is illustrated in Figure 3.3. The state transition operations are described in the next subsections.

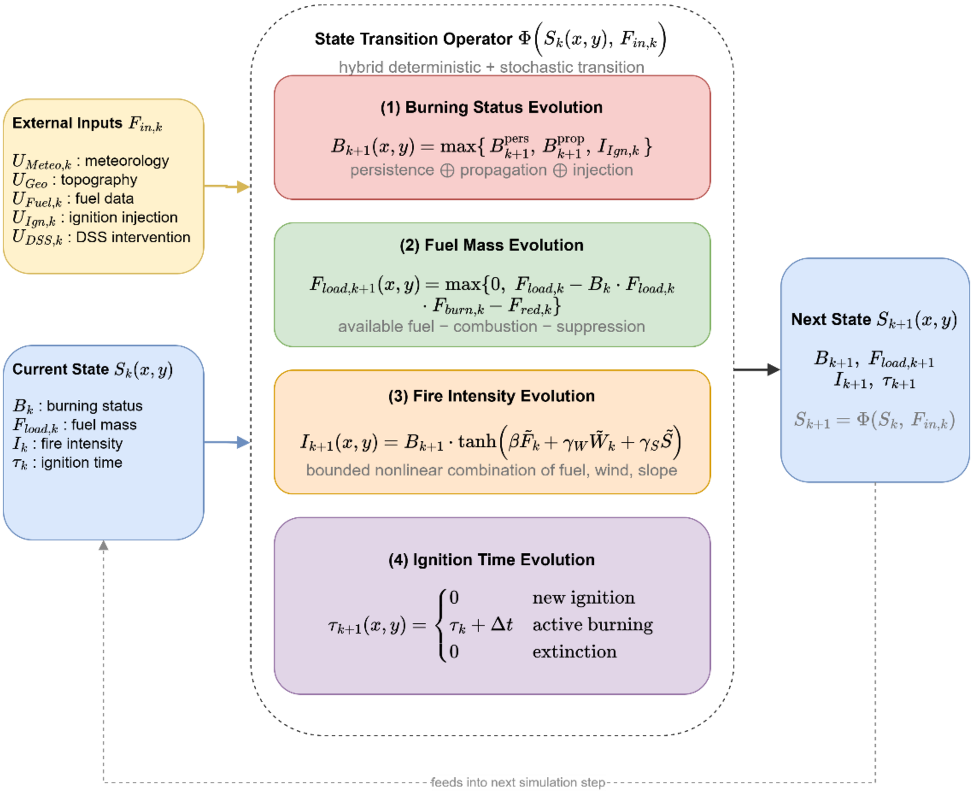


Figure 3.3 State transition block diagram

#### 3.3.2.1 Burning Status Evolution

The burning status $B_k(x, y) \in \{0,1\}$ indicates whether the cell is actively burning at time step $k$. The burning status at the next step $B_{k+1}$ ( 25 ) is determined by four mechanisms: persistence of ongoing burning $B^{pers}_{k+1}$ ( 26 ), propagation from neighboring burning cells, seeded ember-spotting ignition, and external ignition injection $I_{Ign,k}(x, y)$. Combining these mechanisms, the burning status update rule is defined as ( 25 ).

$$B_{k+1}(x,y) = \max\left(B^{pers}_{k+1}(x,y), B^{prop}_{k+1}(x,y), B^{spot}_{k+1}(x,y), I_{Ign,k}(x,y)\right) \quad (25)$$

This formulation ensures that a burning cell persists while fuel remains available; neighboring burning cells may activate a new cell if sufficient spread influence accumulates; external ignition acts as an independent triggering mechanism. The spotting term activates a cell when a seeded ember of the spotting mechanism lands on it and ignites the remaining fuel.

The persistence term $B_{k+1}^{pers}$ ensures that a cell that is already burning remains burning as long as fuel exists and the fuel moisture stays below the extinction moisture, as stated in ( 26 ) where $\mathbf{1}[\cdot]$ denotes the indicator function. Below a few percent of the nominal load a cell cannot sustain flaming combustion, so an extinction threshold $\varepsilon_{fuel_{load}}$ is applied. The second indicator quenches a burning cell once its fuel moisture reaches the fuel-type extinction moisture of the moisture damping factor, consistent with the precipitation mechanism of the hybrid model.

$$\begin{gathered} B_{k+1}^{pers}(x,y) = B_k(x,y) \cdot \mathbf{1}\left[F_{load,k}(x,y) > \varepsilon_{fuel_{load}}\right] \\ \cdot \mathbf{1}\left[F_{moist,k}(x,y) < m_{ext}\left(F_{type}(x,y)\right)\right] \\ \varepsilon_{fuel_{load}} = 0.03 \end{gathered} \tag{26}$$

Propagation to cell $(x,y)$ occurs if the accumulated influence exceeds a critical ignition threshold $\Theta_{ign}$, where $\Theta_{ign}$ is a model parameter representing the minimum spread intensity required to activate combustion in a new cell ( 27 ). In this thesis $\Theta_{ign}$ = 0.125 where $\Theta_{ign} \in (0,1)$ is calibrated threshold relative to base spread rate.

$$B_{k+1}^{prop}(x,y) = \mathbf{1}\left[\Psi_k(x,y) > \Theta_{ign}\right] \tag{27}$$

Wildfire spread to neighboring cells is modeled through local propagation influence. $\mathcal{N}^8(x,y)$ denotes the 8-connected neighborhood of cell $(x,y)$. The total propagation influence received by cell $(x,y)$ at time step $k$ $\Psi_k(x,y)$ is defined in ( 28 ), where the rate of spread $R_{spread,k}(x,y)$ is ( 29 ). The rate-of-spread mapping $f_{ros}(\cdot)$ is instantiated using a semi-empirical Rothermel-type formulation (Rothermel, 1972). Detailed parameterization is provided in Appendix A. $W_{ws,k}(x,y)$ is the wind speed field, $W_{wd,k}(x,y)$ is the wind direction field, $G_{slope}(x,y)$ is the terrain slope, $G_{aspect}(x,y)$ is the terrain orientation, $F_{moist,k}(x,y)$ is the fuel surface moisture, and $F_{type}(x,y)$ is the fuel type.

$$\Psi_k(x,y) = \frac{1}{8} \sum_{(i,j)\in\mathcal{N}^8(x,y)} B_k\,(i,j) \cdot R_{spread,k}(i,j) \cdot\ g_{dir,k}^{(i,j)\to(x,y)} \qquad (28)$$

$$R_{spread,k}(x,y) = f_{ros}\begin{pmatrix} W_{ws,k}(x,y), W_{wd,k}(x,y), \\ G_{slope}(x,y), G_{aspect}(x,y), \\ F_{moist,k}(x,y), F_{type}(x,y) \end{pmatrix} \qquad (29)$$

The scalar magnitude $R_{spread,k}(x,y)$ does not determine spread direction. Directional influence ( 30 ) is introduced when propagating fire from a burning neighbor $(i, j)$ to target cell $(x,y)$. $\varphi_{(i,j)\to(x,y)}$ is the geometric direction between cells. This ensures anisotropic spread aligned with wind direction.

$$g_{dir,k}^{(i,j)\to(x,y)} = \max\big(0, cos\big(W_{wd,k}(i,j) - \varphi_{(i,j)\to(x,y)}\big)\big) \qquad (30)$$

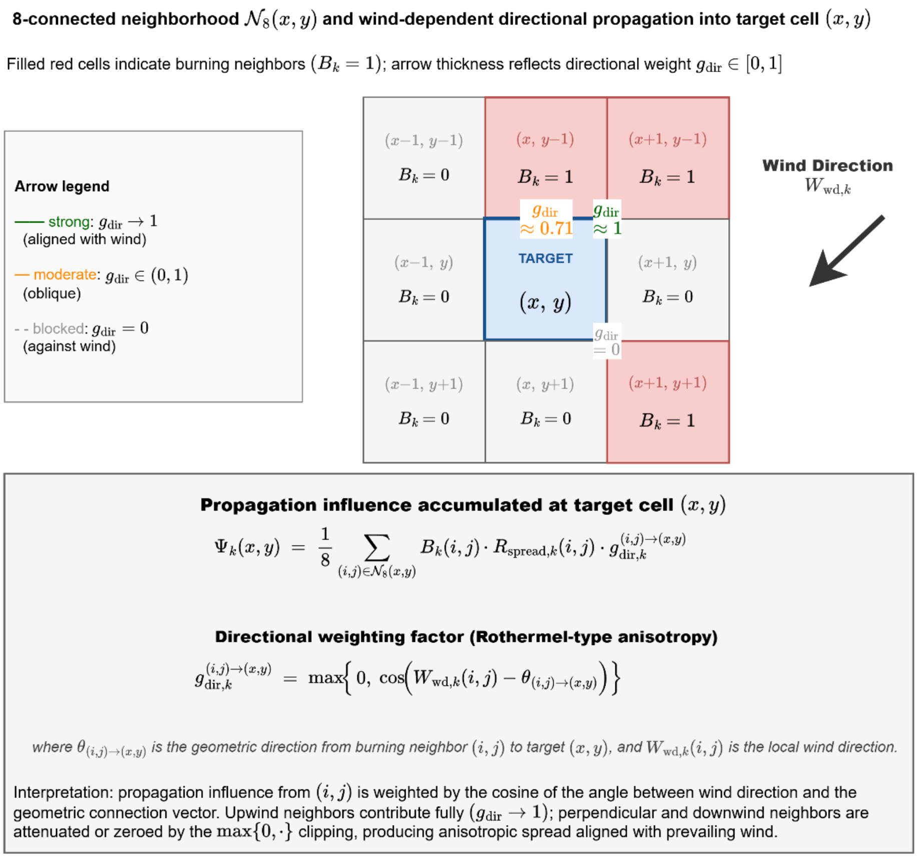


Figure 3.4 Wind-dependent directional propagation into target cell

Figure 3.4 depicts the spatial structure of the propagation influence accumulation. The 3×3 stencil centered on the target cell $(x,y)$ represents the 8-connected neighborhood $N_{8(x,y)}$ defined in ( 28 ); the three cells shaded in red denote burning neighbors with $B_k(i,j) = 1$, while the remaining five are non-burning. The wind direction $W_{wd,k}$, represented by the large arrow in the upper-right region of the figure, establishes the axis of anisotropy.

For each burning neighbor, a propagation arrow is drawn toward the target cell with line weight proportional to its directional weight $g_{dir,k}^{(i,j)\to(x,y)}$: The arrow from the northeast neighbor is aligned with the wind vector and therefore receives full weight $g_{dir} \to 1$; the arrow from the northern neighbor is oblique and receives an intermediate weight $g_{dir} \approx 0.71$; and the arrow from the southeastern neighbor is oriented opposite to the wind and is entirely eliminated, $g_{dir} = 0$, by the $max\{0,\cdot\}$ clipping of the base kernel. The aggregate propagation influence $\Psi_k(x,y)$ computed through ( 28 ) is therefore not a symmetric average over the 8-neighborhood but a wind-aligned weighted sum, and the ignition condition $\Psi_k(x,y) > \Theta_{ign}$ ( 27 ) is consequently satisfied preferentially along the direction of prevailing wind. This mechanism realizes the Rothermel-type directional bias that is central to operationally realistic wildfire propagation. The cosine kernel of ( 30 ) is the base form of this directional weighting. The implementation refines it with a wind-elongated elliptical correction whose length-to-breadth ratio grows with wind speed, keeps a small positive weight for flank and rear spread, scales diagonal influence by the center distance of the neighbor, and advances the front in substeps so that spread never exceeds one cell per substep. These refinements alter rates rather than mechanisms, and the ignition condition of ( 27 ) is unchanged.

External ignition injection $I_{Ign,k}(x,y) \in \{0,1\}$ directly activates combustion independent of neighbor influence. Ignition injection may represent lightning strikes, human-caused ignition, or scenario-based ignition input from the UI layer. Ignition injection is a scenario-level forcing that sets $B_{k+1}(x,y) = 1$ regardless of local conditions; if $F_{load,k}(x,y)$ is insufficient, extinction follows naturally via the persistence condition in the next step. In contrast, the local fire intensity proxy $I_k(x,y)$ represents combustion strength after ignition has occurred and does not participate in the activation condition. This separation preserves conceptual clarity between fire initiation and fire magnitude.

Ember spotting extends ignition beyond the resolved front. Each burning cell whose intensity exceeds 0.6, under wind above 3 m/s and precipitation below 1 mm/h, releases an ember with probability 0.02 per reference step, compounded consistently across substeps. The ember lands six cells downwind and ignites the target cell only if fuel remains there and the local fuel moisture is below the extinction threshold. Spotting is the only stochastic element of the transition operator, is driven by the seeded random stream of the scenario, and allows the fire to cross narrow barriers such as rivers, roads, and containment lines under strong and dry winds.

Precipitation acts on the dynamics through the dead fuel moisture. While it rains, $F_{moist}$ relaxes toward at least 0.35, a level above every extinction threshold, with a time constant of roughly thirty minutes scaled by the rain intensity; under sustained rain of at least 3 mm/h a burning cell is quenched once its fuel moisture has reached that level, so a front dies out gradually rather than instantaneously. When the optional equilibrium moisture mode is active, the dead fuel moisture follows the Simard equilibrium moisture content computed from air temperature and relative humidity, so diurnal weather cycles modulate flammability in the manner of operational fire behavior tools.

Propagation dynamics are governed exclusively by the rate-of-spread mapping $R_{spread,k}(x, y)$. The local fire intensity proxy does not directly influence activation conditions; instead, it modulates suppression effectiveness and decision-layer interpretation. This separation preserves structural clarity between spread mechanics and combustion strength representation. Thus, intensity affects wildfire dynamics indirectly through suppression resistance and decision-layer interpretation, rather than directly controlling fire propagation.

The rate-of-spread mapping $R_{spread,k}(x, y)$ is independent of instantaneous fuel mass $F_{load,k}(x, y)$. Fuel availability influences spread only indirectly through the burning persistence condition, preventing propagation from fuel-depleted cells. This structural separation avoids double- counting mitigation effects.

#### 3.3.2.2 Fuel Mass Evolution

Fuel mass evolution governs the depletion of combustible material within each cell as a consequence of two distinct mechanisms: Combustion-driven consumption and intervention-driven suppression. Unlike the burning status variable, which represents the activation of combustion, the fuel mass variable represents the remaining energy potential available for future fire propagation. Hence, accurate modeling of fuel evolution is essential to ensure physical consistency, numerical stability, and interpretable interaction between wildfire dynamics and decision-driven mitigation. Combustion-driven consumption and suppression-driven reduction are applied within the same discrete step $k$ as a simultaneous update, represented by the combined terms in ( 31 ) where $B_k(x,y) \in \{0,1\}$ is the burning status indicator, $F_{burn,k}(x,y) \in [0,1]$ is the fractional burn rate (Appendix B), $F_{load,k}(x,y)$ is the currently available fuel mass, and $F_{red,k}(x,y)$ is intervention-based fuel reduction.

$$F_{load,k+1}(x,y) = \max\left(0, \left(\begin{array}{c} F_{load,k}(x,y) \\ -F_{load,k}(x,y)\, B_k(x,y)\; F_{burn,k}(x,y) \\ -F_{red,k}(x,y) \end{array}\right)\right) \qquad (31)$$

Fuel cannot become negative and cannot increase unless explicitly modeled, $0 \leq F_{load,k+1}(x,y) \leq 1$. This formulation explicitly separates combustion-induced depletion from suppression-induced reduction while enforcing non-negativity of fuel mass. Fuel consumption due to combustion occurs only when the cell is actively burning. The term $B_k(x,y)\, F_{load,k}(x,y)\, F_{burn,k}(x,y)$ represents the fraction of available fuel consumed during the current simulation step. The multiplication by $B_k(x,y)$ ensures that fuel depletion due to combustion occurs only in actively burning cells. If the cell is not burning, no combustion-driven reduction takes place. This preserves physical realism and prevents unintended fuel decay in inactive regions. The burn fraction $F_{burn,k}(x,y)$ captures the local combustion intensity per time step and can be interpreted as a discretized approximation of heat-release-driven fuel consumption. The suppression-induced fuel reduction term $F_{red,k}(x,y)$ is computed from decision-driven inputs $U_{DSS,k}(x,y)$ and is not an independent input.

On the other hand, operational intervention modifies the fuel state through the suppression term ( 32 ). This mapping converts operational capability and reachability into a normalized fuel reduction effect. The suppression term may represent water or retardant application; mechanical fuel removal; controlled burn operations; and resource-constrained mitigation strategies. $f_{supp}$ is Suppression to fuel reduction map (Appendix B), $R_{cap,k}$ is the allocated suppression capacity, $R_{avail,k}$ is operational availability status of resources, $R_{eff,k}(x,y)$ is suppression effectiveness at cell $(x, y)$, $R_{time,k}(x,y)$ is estimated travel time to reach $(x, y)$, and $G_{access}(x,y)$ is accessibility level at grid cell $(x, y)$.

$$F_{red,k}(x,y) = f_{supp}\begin{pmatrix} R_{cap,k}(x,y), R_{avail,k}(x,y), R_{eff,k}(x,y), \\ R_{time,k}(x,y), G_{access}(x,y) \end{pmatrix} \tag{32}$$

Suppression does not directly overwrite the burning state. Instead, it modifies the available fuel mass, which subsequently influences combustion persistence and propagation through the transition operator. This separation guarantees architectural clarity between physical wildfire dynamics and decision-driven interventions.

#### 3.3.2.3 Fire Intensity Evolution

The local fire intensity proxy evolves according to ( 33 ). The function $f_{intensity}(\cdot)$ approximates local fire strength as a function of available fuel, wind forcing, and terrain gradient. $I_{k+1}(x,y)$ denotes the updated intensity at the next time step; $I_k$ does not appear as an argument of $f_{intensity}$. $f_{intensity}$ is local fire strength function (Appendix C), $F_{load,k}(x,y)$ is the available combustible mass, $W_{ws,k}(x,y)$ is wind speed, and $G_{slope}(x,y)$ is terrain slope.

$$I_{k+1}(\mathrm{x,y}) = f_{intensity}\left(F_{load,k}(x,y), W_{ws,k}(x,y), G_{slope}(x,y)\right) \tag{33}$$

The fire intensity proxy serves as an intermediate physical-descriptive variable with the following roles: It does not control ignition activation; it does not directly modify the spread operator; it influences suppression effectiveness through the intervention mapping; and it provides structured features to the Decision Support System. Thus, intensity is a bounded combustion-strength indicator that links environmental forcing to operational response without introducing thermodynamic complexity.

#### 3.3.2.4 Ignition Time Evolution

The ignition time variable $\tau_k(x, y)$ represents the time elapsed since the most recent ignition event at grid cell $(x, y)$. Unlike fuel mass or intensity, $\tau_k$ does not influence combustion physics directly. Instead, it serves as a temporal memory state that preserves the burning history of the cell. $\tau_k$ is a useful state for decision features and explainability within the DSS layer. A deterministic update is ( 34 ).

$$\tau_{k+1}(x, y) = \begin{cases} 0, & B_{k+1} = 1 \wedge B_k = 0 \\ \tau_k + \Delta t, & B_{k+1} = 1 \wedge B_k = 1 \\ 0, & B_{k+1} = 0 \end{cases} \tag{34}$$

This update has three distinct operational meanings:

1. **Ignition Initialization:** When a cell transitions from non-burning to burning $B_k = 0, B_{k+1} = 1$, the ignition clock is initialized: $\tau_{k+1} = 0$. This marks the start of a new combustion phase.

2. **Active Burning Progression:** When a cell remains in the burning state $B_k = 1$, $B_{k+1} = 1$, the ignition time increases linearly: $\tau_{k+1} = \tau_k + \Delta t$. Thus, $\tau_k$ measures the residence time of active combustion at the cell.

3. **Extinction Reset:** When combustion ceases $B_{k+1} = 0$, the ignition clock resets to 0. It ensures extinguished cells carry no residual temporal memory.

$\tau_k$ does not feed back into the burning activation condition in the present model, preserving the separation between physical fire dynamics and temporal descriptors.

### 3.3.3 Mathematical Formulation of Interaction and Scenario Control

The Interaction and Scenario Control mechanism enables controlled human intervention within the DisasterAware framework while preserving the physical consistency of wildfire dynamics. The external simulation input set is $\mathcal{F}_{in,k} = \{U_{Meteo,k}, U_{Geo}, U_{Fuel,k}, U_{Ign,k}, U_{DSS,k}\}$ and decision-context set is $\mathcal{F}_{DSS,k} = \{U_{Val,k}, U_{Res,k}\}$. The user interaction operator $\Theta_{UI}$ is a structured input transformation mapping that produces admissible modifications of the external input set and decision-context set ( 35 ).

$$\begin{gathered} \Theta_{UI}: \mathcal{F}_{in} \times \mathcal{F}_{DSS} \longrightarrow \mathcal{F}_{in}^{*} \times \mathcal{F}_{DSS}^{*} \\ \left(\mathcal{F}_{in,k}^{*}, \mathcal{F}_{DSS,k}^{*}\right) = \Theta_{UI}(\mathcal{F}_{in,k}, \mathcal{F}_{DSS,k}) \end{gathered} \tag{35}$$

The operator acts independently on four admissible inputs. For each component, the admissible transformation is defined by a bounded perturbation set in Table 3.6. All unmodified components pass through unchanged: $U_{Geo}^{*} = U_{Geo}$, $U_{Fuel,k}^{*} = U_{Fuel,k}$. The decision component $U_{DSS,k}$ is not an operator surface; it is produced by the DSS.

Table 3.6 Admissible transformations of interaction operator

| Component | Admissible transformation | Constraint |
|---|---|---|
| Ignition $\boldsymbol{U_{Ign,k}}$ | $U_{Ign,k}^{*}(x,y) = U_{Ign,k}(x,y) + \delta_{Ign}(x,y)$ | $U_{Ign,k}^{*} \in \{0,1\}$ |
| Meteo. $\boldsymbol{U_{Meteo,k}}$ | $U_{Meteo,k}^{*} = U_{Meteo,k} + \delta_{Meteo}$ | $U_{Meteo,k}^{*} \in \mathcal{U}_{Meteo}$ |
| Value $\boldsymbol{U_{Val,k}}$ | $U_{Val,k}^{*}(x,y) = \sigma_{Val} \cdot U_{Val,k}(x,y)$ | $\sigma_{Val}, U_{Val,k}^{*} \in [0,1]$ |
| Resource $\boldsymbol{U_{Res,k}}$ | $U_{Res,k}^{*}(x,y) = U_{Res,k}(x,y) + \delta_{Res}(x,y)$ | $U_{Res,k}^{*} \in [0,1]$ |

The operator $\Theta_{UI}$ acts exclusively on the external input set $\mathcal{F}_{in,k}$ and decision-context set $\mathcal{F}_{DSS,k}$. The system state $S_k(x,y)$ is not in the domain of $\Theta_{UI}$, $\frac{\partial S_k}{\partial \Theta_{UI}} = 0$. All operator influence propagates through physically consistent state transitions $S_{k+1} = \Phi\left(S_k, \mathcal{F}_{in,k}^{*}\right)$, preserving the causal structure and state immutability.

$\Theta_{UI}$ modifies only future state transitions. It does not alter the current state $S_k$ or any previously realized state $S_j,\ j < k$. This ensures that scenario exploration and operational overrides are prospective, not retrospective. The modified decision-context set influences decision generation $U_{DSS,k} = \pi_{DSS}\left(O_k, \mathcal{F}^*_{DSS,k}\right)$ and the modified simulation input set governs system evolution $S_{k+1} = \Phi\left(S_k, \mathcal{F}^*_{in,k}\right)$

Within this formulation, user interaction supports multiple operational functions: Scenario intervention is modification of ignition inputs $U_{Ign,k}$ to inject or alter fire events. Environmental what-if analysis is controlled perturbation of meteorological inputs $U_{Meteo,k}$. Priority reconfiguration is modification of value-based context $U_{Val,k}$. Resource reallocation is modification of resource availability $U_{Res,k}$. These interaction capabilities allow the system to operate as a **HIL DSS**, where the operator can dynamically influence both the operational context and the evolution of the simulated environment.

Interaction does not bypass the DSS. Any modification applied to $\mathcal{F}_{DSS,k}$ is incorporated into the decision-generation process through $\pi_{DSS}(\cdot)$, ensuring that all intervention signals remain consistent with the decision architecture. This design preserves three key architectural principles. **State immutability:** Wildfire dynamics remain governed exclusively by the transition operator. **Layer separation:** Simulation, decision reasoning, and interaction remain structurally distinct. **Traceability:** All operator interventions are explicitly represented as input-level transformations.

The Simulation Core additionally maintains per-step state snapshots under a bounded memory budget, enabling counterfactual replay. The operator can rewind the incident to any recorded step, modify the conditions or the intervention, and re-simulate. This mechanism underlies the with and without DSS comparisons on identical fire histories reported later. The replay is exact rather than approximate. A snapshot carries every mutable part of the run: the per-cell state vector, the fuel moisture field, the evacuated population, the accumulated cost integrals, and the state of the random number generator, and the exogenous weather driver is replayed unchanged. Two runs started from the same snapshot therefore differ only in the orders under test, and any difference in their outcomes is attributable to those orders alone. The comparison is a controlled experiment on the incident itself, not a statistical estimate over an ensemble.

## 3.4 Decision Support System Core

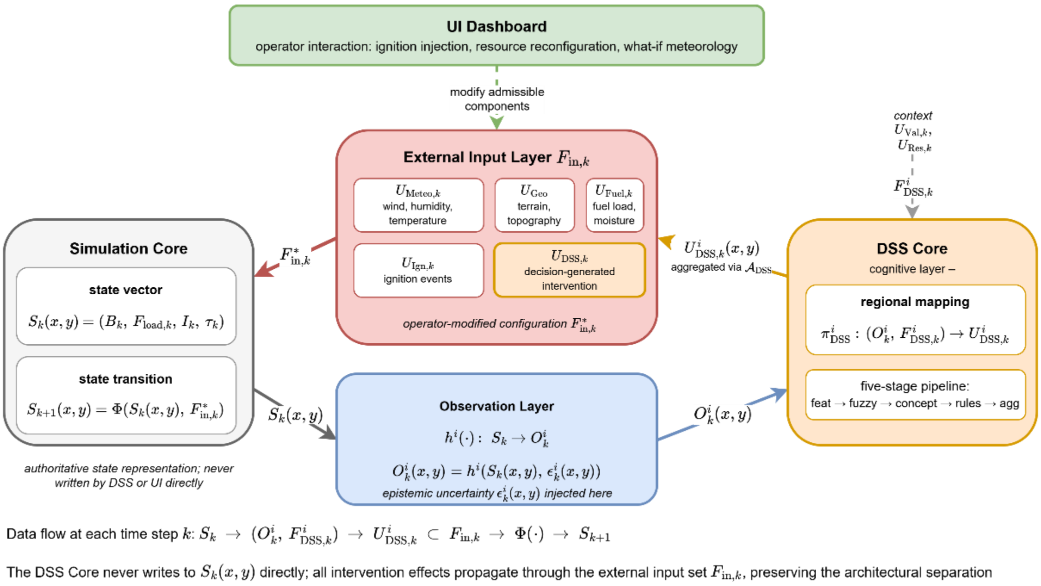


Figure 3.5 DSS Simulation Core Interface

This section defines only the structural role and interface of the DSS within the Simulation Core. Its internal specification and architectural realization are presented in the next chapter. While the Simulation Core governs the physical evolution of wildfire dynamics through the state transition operator $S_{k+1}(x,y) = \Phi(S_k(x,y), \mathcal{F}_{in,k})$, the DSS operates on observable quantities $O_k(x,y) = h(S_k(x,y), \varepsilon_k)$ derived from the simulation state through the observation process, rather than directly accessing the full system state. Observation uncertainty $\varepsilon_k(x,y)$ is incorporated within the DSS through uncertainty-aware reasoning mechanisms operating on $O_k(x,y)$. The DSS evaluates decision-relevant indicators derived from

$O_k(x,y)$ and contextual inputs $\mathcal{F}_{DSS,k}$ to generate structured intervention inputs $U_{DSS,k}$ that influence subsequent system evolution. The DSS interaction follows the ( 36 ). This preserves the architectural separation between state evolution and decision generation. All decision impact must propagate through admissible modifications of the external input set. The information and control flow captured by Equation ( 36 ) is summarized in Figure 3.5.

$$S_k(x,y) \rightarrow \; O_k(x,y), \mathcal{F}_{DSS,k} \rightarrow U_{DSS,k}(x,y) \rightarrow \Phi(\cdot) \rightarrow S_{k+1}(x,y)$$
$$O_k(x,y) = h(S_k(x,y), \varepsilon_k) \tag{36}$$
$$U_{DSS,k}(x,y) \; = \; \pi_{DSS}^{i}\big(O_k(x,y), \mathcal{F}_{DSS,k}\big) \; \subset \; \mathcal{F}_{in,k}$$

The **Simulation Core** maintains the authoritative state vector $S_k(x,y) = \big(B_k,\, F_{\text{load},k},\, I_k,\, \tau_k\big)$ and advances it through the state transition operator $\Phi$. The **Observation Layer** applies the region-specific observation function $h^i(\cdot)$ to project the state onto observable information $O_k^i(x,y) = h^i\left(S_k(x,y),\, \varepsilon_k^i(x,y)\right)$, injecting the epistemic uncertainty $\varepsilon_k^i(x,y)$ at this point and at this point only. The decision-cost trajectory later supplied to the generative adaptation stage is computed from these observed quantities, not from the authoritative state. The **DSS Core** executes the regional mapping $\pi_{\text{DSS}}^i\colon \big(O_k^i,\, F_{\text{DSS},k}^i\big) \rightarrow U_{\text{DSS},k}^i$ on the region-specific observation and the contextual inputs $F_{\text{DSS},k}^i = \big(U_{\text{Val},k},\, U_{\text{Res},k}\big)$, producing the regional decision field $U_{\text{DSS},k}^i(x,y)$; the detailed decision pipeline that realizes this mapping is the subject of Chapter 4. Regional decisions from the $N$ agents are aggregated through the coordination operator $\mathcal{A}_{\text{DSS}}$, realized by the global coordination layer of the next chapter, into the global intervention field, which is injected into the **External Input Layer** $F_{\text{in},k}$ as the $U_{\text{DSS},k}$ component alongside the four exogenous inputs $U_{\text{Meteo},k}$, $U_{\text{Geo}}$, $U_{\text{Fuel},k}$, and $U_{\text{Ign},k}$.

The **UI Dashboard** can modify admissible components of $F_{\text{in},k}$ through the interaction operator $\Theta_{\text{UI}}$, yielding the operator-modified configuration $F_{\text{in},k}^{*}$ that drives the next state transition $S_{k+1}(x,y) = \Phi\big(S_k(x,y),\, F_{\text{in},k}^{*}\big)$.

# CHAPTER 4

## DISASTERAWARE DECISION SUPPORT FRAMEWORK

This chapter presents the DisasterAware DSS, the cognitive layer that operates on top of the simulation core. While the simulation core governs the physical evolution of the wildfire state, the DSS interprets the observable part of that state and produces intervention decisions. All equations of the decision layer are developed in this chapter; together they constitute the complete mathematical specification of the decision pipeline.

The pipeline is organized as a four-layer flow from observation to decision. Regional observations are first reduced to bounded features carrying explicit confidence; the features are abstracted into a compact set of decision concepts; decision rules written over the concepts generate a candidate intervention field; and every candidate is evaluated before acceptance or rejection. Reasoning is hierarchically distributed. A Local DSS agent executes this pipeline over the combinations of its own region, while a single Global DSS coordinates the Local DSS agents without performing inference of its own. The remainder of this chapter is organized as follows:

Section 4.1 presents the DSS block diagram: The four-layer pipeline of a Local DSS agent and its hierarchically distributed organization, in which a non-inferential Global DSS ranks the regions and shares the common resource pool.

Section 4.2 defines Layer 1, the input space: The observable information each region draws from the simulation interface, together with the partial observability and communication effects that make this view incomplete and delayed.

Section 4.3 develops Layer 2, the observation model: The ten observation features and the confidence factors that qualify them, so that every quantity entering the reasoning carries a statement of how much it can be trusted.

Section 4.4 builds Layer 3, the concept space: The feature-concept hierarchy that aggregates the features into five decision concepts, and the confidence gating that blends observation with a persistence prior in proportion to confidence.

Section 4.5 develops Layer 4, decision generation and coordination: The sparse Takagi-Sugeno rule base and its candidate intervention field, the cost and quality measures by which candidates are judged, the satisficing acceptance test, and the staged adaptation that keeps the decision space open.

Section 4.6 presents the closed decision path: The local decision with its graduated fail-safe, the global coordination that turns regional priorities into attention shares and allocations, and the traceability record that makes every order auditable.

## 4.1 Decision Support System Block Diagram

Figure 4.1 presents the complete framework of a Local DSS agent. Unlike Figure 3.1, which emphasizes the simulation-relevant layers, this block diagram focuses on the layers relevant to the DSS. According to this block diagram, **input space (Layer 1)** collects heterogeneous, spatio-temporal measurements; **perception layer (Layer 2)** fuses them into per-combination features with explicit confidence; **concept space (Layer 3)** that abstracts the features into the four-level concept hierarchy; **decision space (Layer 4)** includes rule generate, adaptation reshapes, and evaluation of interventions. The **output of the local DSS ( $U^i_{DSS,k}$)** feeds the simulator core and $S_{k+1}$ is obtained as described in Chapter 3.

In operation, N agents execute this loop concurrently, each restricted to the cells of its own region, while the Global DSS coordinates them by normalizing priorities, allocating the shared resource pool, and enacting cross-region decisions where necessary. The environment band on the right is common to all agents; each agent sees only its own regional projection of it. The cost evaluation of Layer 4 monitors the environment continuously during assessment, so a candidate is always judged against the current situation rather than the one in which it is generated.

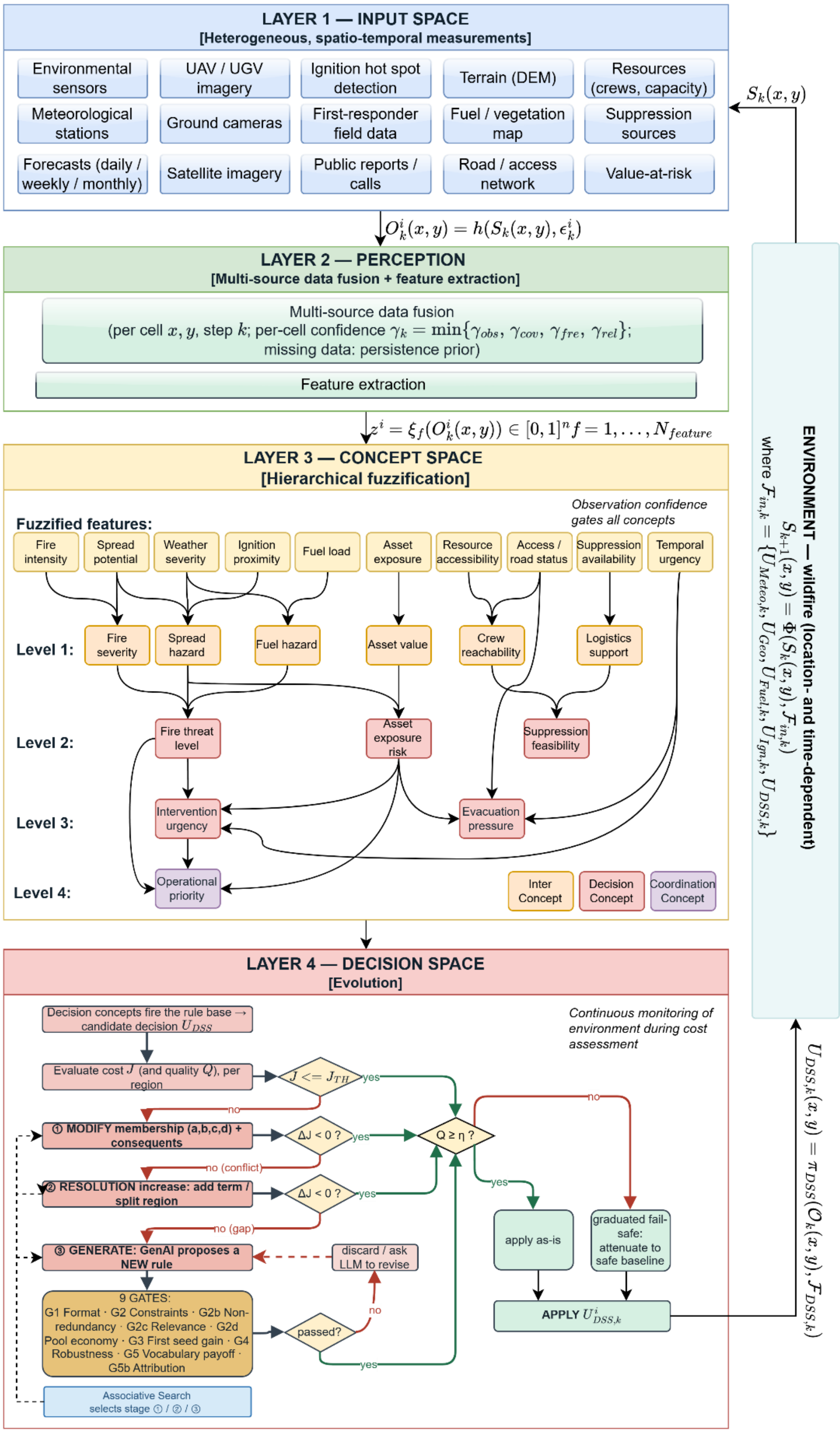


Figure 4.1 DisasterAware system block diagram (DSS focused)

## 4.2 Layer 1: Input Space

The input space of the DSS is the heterogeneous collection of spatio-temporal measurements shown at the top of Figure 4.1: Meteorological stations and forecasts, satellite, aerial, and ground imagery, ignition detections, terrain, fuel, road, and value-at-risk layers, and the resource and field reports of the response organization. These sources coincide with the external data layers formalized in Chapter 3 and are therefore not restated here; the single property that matters for the decision layer is that none of them observes the fire state directly, so **everything the DSS consumes is a partial, differently refreshed projection of a hidden situation.**

## 4.3 Layer 2: Observation, Features, and Confidence

The perception layer turns the raw input space into the quantities on which reasoning operates. Any Local DSS agent $i$ receives at step $k$ the region-restricted observation defined in ( 37 ). The region-specific observation function projects the authoritative state onto the cells of region $i$ and injects the epistemic uncertainty at this point.

The term epistemic denotes that this uncertainty stems from missing knowledge rather than from randomness. A camera may be occluded by smoke, a satellite product may arrive only at revisit intervals of several hours, a road sensor may be offline, and a field report may be delayed or absent; the state is under-observed, not merely noisy. No probability distribution over the error is assumed anywhere in the pipeline, so the treatment that follows is a bounded, confidence-driven one rather than a probabilistic one.

$$O_k^i(x,y) = h^i\left(S_k(x,y),\, \varepsilon_k^i(x,y)\right), \quad (x,y) \in \mathcal{G}^i \qquad (37)$$

Every observed cell carries an observation confidence composed from four factor properties: **Observability ($\gamma_{obs}$), coverage ($\gamma_{cov}$), freshness ($\gamma_{fre}$), and reliability ($\gamma_{rel}$)** as summarized in Table 4.1.

Table 4.1 Confidence factors: Definition, range, and determination

| Term | Determination |
| --- | --- |
| $\gamma_{obs}$ | Degree to which the underlying quantity is sensible at the cell; set from the sensor model of the channel (occlusion, LoS, product availability). |
| $\gamma_{cov}$ | Computed per cell from the covering footprints of satellite, aerial, and in-situ assets and averaged over the region. |
| $\gamma_{fre}$ | Decays exponentially with the age of the most recent report, with a half-life of 90 minutes; one for a report from the current step, decaying toward zero as the report ages. |
| $\gamma_{rel}$ | Trust assigned to the reporting channel from its characterization; calibrated sensors near one, unproven public reports lower. |

The composition is conservative; the confidence equals the weakest factor rather than an average ( 38 ). Satellite products realize wide coverage with coarse freshness, aerial platforms high freshness over narrow footprints, in-situ sensors high reliability at fixed points, and field reports contextual observability with variable reliability.

$$\gamma_k^i(x,y) = min\,\{\gamma_{obs}, \gamma_{cov}, \gamma_{fre}, \gamma_{rel}\} \qquad (38)$$

Confidence bounds the disturbance instead of describing it statistically. The observation error magnitude is limited by the complement of the confidence, as expressed in ( 39 ); a fully confident observation is exact, a vanishing confidence admits the worst-case disturbance, and intermediate values degrade gracefully between these poles. Missing observations are bridged by the persistence prior of the concept layer rather than by imputation.

$$|\varepsilon_k^i(x,y)| \leq (1 - \gamma_k^i(x,y)) \cdot \varepsilon_{max} \qquad (39)$$

From the regional observation, the agent computes the ten features of the input space ( 40 ). Each feature inherits a feature-level confidence from the observation fields its extraction map consumes, so the pair of feature value and feature confidence is the complete interface between perception and reasoning.

$$z_{f,k}^{i} = \xi_f\left(O_k^i\right) \in [0,1], \quad f = 1, \ldots, N_{feature} \tag{40}$$

The ten features are listed in Table 2.1: Fire intensity, spread potential, weather severity, ignition proximity, fuel load, asset exposure, resource accessibility, access and road status, suppression availability, and temporal urgency. Each feature carries established wildfire-science grounding rather than convenience, as documented source by source in the background chapter; by construction the ten features span the threat, exposure, feasibility, and timing dimensions of the operational questions, so no question lacks a supporting feature; and none duplicates information carried by the others, since each is admitted only under the observability, boundedness, operational meaning, and non-redundancy conditions.

**A larger set would not add discriminative information but would enlarge the antecedent space of every downstream mapping, and the analysis of the evaluation chapter exercises the operational parameters of the closed loop. The set is therefore critical, in that removing a feature leaves an operational question unsupported, and sufficient, in that adding one is not justified by any unanswered question.**

## 4.4 Layer 3: Concept Space and the Feature-Concept Hierarchy

Rules cannot be written over the features directly. With ten features and five linguistic terms per feature, a flat rule base spans 9,765,625 antecedent combinations, which can neither be elicited from experts nor audited by operators.

The conceptualization layer maps physically related features into six base concepts, the base concepts into three mid-level and two higher-level decision concepts, and finally into the single coordination concept; decision rules are then written over the five decision concepts only, whose antecedent space contains $5^5 = 3125$ combinations, of which the evolving rule base instantiates only the small subset that observed situations activate. Figure 4.2 illustrates the solution of this problem by hierarchical conceptualization.

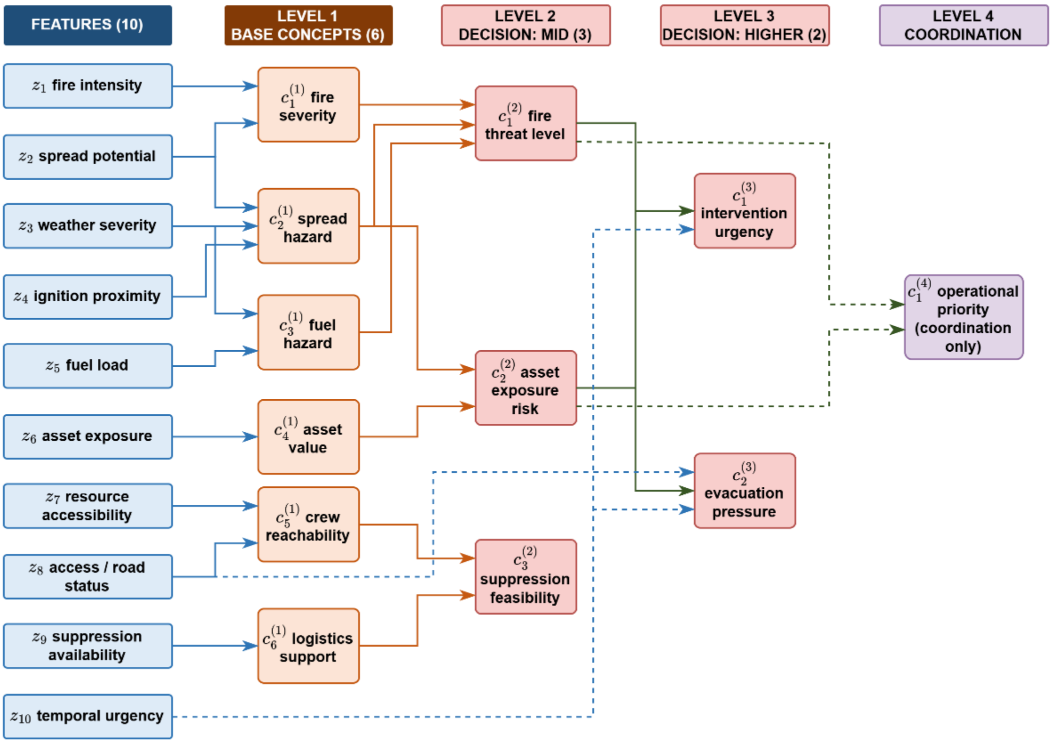


Figure 4.2 Feature-concept hierarchy

The recognition-primed decision literature shows that experienced commanders reason over exactly such condensed situation descriptors rather than over raw measurements. The concept set is sufficient for management because the five decision concepts answer the five canonical questions completely and non-redundantly: **How severe is the threat**, **how exposed are the assets**, **whether an intervention can feasibly be mounted**, **how urgent action is**, and **whether evacuation pressure is building** as listed in Table 2.3.

The mapping between features and concepts is fully specified by the sets of ( 41 ) Each concept is written in the positional notation c with a lower index for its place within a level and an upper index for its level, so that fire severity, spread hazard, fuel hazard, asset value, crew reachability, and logistics support are the six base concepts of level one, fire threat level, asset exposure risk, and suppression feasibility occupy level two, intervention urgency and evacuation pressure occupy level three, and operational priority is the single concept of level four (Table D.1). Table D.1

Base Concepts

$$\mathcal{C}_1^{(1)} = \{z_1, z_2\}$$
$$\mathcal{C}_2^{(1)} = \{z_2, z_3, z_4\}$$
$$\mathcal{C}_3^{(1)} = \{z_3, z_5\},$$
$$\mathcal{C}_4^{(1)} = \{z_6\},$$
$$\mathcal{C}_5^{(1)} = \{z_7, z_8\},$$
$$\mathcal{C}_6^{(1)} = \{z_9\}$$

Decisional Concepts

$$\mathcal{C}_1^{(2)} = \left\{c_1^{(1)}, c_2^{(1)}, c_3^{(1)}\right\}$$
$$\mathcal{C}_2^{(2)} = \left\{c_2^{(1)}, c_4^{(1)}\right\},$$
$$\mathcal{C}_3^{(2)} = \left\{c_5^{(1)}, c_6^{(1)}\right\} \qquad (41)$$
$$\mathcal{C}_1^{(3)} = \left\{c_1^{(2)}, c_2^{(2)}, z_{10}\right\},$$
$$\mathcal{C}_2^{(3)} = \left\{c_2^{(2)}, 1 - z_8, z_{10}\right\}$$
$$\mathcal{C}_1^{(4)} = \{c_1^{(2)}, c_2^{(2)}\}$$

The activation of a base concept is the weighted sum of the term memberships of the features in its set, and the activation of every higher concept is the weighted sum of the activations of the concepts in its set, with the nonnegative weights of each concept normalized to one ( 42 ).

$$a_{n,k}^{(1),i} = \sum_{f \in \mathcal{Z}_n} \omega_{n,f} \cdot \mu_{\mathcal{A}_{n,f}}\left(z_{f,k}^i\right)$$
$$a_{n,k}^{(l),i} = \sum_{m \in \mathcal{C}_n^{(l)}} \omega_{n,m}^{(l)} \cdot a_{m,k}^{(l-1),i}, \quad l = 2,3,4 \qquad (42)$$

In ( 42 ), $a$ denotes a concept activation, with the subscripts giving the concept index $n$ and the step $k$ and the superscripts giving the level $l$ and the region $i$; $\omega$ are the nonnegative aggregation weights, normalized to sum to one for each concept; $\mathcal{Z}$ is the index set of the features assigned to a base concept, and $\mathcal{C}$ the index set of the concepts one level below that feed a higher concept; μ is the term membership of the fuzzified feature introduced in the background chapter; and $z$ is the feature value. Throughout the design an activation is carried as a five-term membership vector, one entry per linguistic term, and ( 42 ) applies term by term, so the aggregation yields a membership distribution rather than a single number. A scalar reading of a concept, the expected value of its vector over the term centers, is used only where one value per concept is required, by the coordinator and by the quality test. The input set of a concept is not restricted to the level immediately below: it may include a raw feature directly, entering through its term memberships or through its complement, and it may draw on concepts more than one level down; the tables of Appendix D list the exact input set of every concept.

An example is illustrated in Figure 4.3. At level one, consider fire severity with feature set $\{z_1, z_2\}$, aggregation weights 0.6 and 0.4, and the severity-oriented term high of each feature. A fire intensity reading of 0.70 lies on the core of high, giving membership 1.00, and a spread potential reading of 0.62 lies on its lower shoulder, giving membership 0.47; the aggregation of ( 42 ) therefore yields the activation $0.6 \times 1.00 + 0.4 \times 0.47 = 0.79$. Level two repeats the same operation over activations instead of memberships. Fire threat level draws on fire severity, spread hazard, and fuel hazard with weights 0.5, 0.3, and 0.2; with the fire severity of 0.79 just computed, a spread hazard of 0.55, and a fuel hazard of 0.30, the activation is $0.5 \times 0.79 + 0.3 \times 0.55 + 0.2 \times 0.30 = 0.62$. Intervention urgency draws on fire threat level, asset exposure risk, and the feature temporal urgency with weights 0.4, 0.35, and 0.25; with the 0.62 just computed, an asset exposure risk of 0.61, and a temporal urgency reading of 0.72 whose high membership is 1.00, the activation is 0.71. One sensor-level number has now propagated through three levels, and at every step the intermediate value remains readable as an operational statement.

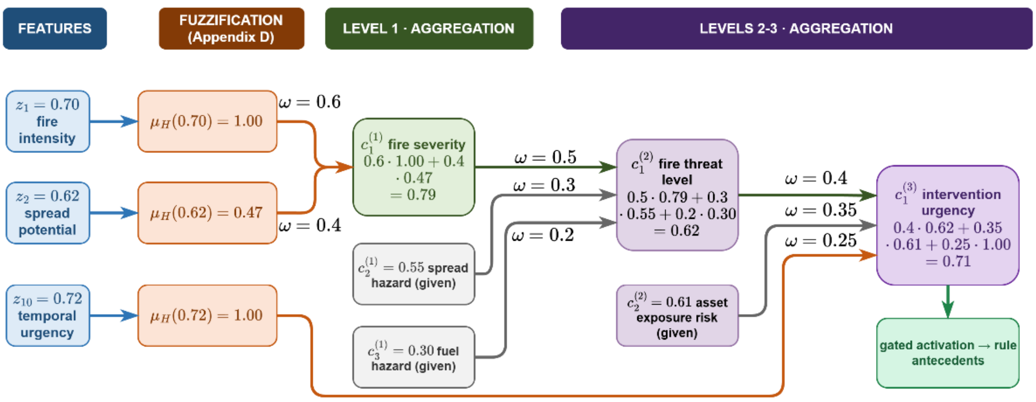


Figure 4.3 Hierarchical concept activation example

Raw activations are never consumed directly. Every activation is gated by its observation confidence and blended with a persistence-carried prior ( 43 ). A reliable observation passes through unchanged, while an uninformative one falls back to the decayed previous estimate.

$$a_{n,k}^{eff,i} = \gamma_{n,k}^{i} \cdot a_{n,k}^{obs,i} + \left(1 - \gamma_{n,k}^{i}\right) \cdot \rho \cdot a_{n,k-1}^{eff,i},$$
$$\gamma_{n,k}^{i} = min \{\gamma_{f,k}^{i} : f \in \mathcal{Z}_n\} \qquad (43)$$

In ( 43 ), the superscripts distinguish three activations of the same concept: The observed activation computed through ( 42 ), the persistence prior carried from the previous step, and the effective activation that the decision rules consume. The gate γ is the observation confidence of ( 38 ), composed at concept level as the minimum of the feature confidences feeding the concept, and the decay factor $\rho \in [0, 1)$ discounts the previous effective activation, so a concept that ceases to be observed fades over successive steps instead of persisting indefinitely.

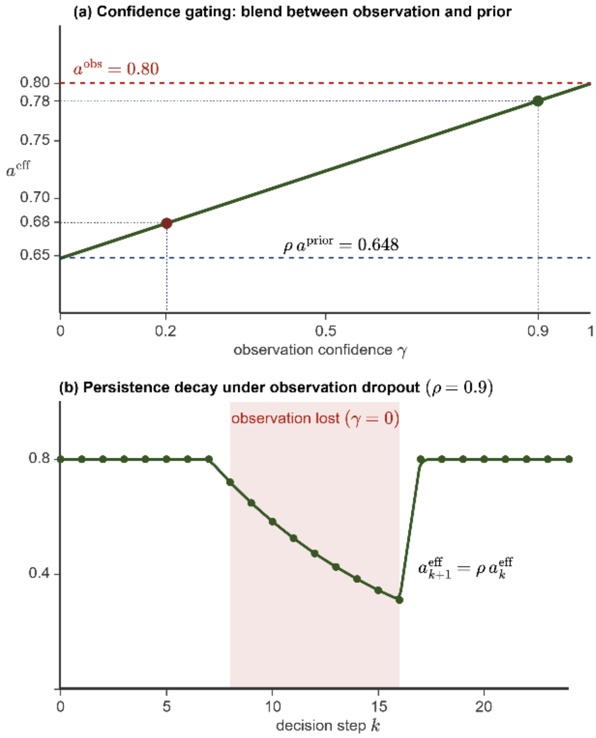


Figure 4.4 Confidence gating (a) persistence decay (b)

In Figure 4.4, the plot (a) shows the effective activation against the confidence for an observed activation of 0.80, a carried prior of 0.72, and $\rho = 0.9$: At $\gamma = 0.9$ the effective value is 0.78, at $\gamma = 0.2$ it is 0.68, and the transition between trust and caution is linear, with no threshold to tune. The plotted values follow one component of the vector; the gate applies to every term alike. The plot (b) shows the temporal role of the prior. When observation is lost, effective activation decays geometrically by $\rho$ per step and snaps back to the observation as soon as confidence returns.

## 4.5 Layer 4: Decision Generation and Coordination

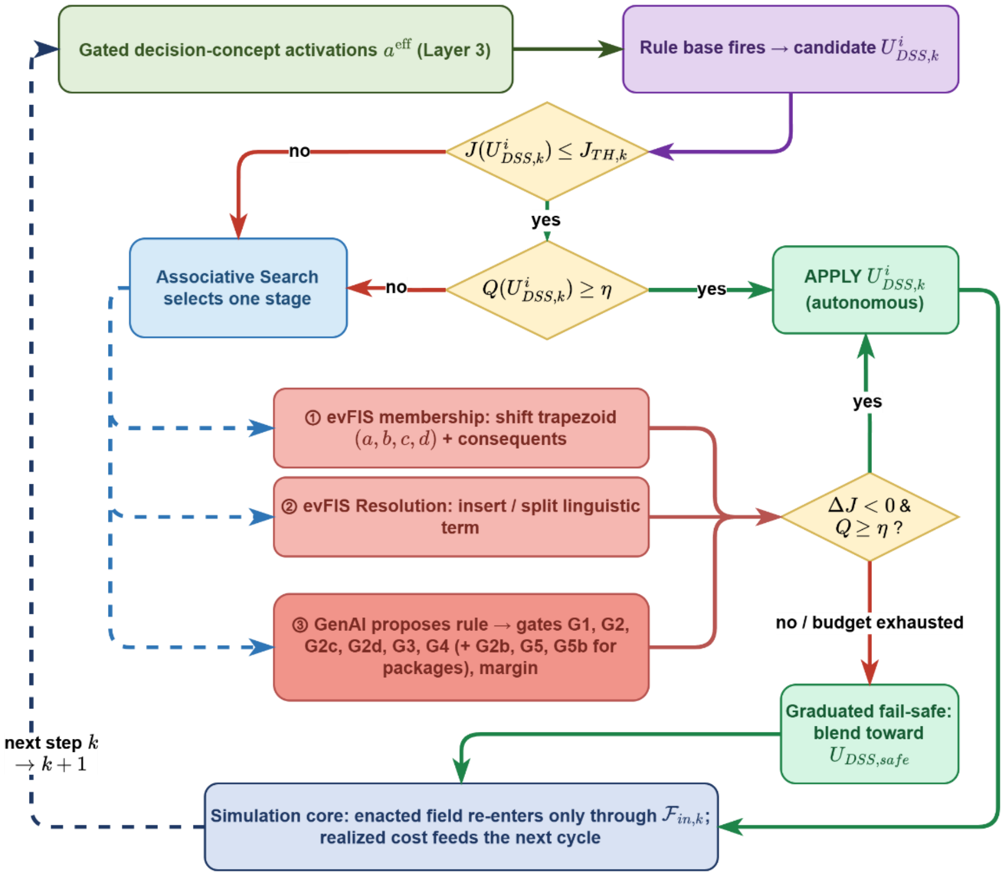


Figure 4.5 Layer 4 decision and adaptation flow

The decision space is where the gated concept activations of Table 2.3 become the executable interventions of Table 2.4. The layer operates once per decision cycle and follows a fixed order. The rule base fires on the gated activations and produces a candidate intensity vector per region; the candidate is forecast on a shadow copy of the simulation and accepted if it satisfices; the quality measure then checks the accepted candidate against the concepts that demanded it, and the graduated fail-safe attenuates an unbalanced candidate; the coordinator converts the regional operational priorities into attention shares; a rejected forecast engages exactly one stage of the staged adaptation loop, selected by a learned controller; and a no-harm rule withholds the offensive part of any allocation that is forecast to worsen the physical outcome. Each element is developed in the following subsections, and the value of every threshold is stated where its mechanism is defined.

As illustrated in Figure 4.5, Layer 4 receives the gated decision-concept activations $a^{eff}$ of Layer 3, five term-membership vectors, one per decision concept with entries in [0,1] that already carry the confidence gating and the persistence prior of the concept layer, and it returns $U_{DSS,k}^{i}$, an intensity vector over the six intervention families for region $i$ at cycle $k$. Nothing else crosses the boundary. The layer never reads the physical state directly and never writes to it.

**The rule base fires first.** Antecedents are written over the five gated decision concepts, each partitioned into the five linguistic terms of Layer 3, and consequents are numeric intervention intensities in $[0,1]$, that is, zero-order Takagi-Sugeno singletons, which is the form the adaptation loop can tune. Intensities are combined per intervention family by an activation-weighted average with a unit floor on the denominator, so a weakly supported conclusion yields a weak order and only a fully fired rule delivers its full intensity.

The base is deliberately sparse and evolving. The antecedent space admits $5^5 = 3125$ combinations, yet only situation-activated rules are ever instantiated, and the base grows through the adaptation loop instead of being enumerated in advance. The seed rules and a representative subset of the adaptation-born rules, with their provenance, are catalogued in Appendix E. A situation whose strongest firing stays below the coverage floor $\alpha_{min} = 0.05$ is flagged as a coverage gap, which falls back to the watchful posture and at the same time triggers adaptation.

**The candidate is then judged by what it is expected to cost.** The candidate allocation is applied over the forecast horizon, and the normalized decision cost $J$ of the end state is compared against a no-action shadow run started from the same state. The acceptance threshold $J_{TH}$ is not entered by hand for an individual decision. It is recomputed every cycle as the tighter of a fixed normalized ceiling of $0.35$ and a required relative gain over the no-action forecast, so a situation in which doing nothing is already cheap automatically demands a stronger candidate. If $J \le J_{TH}$ the cycle proceeds to the quality gate, and if not it enters the staged adaptation loop.

**The quality gate measures not how much the candidate costs but whether it serves the concepts that demanded it.** Each scored decision concept is paired with the intervention family that answers it, fire threat level with suppression effort, asset exposure risk with asset protection, intervention urgency with resource deployment, and evacuation pressure with evacuation and public warning. Suppression feasibility is the gatekeeper's concept and modulates others rather than being scored against a family of its own.

A failed cost test hands the cycle to the value-based stage controller, which invokes exactly one adaptation stage per cycle, selected from three:

- **The adaptation stage ① is evFIS membership modification,** which shifts the trapezoidal parameters (a, b, c, d) of the active terms together with the rule consequents.
- **The adaptation stage ② grows the rule base.** It writes a rule on an antecedent combination that no rule covers and, where a covered combination is linguistically ambiguous, it inserts a narrow term to separate the two situations that share the combination.
- **The adaptation stage ③ is the generative proposal**, in which a language model drafts one new rule in the framework's own vocabulary of concepts, terms and admissible interventions.

The value-based stage controller is epsilon-greedy over a state that buckets the cost deficit and flags a coverage gap, its reward is the realized improvement of the physical forecast, and it therefore learns which stage pays off in which situation. **The result of a trial is kept only when the shadow forecast improves.**

**A generated rule is a suggestion and never acts on its own.** It must pass the applicable verification gates before it can enter the base. $G1$ checks format, that the antecedent and the consequent are well formed. $G2$ checks constraints, that the arity, the vocabulary, and the numeric ranges are admissible and that the rule does not duplicate an antecedent combination that is already occupied. G2c checks relevance, that every antecedent of the rule fires in the present situation. G2d checks pool economy, that the rule does not order capacity beyond the funded budget. $G3$ checks simulation, that the rule lowers the forecast cost, $\Delta J < 0$, on the shadow copy. $G4$ is an $A/B$ check that the improvement holds on two independent seeds rather than on a single lucky draw. A proposal that fails any gate is discarded and the failing gate is recorded, so a rejected suggestion remains auditable. A plain rule that clears all six gates is admitted with its provenance into the base; a package additionally faces G2b for a new concept and G5 with G5b for its vocabulary growth. Twenty representative admitted rules are catalogued in Appendix E.

The loop then returns the repaired candidate to the same two tests. If both tests pass, the candidate is applied. Otherwise, or when the adaptation budget is exhausted, the graduated fail-safe blends the candidate toward $U_{DSS,safe}$ in proportion to the quality deficit. The offensive families, suppression effort, resource deployment, containment line and asset protection, are attenuated by the factor $Q/\eta$, while the life-safety orders, evacuation and public warning, are never reduced. This scaling is the fail-safe blend of ( 11 ) written per family: $U_{DSS,safe}$ holds the offensive families at zero and carries the life-safety intensities forward, so the blend with $\lambda = Q/\eta$ reduces to the attenuation above.

If even the adapted candidate is forecast to end worse than doing nothing on the physical part of the cost, that is, burned area, asset loss, and population exposure, without crediting the response and delay bookkeeping, the no-harm rule withholds the offensive allocation for the cycle. No capacity is fielded, no response cost is paid, and only the life-safety orders stand. Wetting and line construction take effect thirty to sixty minutes ahead; hence, this comparison is always re-run over a lookahead, and orders are withheld only when the candidate is forecast clearly worse rather than merely equal. Clearly worse means that the forecast physical cost of the candidate exceeds the no-action forecast by more than a fixed tolerance of one part in a thousand of the normalized cost. The rule bounds harm as predicted by the shadow forecast; its validity in the field is conditional on the forecast fidelity assessed in Chapter 5. Whichever exit is taken, the enacted field re-enters the simulation core only through the external input layer $F_{in,k}$, and the realized cost of the step feeds the next cycle.

### 4.5.1 Rule Base and Candidate Intervention

Layer 4 converts concept activations into orders through a rule base written exclusively over the five gated decision concepts. Every rule takes the canonical conjunctive form of ( 44 ). Each antecedent clause tests one decision concept against one linguistic term, and the consequent assigns a singleton intensity to each intervention family, so a concept triggers an intervention precisely by the degree to which it satisfies the clause that names it.

$$R_r: IF\ a_{1,k}^{eff,i}\ is\ \mathcal{A}_{r,1}\ AND\ \ldots\ AND\ a_{5,k}^{eff,i}\ is\ \mathcal{A}_{r,5}\ THEN\ U_{DSS,k}^{i}\ is\ \mathcal{B}_r \qquad (44)$$

In ( 44 ), $R_r$ denotes rule r, $a_{n,k}^{eff,i}$ the gated activation of decision concept $n$ in region $i$ at cycle $k$, $A_{r,n}$ the linguistic term the rule requires of that concept, and $B_r$ the consequent, read as the singleton intensity profile of the rule together with its optional effects. The candidate of the region is the vector $U_{DSS,k}^{i}$, whose components $u_{m,k}^{i}$ are the intensities ordered for the six intervention families indexed by m. A concept absent from an antecedent is left unconstrained, so a rule may refer to a single concept or about all five.

A rule fires with the strength of its weakest satisfied clause, and the candidate is assembled family by family from every rule that fires, as stated in ( 45 ). Here $\alpha_r$ is the firing strength of rule $r$, taken as the minimum membership over its antecedent clauses, $\mu_{A_{r,n}}$ is the reading of the clause: For a native term it is the component of the effective activation vector at the term the clause names, the aggregated membership of ( 42 ), and only an inserted term of stage ② is evaluated on the crisp reading of its concept against its own trapezoid, and $v_{r,m}$ is the intensity that the rule assigns to family $m$. The denominator is floored at one, so a weakly supported conclusion yields a weak order rather than a full-strength one, and only rules that together fire with unit strength deliver their full intensity.

$$\begin{aligned} u_{m,k}^{i} &= \frac{\sum_r \alpha_r \, v_{r,m}}{max(1, \sum_r \alpha_r)} \\ \alpha_r &= \min_{n \in ant(r)} \mu_{A_{r,n}}(a_{n,k}^{eff,i}) \end{aligned} \quad (45)$$

The base is sparse by construction. The antecedent space admits $5^5$ = 3125 combinations, but this is the catalogue of conditions the vocabulary can express, not a list of rules to be authored. The base holds the seeded doctrine and whatever the adaptation loop later adds. Sparsity costs no coverage because adjacent linguistic terms overlap. Equation ( 46 ) counts what a single rule already covers, where $N_{term}$ is the number of linguistic terms partitioning each concept, $N_{concept}$ is the number of decision concepts and $N_{antecedent,r}$ is the number of antecedent clauses of rule $r$. While 2-clause rule covers $5^{5-2} = 125$ combinations, full-clause covers $5^{5-5} = 1$.

$$N_{cover}(R_r) = N_{term}^{N_{concept} - N_{antecedent,r}} \quad (46)$$

The seeded base must satisfy, where $a^{eff} \in [0,1]^5$ is any reachable vector of gated concept activations, $\alpha_r(a^{eff})$ is the firing strength of rule $r$ at that reading, and $\alpha_{min} = 0.05$ is the coverage floor. The condition states full graded coverage: Even the worst reading of the whole space leaves some rule firing with strength at least $\alpha_{min}$. The full doctrine of Appendix E satisfies it. The minimal evaluation profile of five seed rules deliberately does not: Its rules sit on the high terms, so low-activation readings fall below the floor by design. Such a reading marks a coverage gap. The layer falls back to the watchful posture and adaptation is triggered.

A rule with $N_{antecedent,r}$ clauses refers to the combinations of the catalogue counted in ( 46 ), those that agree with it on its clauses and are free on the rest, and the full seed doctrine satisfies the coverage condition of ( 47 ), so at every reachable reading some rule fires with strength at least $\alpha_{min}$ = 0.05. The evaluation campaigns start from the minimal profile instead, which surrenders this property on purpose: A coverage gap is then not a defect but the signal that summons stage ②, and until a rule arrives the layer holds the watchful posture.

$$min_{a^{eff} \in [0,1]^5}\ max_r\ \alpha_r(a^{eff}) \geq \alpha_{min} > 0 \qquad (47)$$

As an example, a fire threat activation of $0.60$ belongs to the term medium with membership $0.667$ and to the term high with membership $0.333$. If the base prescribes suppression effort $0.5$ for the first and $0.8$ for the second, both rules fire and ( 45 ) returns $0.600$, a value no rule states explicitly. Coverage is therefore graded rather than binary, and it carries two calibrated thresholds. Below $0.05$, no rule runs meaningfully and the layer falls back to the watchful posture, while a strongest firing that stays under the gap threshold marks a coverage gap and triggers adaptation.

Full enumeration is rejected on principle, and the comparison of Table 4.2 is the argument. A catalogue of 3125 rules cannot be audited from an expert and freezes a guess into every combination that operations have never visited. It also fixes the alternatives in advance, which is precisely the closed decision space this thesis sets out to leave behind. The sparse base is authored where doctrine exists, interpolates where it does not, and grows only where realized cost proves that it must. Table 4.3 states how each concept has a primary family that answers its operational question and a set of secondary families that it modulates. A run starts from the minimal profile of that base in Appendix E: Five seed rules that together cover the six intervention families, so the system starts without doctrine and must learn its base by trial.

Table 4.2 Full enumeration versus the sparse evolving base

| **Property** | **Full enumeration** | **Sparse evolving base** |
|---|---|---|
| **Authorship** | Cannot be elicited by experts; most rows describe unrealizable situations | Expert-seeded doctrine plus adaptation-born rules, each individually validated |
| **Auditability** | Thousands of rows, no operator can inspect them | Tens of rules, every one traceable and readable |
| **Coverage** | Exhaustion by construction | Complete through overlap and interpolation, guaranteed by ( 47 ) |
| **Unvisited regions** | Frozen guesses written in advance | Left to interpolation until the field demands a rule there |
| **Novel situations** | Cannot be represented; the catalog is closed | Adjusted, refined, or generated on demand through the staged adaptation loop |

Table 4.3 Mapping between decision concepts and intervention families

| **Concept** | **Primary intervention** | **Secondary interventions and modulation role** |
|---|---|---|
| **Fire threat level** | Suppression effort | Containment line when direct attack is insufficient; resource deployment scaled with the threat level |
| **Asset exposure risk** | Asset protection | Containment line placed to shield exposed assets; evacuation when exposure combines with high threat |
| **Suppression feasibility** | Resource deployment | In the gatekeeper role, high feasibility commands resource deployment toward the front, while low feasibility caps suppression effort and shifts intensity toward containment line and evacuation. |
| **Intervention urgency** | Resource deployment | Scales the intensities of all active families; triggers cross-region priority preemption at very high activation |
| **Evacuation pressure** | Evacuation | Public warning raised jointly; offensive intensities withdrawn at very high activation |

### 4.5.2 Cost, Quality, and the Satisficing Test

Every candidate is judged by two complementary measures, both computed on a shadow copy of the simulation before anything reaches the field. The cost asks what the candidate is expected to cost the incident. The quality asks whether it serves the concepts that demanded it. A candidate must clear both, and the two tests appear as the two decision diamonds of Figure 4.5.

#### 4.5.2.1 Cost Test

The cost is not evaluated on the current state. The candidate allocation is held constant on the simulation over a horizon of at least $\tau_{horizon}$, stepped at $\tau_{resolution}$, and the cost of the forecast end state is compared with a no-action shadow run started from the same state. The cost itself is the weighted five-term ( 48 ) with the exact normalizations of the implementation so the weighting is a calibration surface rather than an assumption.

$$J = \frac{\sum_j w_j J_j}{\sum_j w_j}, \qquad j \in \{burn, asset, pop, resp, delay\}$$

$$J_{burn} = \frac{\dfrac{N_{burned}}{\rho_{burn} \cdot N_{burnable}}}{1 + \dfrac{N_{burned}}{\rho_{burn} \cdot N_{burnable}}} \qquad J_{pop} = \min\left(1, \frac{P_{steps}}{P_{total} \cdot N_{decisions}}\right)$$

$$J_{resp} = \min\left(1, \frac{C_{steps}}{C_{ref} \cdot N_{decisions}}\right) \tag{48}$$

$$J_{asset} = \frac{V_{lost}}{V_{total}} \qquad J_{delay} = min(1, \frac{\bar{\tau}}{\tau_{reference}})$$

In ( 48 ), $w_j$ are the non-negative priority weights, and the sum is divided by their total so that $J$ remains in the unit interval and the weights encode relative priority rather than scale. Each term is an actual quantity divided by a scenario reference, which is what makes five quantities of different units summable.

$J_{burn}$ charges the burned area against a major-fire reference rather than against the whole landscape. $N_{burned}$ is the number of cells that have burned and $N_{burnable}$ the number of burnable cells of the scenario, with $\rho_{burn} = 0.05$. The product $\rho_{burn} \cdot N_{burnable}$ is the reference area whose loss is treated as a full-scale fire. The rational form $x/(1+x)$ with $x = N_{burned}/(\rho_{burn} \cdot N_{burnable})$ is linear near the origin, reads exactly 0.5 at the reference fire, and approaches one without reaching it; its slope decays quadratically.

$J_{asset}$ is the fraction of exposed value that the fire has taken, where $V_{lost}$ is the value of the structures and critical infrastructure standing on burned cells and $V_{total}$ the total value exposed in the scenario.

$J_{pop}$ integrates exposure rather than counting heads. $P_{steps}$ accumulates the persons present inside actively burning cells (not evacuated people) at every decision step, $P_{total}$ is the total population of the scenario at risk and $N_{decisions}$ the number of decision steps of the horizon, so the denominator is the worst case in which the entire population is exposed for the whole horizon. A brief exposure of many people and a long exposure of a few therefore become commensurable.

$J_{resp}$ prices effort in the same way. $C_{steps}$ is the fielded capacity summed over the decision steps, which is the time integral of the committed force, and $C_{ref}$ is the reference capacity of the scenario, so the denominator is the reference force kept in the field for the whole horizon. An army held on the line for hours is therefore charged more than a short decisive strike of the same size.

$J_{delay}$ prices timeliness. $\bar{\tau}$ is the capacity-weighted mean dispatch time of the resources actually fielded, so a large force that arrives late weighs more than a single late unit, and $\tau_{reference}$ is the reference response time of sixty minutes. Each of the last three terms is capped at one, so no single term can dominate the sum once its reference is exceeded.

The five-term sum is the criterion by which a candidate is accepted, but it is not the measure by which two runs are compared. The response and delay terms price the decision itself, so a run that never intervenes carries neither of them, nor a comparison on the total cost would reward inaction. Cross-run comparisons, the with and without DSS studies of Chapter 5 and the trial verdicts of the adaptation stages, therefore read the physical cost, the burned-area, asset, and population terms with their weights renormalized. The physical cost measures only what the fire did to the land, the assets, and the people, and it is indifferent to how much effort the decision spent, which makes it a fair basis between a run that fought and a run that watched.

#### 4.5.2.2 Acceptance Test

Acceptance on cost is satisficing rather than optimizing. The candidate is held on the shadow copy over the forecast horizon, and the resulting cost $\hat{J}_k\left(U_{DSS,k}\right)$ is compared with the forecast of taking no action from the same state, $\hat{J}_k(0)$.

$$\hat{J}_k\left(U_{DSS,k}\right) \leq min(J_{TH}, (1-\delta_{gain}) \cdot \hat{J}_k(0)) \quad \delta_{gain} = 0.05 \tag{49}$$

The acceptance bound is the smaller of two quantities, and each one protects against a different mistake. The first is the fixed ceiling $J_{TH}$. It sets the highest cost at which the incident may be left. On a large fire this is the binding condition. A candidate is rejected if it leaves the incident expensive, even when it is far better than doing nothing. The second is $(1 - \delta_{gain}) \cdot \hat{J}_k(0)$, a fixed fraction of the no-action forecast. On a small fire this is the binding condition. When doing nothing already costs less than the ceiling, the ceiling alone would accept every candidate, including one that changes nothing. The margin $\delta_{gain}$ prevents this by requiring the candidate to beat the no-action forecast. Without it, the incidents that are easiest to improve would be the ones the adaptation stages never touch.

Failing the cost test does not suspend action. The orders already produced by the standing rule base remain in force and are applied while adaptation runs, so an acceptable decision is never withdrawn in the hope of a better one.

**The quality measures what the cost cannot.** A concept demands an action; the quality asks whether the candidate supplied it. For each scored concept, gated activation is compared, read as the scalar expected value of the activation vector, with the strongest intensity the candidate assigns to the intervention family that answers that concept, and the quality is one minus the mean of these mismatches ( 50 ). $F(n)$ is the intervention family that answers concept $n$, ${a_n}^{eff}$ is the gated activation of that concept and $u_f$ the intensity the candidate orders for family $f$.

$$Q_k = 1 - \frac{1}{4}\sum_{n \in C_Q} |{a_n}^{eff} - max_{f \in F(n)}\, u_f|\,,\ \ accept\ \Leftrightarrow\ Q_k \geq \eta \qquad (50)$$

$C_Q$ is the set of the four scored decision concepts. Suppression feasibility is excluded from $C_Q$. It is a modulator rather than a demander. Its activation states whether action is possible, not how much action is required. A high feasibility permits a strong response without calling for one, so a region with workable terrain and no fire would be penalized by a match test even though a null response is correct. A low feasibility calls instead for a reduction of the intensities that the other concepts demand, and a reduction is not something a match test can score. The family through which it acts, resource deployment, already answers intervention urgency in ( 50 ).

**A candidate that "suppresses heavily while the fire threat is low" is penalized as much as one that ignores a high evacuation pressure, even though only the second failure would raise the cost within the horizon.** Quality is kept separate from cost. It detects a misdirected response before its consequences appear. The two tests are in Table 4.4. When the quality falls below the gate, the graduated fail-safe attenuates the offensive intensities in proportion to the deficit, and a quality failure is itself one of the symptoms that summon adaptation.

Table 4.4 The two acceptance tests and the consequence of failure

| **Test** | **Condition** | **If it fails** |
|---|---|---|
| **Cost** | $\hat{J}_k(U) \leq \min\left(J_{TH}, \left(1 - \delta_{gain}\right) \cdot \hat{J}_k(0)\right)$<br>$J_{TH} = 0.35, \delta_{gain} = 0.05$ | Staged adaptation is engaged; the standing orders remain in force. |
| **Quality** | $Q_k \geq \eta$<br>$\eta = 0.60$ | The graduated fail-safe attenuates the offensive intensities by $\frac{Q_k}{\eta}$. |

#### 4.5.2.3 Cost and Quality Acceptance Test Example

The cost-quality cycle is taken for a defined horizon into an incident. A decision candidate may fail the cost test, or pass it and fail the quality test, or pass both. The last two outcomes occur in this cycle, in different regions. As an example, there are two forecasts, namely one with the candidate allocation held constant and one with no action. Each is priced at the five-term cost of ( 48 ). Table 4.5 gives the terms of both, already normalized, with the weights that combine them.

Table 4.5 Forecast cost terms of the candidate and of no action

| **Term** | **Weight $w$** | **Candidate $\widehat{J_k}$ $(U)$** | **No action $\widehat{J_k}$ (0)** |
|---|---|---|---|
| **Burned area, $J_{burn}$** | 1.0 | 0.5218 | 0.5376 |
| **Asset loss, $J_{asset}$** | 1.0 | 0.2226 | 0.3061 |
| **Population, $J_{pop}$** | 1.0 | 0.0004 | 0.0008 |
| **Response, $J_{resp}$** | 0.2 | 0.0549 | 0.0300 |
| **Delay, $J_{delay}$** | 0.2 | 0.2230 | 0.0000 |
| **Weighted sum** | | **0.8004** | **0.8506** |

Each cost is that weighted sum divided by the total weight, 3.4.

$$\widehat{J_k}(U) = \frac{0.8004}{3.4} = 0.2354$$
$$\widehat{J_k}(0) = \frac{0.8506}{3.4} = 0.2502 \tag{51}$$

**Cost Test:** These two numbers are what the acceptance test of ( 49 ) compares.

$$0.2354 \leq \min(0.35, (1 - 0.05) \times 0.2502) = 0.2377$$
$$J_{TH} = 0.35, \delta_{gain} = 0.05 \tag{52}$$

The candidate is accepted. The ceiling is not the binding condition: The forecast cost of no action, 0.2502, already lies below the ceiling of 0.35, so the ceiling alone would accept every candidate. The margin sets the requirement. A candidate that fails is not withdrawn: The standing orders remain in force while the adaptation stages of Section 4.5.3 search for a better one.

Acceptance on cost applies to the incident as a whole. The quality test of ( 50 ) is evaluated in each region, over the four scored concepts. Table 4.6 lists their terms. The activation is what the concept demands, the intensity is what the candidate orders from the family that answers it, and ordering more than is asked for counts as much as ordering less.

Table 4.6 Terms of the quality test in the two regions of the cycle

| **Region** | **Concept** | **Activation $\boldsymbol{a_{\mathrm{n}}^{\mathrm{eff}}}$** | **Ordered intensity $\boldsymbol{u_{\mathrm{f}}}$** | **Mismatch $\boldsymbol{\lvert a_{\mathrm{n}}^{\mathrm{eff}} - u_{\mathrm{f}} \rvert}$** |
|---|---|---|---|---|
| **Region 1** | fire threat | 0.597 | 0.360 | 0.237 |
| **Region 1** | asset exposure | 0.541 | 0.049 | 0.492 |
| **Region 1** | intervention urgency | 0.639 | 0.554 | 0.085 |
| **Region 1** | evacuation pressure | 0.497 | 0.000 | 0.497 |
| **Region 2** | fire threat | 0.687 | 0.360 | 0.327 |
| **Region 2** | asset exposure | 0.831 | 0.000 | 0.831 |
| **Region 2** | intervention urgency | 0.816 | 0.920 | 0.104 |
| **Region 2** | evacuation pressure | 0.674 | 0.000 | 0.674 |

The mismatches of Region 1 are small. The quality result of ( 53 ) stands above the gate, 0.672 ≥ 0.60, so the orders of that region are applied as written.

$$Q_1 = 1 - \frac{1}{4}(0.237 + 0.492 + 0.085 + 0.497) = 0.672$$
$$Q_1 = 0.672 > \eta = 0.60 \qquad (53)$$

The second region orders slightly more than intervention urgency asks for, and answers two demands with nothing: Asset exposure asks for 0.831, and evacuation pressure asks for 0.674. The cost test cannot see this, because assets that are not burning and people who are not exposed carry no cost inside the forecast horizon.

The quality test of ( 54 ) falls below the gate, so the fail-safe engages. It does not cancel the orders; it attenuates the offensive ones in proportion to the deficit ( 55 ).

$$Q_2 = 1 - \frac{1}{4}(0.327 + 0.831 + 0.104 + 0.674) = 0.516 \tag{54}$$
$$Q_2 = 0.516 < \eta = 0.6$$

$$u_\mathrm{f}' = \frac{Q_2}{\eta} u_\mathrm{f} = 0.86 u_\mathrm{f} \tag{55}$$

Suppression falls from 0.360 to 0.310 and resource deployment from 0.920 to 0.791. Evacuation and public warning are never attenuated. Neither is ordered in this cycle, which is the omission the quality test records.

The cost test passes for the incident as a whole, and a single number cannot show where inside the response is misdirected. The quality test judges each region and fails in one. **The cost asks what the orders are expected to achieve; the quality asks whether they answer what is demanded.**

### 4.5.3 Open Decision Space: Staged Adaptation

Adaptation is organized as three stages of strictly increasing invasiveness, namely ① **parameter adjustment inside the existing rules**, ② **growth of the rule base into ground it does not cover**, and ③ **generative proposal of a new rule** and, where a package is admitted, of a new element of the vocabulary. The first two stages are the evolving fuzzy (evFIS) mechanism, which tunes and grows the rule base; the third stage is generative. A stage engages only when the standing decision demonstrably falls short. Every trial it produces is judged by a single criterion, the physical cost: The burned area, asset, and population terms with their weights renormalized, evaluated on a shadow forecast. The total cost of ( 48 ) continues to assess the standing decision, but it is deliberately not the criterion by which a trial is judged. On a small fire the response cost of any additional commitment exceeds the physical gain that a short horizon can reveal, so a trial criterion built on the total cost would reject every proposal at the very moment when the base most needs to grow.

In Table 4.7, the rows follow the invasiveness order of the ladder, and the columns state the guarantee that keeps the ladder auditable, namely that a consequent is tuned, a rule is written, and a name enters the vocabulary.

Table 4.7 Adaptation stages, the evidence that admits each one, and its reach

| Stage | Evidence that admits it | What it may change |
|---|---|---|
| ① | **Cost deficit:** The forecast cost exceeds the satisficing bound while a rule still fires above the gap threshold. | The stage adjusts the consequents of the two strongest-firing rules by one step of 0.05 and may move one shared term boundary. It creates nothing new. |
| ② | **Coverage gap:** Fire is alive while no rule fires above the gap threshold, or the area grew under standing orders. | The stage writes one rule on an antecedent cell that carries none and may insert one narrow term when no term reaches 0.62. It edits no existing rule. |
| ③ | Either symptom qualifies once the cheaper stages have been tried in this context and the shortfall stands. | The stage writes one rule and at most one new vocabulary object that this rule uses. It adds objects and never edits existing ones. |

Consecutive trials are separated by a cooldown period, so the shadow forecasts do not delay the decision cycle. When a coverage gap appears alone, without a cost deficit and without spread, only the rule-creating adaptation stages are allowed: An empty combination has no rule to tune, so a rule must be created first.

#### 4.5.3.1 EVFIS: Parameter Adaptation

**The adaptation stage ① tunes the rules already in the base and adds none.** It has two moves, namely consequent step, and boundary shift. While the consequent step changes the output value of a rule, the boundary shift moves the edge that two neighboring antecedent terms share.

**The Consequent step:** The consequent of a rule $v_{r,m}$ is the intensity that the rule assigns to intervention $m$. The consequent step adds or subtracts δ, fixed at 0.05, from that number and clips the result to the unit interval ( 56 ).

$$v_{r,m} \leftarrow clip_{0,1}\left(v_{r,m} \pm \delta\right), \quad m \in cons(r) \tag{56}$$

It changes one rule only and is easy to undo, so it is tried first. It is applied to the two rules with the highest firing strength, not to the strongest rule alone, because with a small base one rule dominates every cycle and the second would otherwise never be tuned. The field decision is the firing-weighted average of the rule consequents, so moving one consequent slides that average toward the new value in proportion to the firing strength of the rule.

**Boundary shift step:** Two neighboring antecedent terms share their transition interval. Where the left term stops being fully true, the right one starts, so the falling edge of the left term $A^{-}$ is the rising edge of the right term $A$, and the two coincide, $(a_A, b_A) = (c_{A^-}, d_{A^-})$ ( 57 ). The boundary shift moves this shared edge. The edge belongs to both terms, both move together, so the memberships still sum to one at every reading and the partition of unity required by ( 45 ) is preserved. Moving the edge changes the reading of every rule that uses either term, which is why this move is tried only after the consequent trials are spent. The step is bounded by the width of the core it narrows, $\delta \le c_{A^-} - b_{A^-}$, because a larger step would erase that core and turn the trapezoid inside out.

$$\begin{gathered}(a_A, b_A) \leftarrow (a_A - \delta, b_A - \delta), (c_{A^-}, d_{A^-}) \leftarrow (c_{A^-} - \delta, d_{A^-} - \delta) \\ (a_A, b_A) = (c_{A^-}, d_{A^-}), \quad 0 < \delta \le c_{A^-} - b_{A^-}\end{gathered} \tag{57}$$

Every trial is a hypothesis. It is judged on the forecast physical cost on a basis, it is kept only when that cost strictly falls, and it is rolled back otherwise ( 58 ). This strict criterion governs the tuning trials of stage ① and the admissions of stage ③. The single deliberate exception is the void-filling instantiation of stage ②, which is admitted when non-inferior, because an uncovered cell must first carry a rule before anything can be tuned.

$$kept \Leftrightarrow \Delta\hat{J}_{phys} < 0 \tag{58}$$

All trials act on a working copy of the rule base, so the seed base of Appendix E is never modified. Both adaptations are shown in Figure 4.6 .

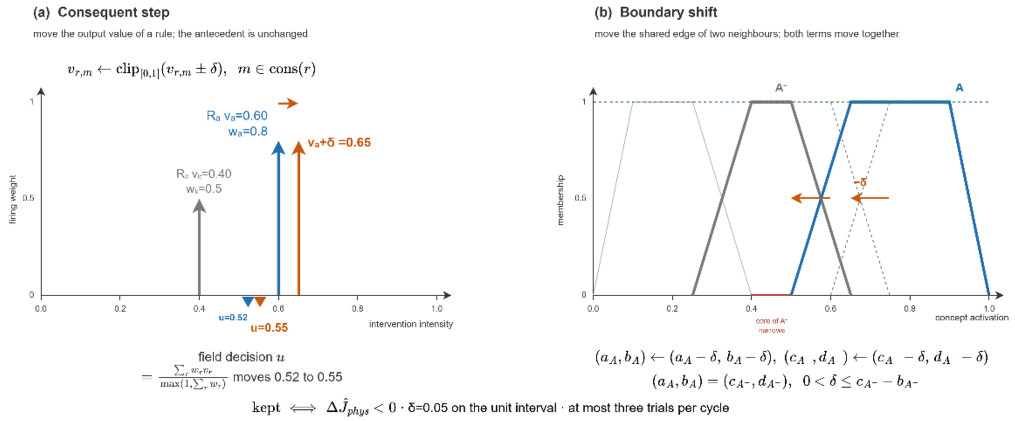


Figure 4.6 Consequent step and boundary shift

### 4.5.3.2 EVFIS: Rule Instantiation and Resolution Increase

**The adaptation stage ② grows the rule base**. It works in two modes. The first mode writes a rule on an empty combination. The two decision concepts with the highest activation, $c^{(1)}$ and $c^{(2)}$, with their dominant terms $t^*$, define the combination that describes the current situation, as given in ( 59 ).

$$\begin{gathered} A_{new} = \left\{ \left(c^{(1)}, t^*\left(c^{(1)}\right)\right), \left(c^{(2)}, t^*\left(c^{(2)}\right)\right) \right\} \\ t^*(c) = \underset{t \in T_c}{argmax} \mu_t({a_c}^{eff}) \end{gathered} \tag{59}$$

When a combination of concept values has no active rule, a new rule is added for it, and its outputs are set to what the strongest concepts ask for. Since the slot was empty, the new rule is accepted as long as it does not make the forecast worse, so the rule base grows to cover more cases. If a combination already has a rule, no second rule is added for it. Adjusting the numbers of an existing rule is the job of the **adaptation stage** ①, so a proposal of that kind is rejected here.

The second mode grows the term set itself. A situation is ambiguous when no term of the strongest concept reaches a membership of 0.62. The reading then lies between two term cores, and no change of consequents can separate the two situations that share the same combination. A narrow term $X$ is inserted into the partition of that concept, centered on the crisp reading $x$, as given in ( 60 ). The rule is then written on the refined combination.

$$\max_{t \in T_{c^{(1)}}} \mu_t\left(a_{c^{(1)}}^{eff}\right) < 0.62 \quad \Rightarrow \quad T_{c^{(1)}} \leftarrow T_{c^{(1)}} \cup \{X\}$$
$$X = \mathrm{trap}(x - 0.12,\ \ x - 0.05,\ \ x + 0.05,\ \ x + 0.12), \qquad (60)$$
$$x = a_{c^{(1)}}{}^{eff}$$

The number of expressible combinations is the product of the term counts of the decision concepts, and the first insertion raises it from 3125 to 3750 ( 61 ).

$$|C| = \prod_{c \in dec} |T_c| : \quad 5^5 = 3125 \quad \rightarrow \quad 6 \cdot 5^4 = 3750 \qquad (61)$$

An inserted term stays outside the aggregation of Layer 3. It never enters the activation vectors, and it is read on the crisp activation of its own concept against its own trapezoid, so the five-term algebra of Layer 3 is untouched, and the growth remains on the rule side. Inserted terms are dropped when the vocabulary is reset and are restored from the lineage store otherwise. Both modes are shown in Figure 4.7.

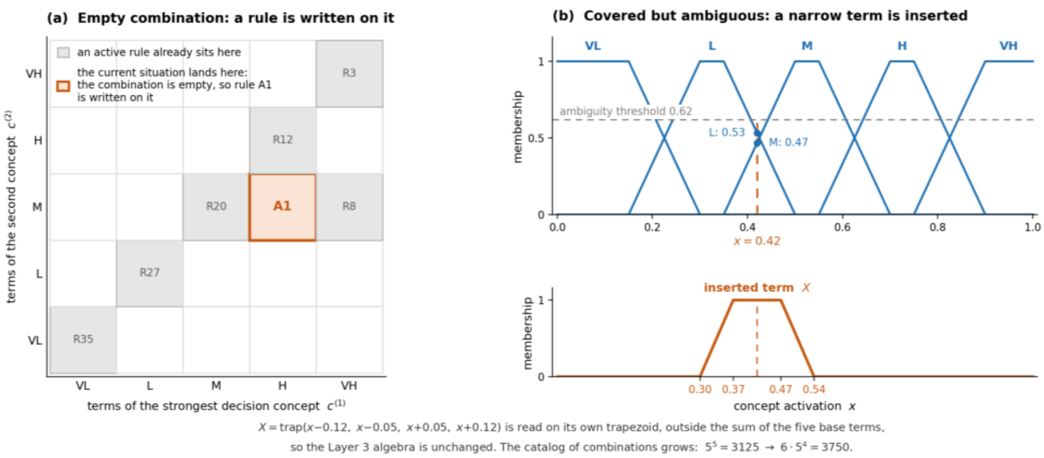


Figure 4.7 Resolution increase, rule instantiation, and term insertion

### 4.5.3.3 Generative Proposals and Vocabulary Packages

**The adaptation stage ③ adds knowledge that neither tuning nor instantiation can produce**. It engages when the evidence of the cycle shows that the base itself has fallen short. The standing decision falls short in one of two ways. Either its forecast cost exceeds the satisficing bound, or fire is alive while no rule fires above the gap threshold. When one of these is true, the stage runs as shown in Figure 4.8.

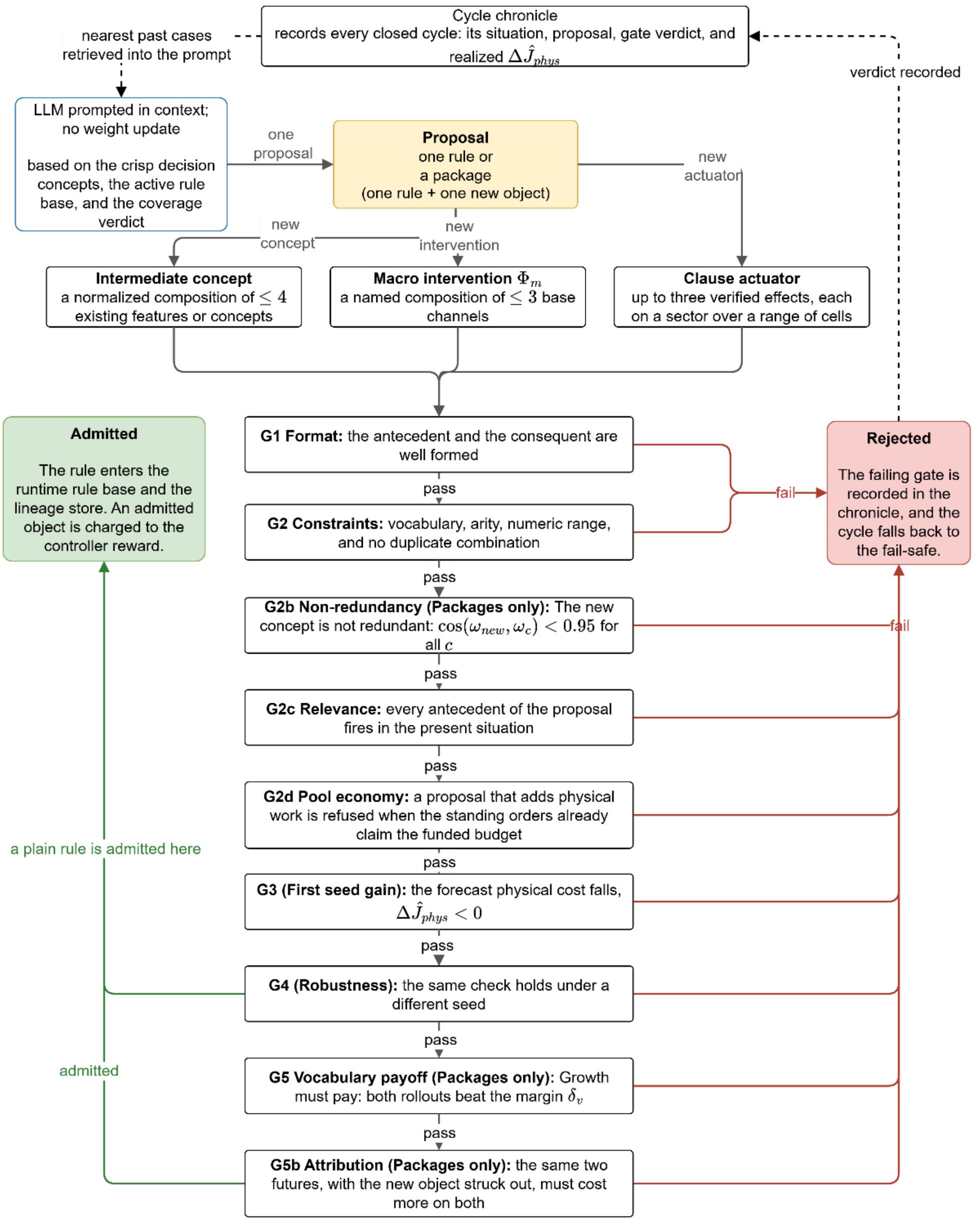

Figure 4.8 Generative proposal and the verification gates

The stage produces four kinds of output: A plain rule, a macro intervention that blends existing interventions, a clause actuator that composes a new intervention from the verified physical effects, and an intermediate concept.

The large language model (LLM) is the proposer. It is a pretrained, general-purpose model, used in context only, with no task-specific training, no fine-tuning, and no weight update. When this stage runs, the model receives the situation of the cycle, which holds the crisp value of every decision concept, the whole active rule base in its IF-THEN form, and the coverage verdict. It is conditioned in three ways: A system instruction fixes its role, the structured summary states the situation, and a schema-constrained output forces every proposal into the framework's own vocabulary, that is, the decision concepts, the admissible interventions, and the VL to VH terms. From this input the model returns exactly one proposal, written in the canonical rule form. The proposal is untrusted by construction: It may act only after it clears the six gates that every proposal faces (G1, G2, G2c, G2d, G3, G4), the further gates that admit a package (G2b for a new concept, G5 and G5b for vocabulary growth), and the satisficing test, so the guarantee comes from validation rather than from training.

A proposal takes one of two forms. It is a single rule, or it is a package, which is one rule together with one new vocabulary object that the rule uses. An object that no rule cites is inert and cannot be verified, so an object is never admitted on its own. Every proposal, whether a rule or a package, enters the gate sequence as a whole, and Figure 4.8 shows that sequence with the three kinds of object entering it side by side. The three kinds of object described below are the reach of the stage, not its policy. The form of a proposal follows from the reason the stage is called, and that reason is stated in the situation the model receives. Table 4.8 lists the cases, including the form the stage may not produce. The stage never modifies a rule. A proposal landing on a covered antecedent cell is refused at G2, since retuning an existing rule is the work of stage 1 and costs no model call. This separation keeps the ledger readable, since a consequent that moved was moved by the tuning stage and a rule that appeared was written by a growth stage.

Table 4.8 Proposal forms of stage ③ and their admitting conditions

| Form | What calls it |
|---|---|
| **Plain rule** | The concepts can express the situation, and an existing action answers it, but no rule sits on the cell that holds now. |
| **Rule that edits a rule** | This form is never produced. A cell that already carries an active rule is refused at G2, since retuning an existing rule is the work of stage ①. |
| **Rule with an intermediate concept** | The situation recurs and the five decision concepts cannot name it, so every rule the stage can write lands on a used cell. The object is a weight vector over at most four existing inputs. |
| **Rule with a macro intervention** | Two or three base families must act as one at a fixed ratio, which separate orders would let the aggregation dilute. |
| **Rule with a clause actuator** | The tactic is a placement rather than a mixture: It needs a where and a how that no base family carries. |

#### 4.5.3.3.1 Plain Rule

A plain rule is the simplest product of the stage: An antecedent over existing decision concepts and a consequent over existing interventions, written on an antecedent cell that carries no active rule. It is called when the concepts can express the situation and an existing action answers it, and it grows the rule base without growing the vocabulary, so it faces the six universal gates and brings no vocabulary object.

#### 4.5.3.3.2 Macro Intervention

A package may carry one of three kinds of object, and all of them stay inside the semantics that the simulator can execute, and the cost model can price. As illustrated in Figure 4.9 (a), a macro intervention $\Phi_m$ is a named composition of at most three admissible channels, drawn from the six intervention families of Table 2.4, defined in ( 62 ). A consequent on the macro with intensity $v$ expands, inside the aggregation of ( 45 ), into base-intervention consequents with intensities $\beta_j v$. As illustrated in Figure 4.9 (c), a third kind, the clause actuator, carries its own geometry rather than a weight vector and is defined through the verified effect vocabulary described next.

$$\begin{aligned}&\text{Macro intervention: } \Phi_m = \langle (b_1, \beta_1), \dots, (b_p, \beta_p) \rangle, p \leq 3,\\ &\qquad (m, v) \mapsto (b_j, \beta_j v)\\ &\text{Intermediate concept: } a_{c'} = \textstyle\sum_i \omega_i a_i, \quad \sum_i \omega_i = 1,\ \ i \leq 4\end{aligned} \tag{62}$$

#### 4.5.3.3.3 Clause Actuator

The vocabulary the generative stage receives is deliberately primitive. It is not a menu of ready tactics but a basis of verified physical effects that a clause may invoke, a small set of them (wet, clear, ignite, coat, draft, evacuate, and prime), a geometry that places an effect at a target sector and a distance band from the front, and a reading of the map that reports the affordances present, such as a nearby water body, the closest asset, and the resources on hand. A new intervention is therefore not selected from a list; it is composed from this basis in response to the situation.

When the map reports a lake near the front, the stage can compose a clause actuator that drafts from the water body and wets the fuel ahead of the head, an intervention that no seed rule contains and no menu offers. The discovery lies in the composition and in its grounding to the specific map, not in the invention of a physical effect: Every clause invokes an effect the simulator already validates, and the deterministic compiler resolves the actual water body, the resource that supplies it, and the distance to reach it, before any gate is run. The action space stays closed at the level of physical effects yet open at the level of interventions.

A backburn is an example. It is not new physics, but containment-weighted cut ahead of the front with suppression on the flanks, so it runs through the existing resource fields and is priced by the response term.

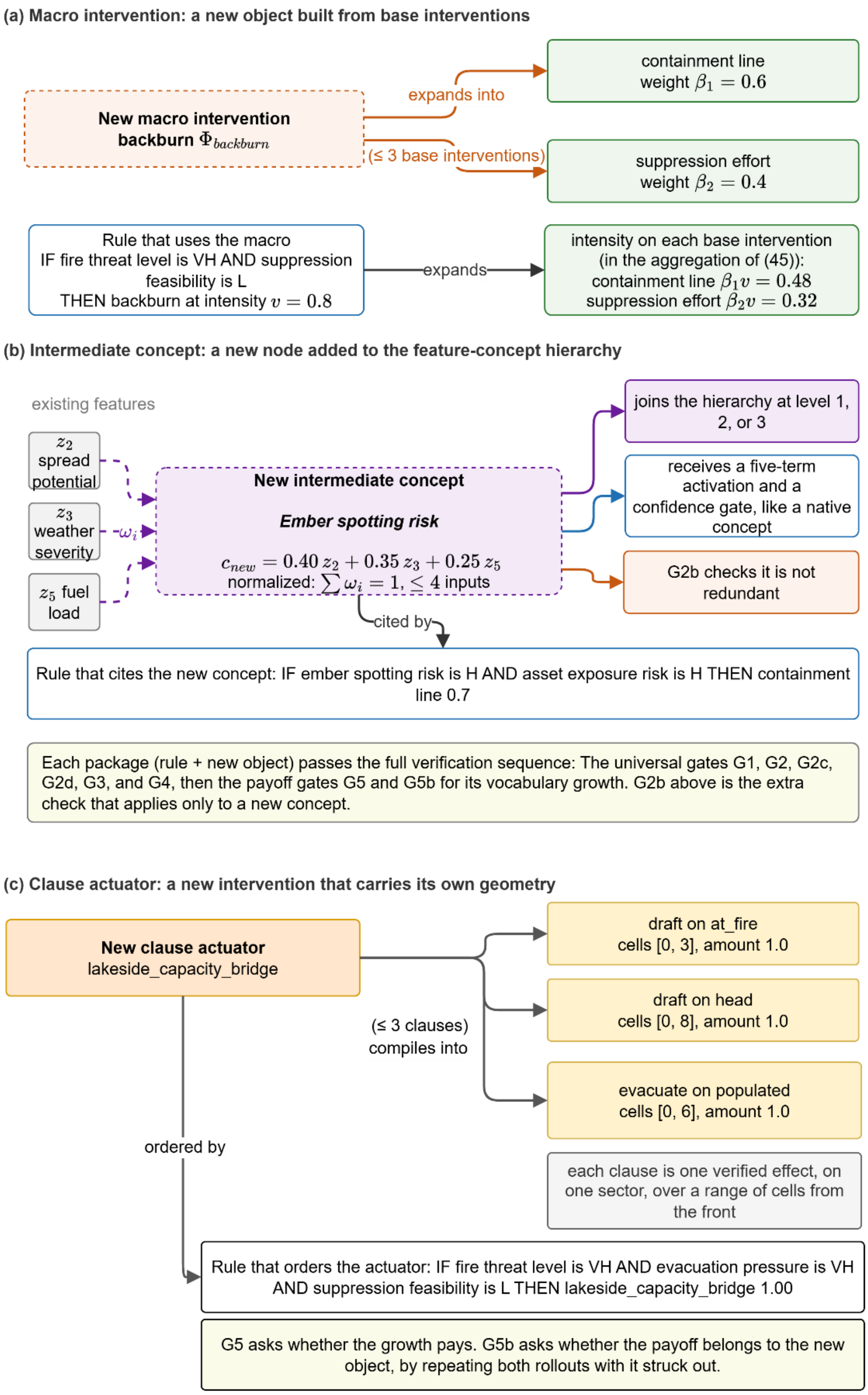


Figure 4.9 Package objects: (a) macro intervention, (b) intermediate concept and (c) clause actuator

Only the last form carries geometry. A macro fixes a ratio between orders that already exist and a concept fixes a direction in the feature space, but neither can say where on the map an order should act. A clause names one of the seven verified effects of Table 4.9, one of seven sectors, a range of cells from the front, and an amount, and an actuator is at most three clauses. The sectors are head, flank, and rear, which cut the band by bearing against the direction of spread; ring, which is the band without a bearing test; at_fire, which is the burning cells and the two rings around them; assets, which covers the assets within fifteen cells of the fire; and populated, which is every cell holding population. The list of effects is short because each entry is a field the engine already writes and the cost model already prices. An actuator can be new; the physics it invokes cannot.

Table 4.9 The verified effect vocabulary a clause may invoke

| Effect | Effect in the engine | Limiting condition |
|---|---|---|
| **wet** | Raises capacity and crew availability: A pre-wetted band. | Workable ground; draws on the pool. |
| **clear** | Marks cells as cut and raises capacity: A containment line. | Workable ground, never built-up or water; draws on the pool. |
| **ignite** | Sets the cells alight: A firing operation. | Never built-up or water; adds fire the forecast then judges. |
| **coat** | Raises the moisture floor to 0.45 of the amount laid: Retardant or soil. | The only thresholded effect; acts only where that floor reaches the extinction moisture of the fuel. |
| **evacuate** | Raises the evacuation order. | Only where population is present; spends no pool capacity. |
| **prime** | Raises the public warning. | Only where population is present; spends no pool capacity. |
| **draft** | Multiplies capacity already on the cells, by a factor falling with the distance to water. | Needs a water body; acts only where capacity exists, so it raises the yield of committed crews. |

The third column carries operational content that the names do not show. Two effects draw on the shared crew pool and five do not, so on an incident whose pool is spent the first two only redivide capacity already committed. One is thresholded, so a coating below the extinction moisture of the fuel underneath slows the fire without stopping it and is measured as a tactic that acted and bought nothing. One multiplies capacity instead of requesting it, which makes it the only effect that can add real work to a saturated incident. These facts are assembled from the fuel table and the engine constants and placed in the situation the model receives. Geometry matters as much: A sector resolves against the fire as it stands in this cycle, so a band written far ahead of a small fire resolves to no cells, and the written amount is multiplied by the intensity the rule realizes, so a rule firing at half strength with a consequent of 0.9 delivers about 0.45.

The engine never executes text the model produces. An actuator arrives as a named list of clauses, its name is checked for collision, each clause is checked against the effect and sector lists, and anything outside them is refused at G2 before a simulation is spent. The definition is then installed on a working copy of the vocabulary and the rule on a working copy of the base, so the seed base and the standing vocabulary are untouched while the proposal is judged.

When the rule fires, the framework resolves each sector to a mask on the current map, intersects it with the region and with the ground the effect may touch, multiplies the amount by the realized intensity, and writes the corresponding field of the resource plan. A clearing or igniting clause is intersected with the diggable mask as well, so a generated actuator cannot cut a line through a settlement or light a fire on water, which are the orders the base containment family is forbidden to give. The plan that leaves this path is the same object the base families produce, so the simulator, the cost model and the audit trail treat a generated actuator and a doctrinal order identically. The decision space is therefore open at the level of interventions and closed at the level of physics.

#### 4.5.3.3.4 Intermediate Concept

An intermediate concept is a normalized composition of at most four existing features or concepts, also defined in ( 62 ). As illustrated in Figure 4.9 (b), it may join the hierarchy at any of its three levels, according to the inputs it aggregates, and it receives a five-term activation and a confidence gate like every native concept, and it reaches the decision only through rules that cite it. Rules are defined between concepts and interventions regardless of provenance: A generated concept may be cited by the rule of a later proposal exactly as a native concept is, and that rule may order a generated intervention, so a generated concept can be bound by a rule to a generated intervention. A generated concept always arrives as a package, together with the rule that cites it, such as the interface exposure concept among the admitted rules of Appendix E.

The vocabulary grows along one axis and is fixed along the other. Every package that clears the gates adds one name, and an admitted name is written to the lineage store, so later rules in the same incident may cite it and a later incident restores it unless the vocabulary is reset. An admitted macro or actuator is ordered by a later rule exactly as a family is ordered, and a later package may draw on the same action-list channels, so the products of the stage enter the working vocabulary rather than remaining one-off patches.

The basis itself is not proposable. Adding an effect means implementing and validating a field in the physical model, which is a change made by the authors of the framework and reported in Chapter 3, not an action available to the proposer, because a system that let the proposer extend the basis would be asking a language model to certify physics. Growth is also priced. Each admitted concept adds an axis to the antecedent space and therefore to the search, which is why G2b refuses a concept pointing in a direction an existing concept already covers, and each admitted intervention adds a consequent the coordinator must price. The controller reward is read on the reduction in forecast physical cost, so a stage that buys names without buying outcomes loses value in its own table and is selected less often. The admission tests and the controller reward read the same shadow forecast, so the adaptation loop is calibrated only as well as that forecast; this dependence is assessed with the validation of Chapter 5.

#### 4.5.3.3.5 Verification Gates

**G1 checks format.** The reply must be parsed into an antecedent list of concept-term pairs and a consequent list of intervention-intensity pairs. A well-formed reply reads, for example, *IF fire threat level is VH AND suppression feasibility is L THEN evacuation 0.80 and public warning 0.60*. Some examples show what G1 rejects:

- *A reply that arrives as free prose, such as 'the fire is severe, so the crew should begin evacuating'.*
- *A reply that names an antecedent combination with no consequent.*

G1 exists because the proposer is a text generator, so a reply must first be a rule before anything else can be checked.

**G2 checks the constraints, the content of a well-formed rule against the closed vocabulary and its boundaries.** Every antecedent must name an existing decision concept and one of its linguistic terms. Every consequent must name one of the six base intervention families listed in Table 2.4, one of the action-list base channels offered for the current map, or a previously admitted macro, with an intensity in [0, 1]. The number of antecedents and consequents limits stated above must hold. And the antecedent combination must not already carry an active rule. Neither G1 nor G2 costs a simulation step. Some examples show what G2 rejects:

- *A proposal that orders an intervention outside the vocabulary,* ***such as cloud seeding, fails on vocabulary,*** *because the physics implements no such effect. An order for an action-list base channel such as an aerial retardant drop does not fail here: The channel is offered to the proposer whenever the map supports it, even though no seed rule uses it.*

- *A proposal of suppression effort **1.3** fails on range.*
- *A proposal that re-issues the combination of the active rule R26 (fire threat level is VH) with different numbers fails on duplication, **since renumbering an existing rule belongs to the adaptation stage 1, not to growth**.*

**G2b applies only to a package that proposes a new intermediate concept and checks non-redundancy.** A new concept is a normalized weighted combination of existing features or concepts, so it is fully described by its weight vector $\omega_{new}$ over those inputs. G2b compares this vector against the weight vector $\omega_c$ of every existing concept defined over the same inputs, through the cosine of the angle between them, as given in ( 63 ).

$$G2b:\ cos(\omega_{new}, \omega_c) < 0.95 \quad \forall c \tag{63}$$

If the cosine reaches 0.95 for any existing concept, the two vectors point in the same direction. The new concept is then a near-duplicate, and the package is rejected. Such a concept adds no new direction. It only enlarges the antecedent space and the search, without adding information. This is the concept-level counterpart of the duplicate-combination check in G2. Like G1 and G2, G2b is an algebraic test and costs no simulation step. An example of G2b rejection:

- *A proposed concept built from fire threat and asset exposure with weights (0.60, 0.40) is checked against an existing concept with weights (0.62, 0.38). Their cosine is almost 1.0, so the proposal is rejected at G2b.*

**G2c requires relevance.** Every antecedent of the proposal must fire in the present situation. A rule written for some other, imagined situation cannot answer the shortfall that called the stage. If a term fails only by its form, it is repaired to the nearest reading that still holds, as long as the repair keeps the protective intent, and every repair is recorded in the provenance note of the new rule. As a negative example, a proposed rule needs spread potential to be High, but the present reading holds it at 0.15, inside the Low term. The antecedent does not fire, so the proposal is rejected at G2c. An example of G2c rejection:

- *A proposal whose antecedent requires spread potential to be High is checked against the present reading, where spread potential is 0.15, inside the Low term. The antecedent does not fire, so the proposal is rejected at G2c.*

**G2d enforces pool economy.** If the current orders already ask for more capacity than the budget allows, a proposal that adds more physical work is refused, and the gate names both the demand and the budget. Such an order cannot add real effort since it only splits the same capacity and weakens the work already under way. Orders that use no capacity, such as evacuation and public warning, and orders that add capacity, such as water drafting, still pass. An example of G2d rejection:

- *A proposal adds a suppression order of 0.30 while the standing orders already claim the full funded budget of 1.0. The extra order could only re-divide committed capacity, so the proposal is rejected at G2d, while an evacuation order, which spends no capacity, would pass.*

**G3 is the first gate that costs a simulation and checks the first seed gain.** The candidate rule is appended to a working copy of the base, and the decision that this extended base produces is played on a shadow copy of the simulator. Its forecast is evaluated on the common trial basis of the adaptation loop, under the physical cost (burned area, asset, and population), taken at the trial horizon. G3 runs the same rollout again without the candidate and compares the two, as given in ( 64 ). The forecast physical cost must strictly fall. A stage-③ rule always lands on an uncovered combination, since any duplicate has already died at G2; even so, admission demands a strict improvement on both reseeded rollouts, so the base grows only where growth demonstrably pays. An example of G3 rejection:

- *A candidate that orders a heavy resource deployment on a cell where the fire is already contained leaves the burned area, the asset loss, and the population exposure unchanged, so the forecast physical cost does not fall and G3 fails; the effort the order would add is priced by the pool-economy gate G2d and by the total cost of the standing decision, not by this gate.*

**G4 repeats the same comparison once more with a different random seed driving the shadow weather to check robustness.** The two runs are the two cases $V_a$ and $V_b$ of ( 64 ), an A/B test on two independently drawn futures. A single rollout can favor a candidate by chance, which is stochastic flattery. G3 and G4 are kept as two gates, not one, because each records a distinct verdict. A failure at G3 means the candidate did not help even once. A failure at G4 means it helped on one future but not on a fresh one. An example of G4 rejection:

- *A candidate that passes G3* but fails to lower the cost on the second seed fails at G4, *because a gain that does not survive a fresh future is not admitted.*

$$\Delta\hat{J}^{(\nu)}_{phys} = \hat{J}^{(\nu)}_{phys}(\mathcal{R} \cup \{r'\}) - \hat{J}^{(\nu)}_{phys}(\mathcal{R}), \quad \nu \in \{\mathrm{V_a}, \mathrm{V_b}\}$$
$$G3\colon\ \Delta\hat{J}^{(\mathrm{V_a})}_{phys} < 0;\ G4\colon\ \Delta\hat{J}^{(\mathrm{V_b})}_{phys} < 0 \tag{64}$$

In ( 64 ), $\mathcal{R}$ is current rule base, $r'$ is the candidate rule while $V_a$ and $V_b$ are random seeds. The seeds $V_a$ and $V_b$ are two independently drawn futures. Seed $V_a$ draws one future, in which the wind holds from the west at a moderate speed and the fire keeps to its predicted line. Seed $V_b$ draws a different future, in which the wind swings to the north and gusts, and the fire runs in a new direction. G3 scores the candidate on seed $V_a$, and G4 scores it again on seed $V_b$.

**G5 applies only to a package and checks the vocabulary payoff.** A package must earn its vocabulary growth. G5 requires that both reseeded rollouts reduce the physical cost by more than the vocabulary margin $\delta_v$, not merely reduce it, as given in ( 65 ). As a negative example, a package whose new object lowers the cost by less than $\delta_v$ on either seed fails here, since the growth does not pay for itself. Every admitted object is also charged the controller reward, so the vocabulary grows only when growth clearly pays.

$$G5\colon\ \Delta\hat{J}^{(a)}_{phys} < -\delta_v \quad \wedge \quad \Delta\hat{J}^{(b)}_{phys} < -\delta_v \quad \delta_v = 0.0001 \tag{65}$$

**G5b** is the **last gate and applies to a package that grows the intervention vocabulary**. G5 weighs the package against no package, and the rule carrying a new object may order ordinary interventions in the same consequent list, so a package can clear G5 on the strength of an order the framework already had. G5b runs the same two futures again with the consequent on the new object removed and the ordinary orders of that rule left standing, and requires the cost to be higher without the object on both. It is not run when the new object is the only consequent, since G5 has then already measured the object alone, and a new concept is not ablated because the rule is written on it. As a negative example:

- *A package that lowers the cost while its own rule stripped of the new actuator lowers it by the same amount fails here: The gain belongs to the ordinary order, and the vocabulary does not grow on borrowed credit.*

A proposal is rejected the moment it fails a gate that applies to it, and the failing gate is named. A plain rule faces G1 and G2, then the relevance gate G2c and the economy gate G2d, and finally the forecast gates G3 and G4. A package faces the same gates and, in addition, G2b when it proposes a new concept and G5 with G5b for its vocabulary growth. A proposal is admitted only when it clears every gate that applies to it. A rejection at G1, G2, G2c, or G2d costs no simulation, so the proposal is returned to the model with the failing gate named, within a bounded revision budget. A rejection at G3, G4, G5, or G5b has already spent a rollout, so it ends the stage for this cycle, and the controller may select the stage again later.

Either way the cycle falls back to the graduated fail-safe. Every closed cycle is written in the cycle chronicle. Each record holds the situation that entered the cycle, the proposal it produced, the gate that admitted or rejected it, and the realized change in the physical forecast. The chronicle is not only a log. Before the next proposal is requested, the records whose situations are nearest to the current one are retrieved and placed in the prompt, each with its verdict, so the prompt carries what has already worked and what has already been refused in similar situations, which suppresses the repetition of a recorded rejection. Nothing in this channel is opaque since the retrieved records are the same objects the audit trail exposes.

Grounding by construction, not by retrieval. The generative stage receives a deterministic context assembled from the simulation such as the decision-concept activations, the observation confidence, the recent cost trajectory, the rule-base statistics, and the location where residual cost concentrates together with the complete closed vocabulary of the framework (i.e., every decision concept, every admissible intervention, and the five fuzzy terms) as in Appendix F.

The clause grammar is compact to write and large to enumerate. Across the seven effects, the seven sectors, the admissible ranges of the reach, and the graded amounts, an actuator of up to three clauses spans on the order of $10^{13}$ definitions. Each candidate costs two reseeded rollouts of the physical forecast, and most of the space returns a forecast identical to the baseline, since an empty sector, a coating below the extinction moisture of its fuel, or an effect rationed away on a spent pool changes nothing the forecast reads. Enumeration and local search are therefore impractical on this space. The proposer instead restricts attention through the stated facts of the situation, namely the map affordances, the pool state, and the effect notes, and the gates, which contain no model, decide what is kept. Section 5.5.3 reports how these statements changed the proposal mix.

#### 4.5.3.4 Value Based Stage Selection

The value-based stage controller works by associative search, which is a contextual bandit that reads the current state and chooses one of the adaptation stages (Sutton & Barto, 2018) (Sutton & Barto, 2018). It keeps a small table of action values, one value for each pairing of a state with a stage, and each value estimates the reward that the stage earns in that state. In each cycle the controller reads the state, looks up the action value of every allowed stage for that state, and applies a greedy rule, so the stage with the highest value is run with probability $1 - \varepsilon$ and a random allowed stage is run with probability $\varepsilon$. Once the stage is selected and evaluated its trial, an immediate reward is read and the value of that state and stage is moved a small step toward the reward, ( 66 ).

$$
\begin{gathered}
q_k = \underset{q \in \mathcal{A}(e_k)}{\operatorname{argmax}} H(e_k, q) \textit{ with probability } 1 - \varepsilon \\
\textit{otherwise a uniformly random stage in } \mathcal{A}(e_k) \textit{ with probability } \varepsilon \\
H(e_k, q_k) \leftarrow H(e_k, q_k) + \alpha\big(r_k - H(e_k, q_k)\big);\ \alpha = 0.05, \varepsilon = 0.10
\end{gathered}
\tag{66}
$$

The context that the selector reads is small. It combines how large the current shortfall of the standing decision is, placed in a low, mid, or high band, with a flag that marks whether fire is still active on the ground while the rule base stays effectively silent. For an ordinary shortfall, all three adaptation stages are available, which are the evolving fuzzy adjustment, the resolution increase, and the generative proposal. When the context is a pure coverage gap, the choice is limited to the two stages that can create rules, since an empty situation has nothing to tune. Any stage that has not yet been tried in a context is tried once before the greedy rule takes over, so no stage is starved at the start of an incident.

The reward that a stage earns is the reduction in the forecast physical cost. This keeps the selector from favoring stages that enlarge the vocabulary without a clear gain. After each trial, the value of the chosen stage in the current context is moved toward this reward, so across the few trials an incident allows the selector to settle on the stage that helps most in each context.

## 4.6 Local Layer, Graduated Fail-Safe, Global Coordination

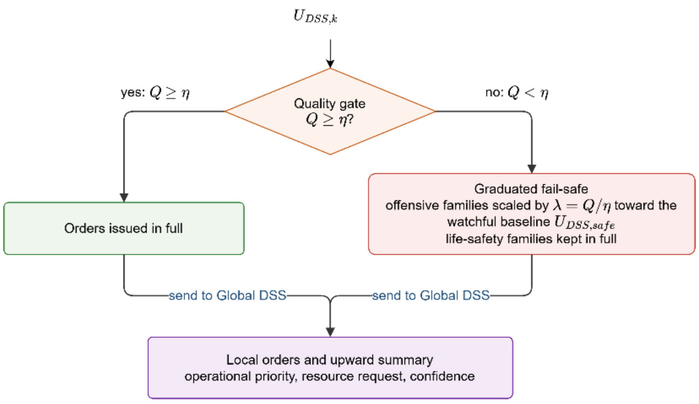


Figure 4.10 Local decision and graduated fail-safe

A **Local DSS** agent, shown in Figure 4.10, decides only the cells inside one region. It runs the whole decision path of the previous sections in its own region and turns the gated concept activations into a candidate intervention. All inference stays inside the region, so the decision path is short, and every order can be traced back to the concepts that raised it.

The **graduated fail-safe** checks the candidate and scores it with a decision quality that measures how well it serves the concepts that demanded it, then compares the quality with a threshold. If it is above the threshold, the candidate is issued in full; otherwise, the offensive families are scaled to the watchful baseline in proportion to the shortfall, ( 11 ), *while evacuation and public warning are never reduced*. The response is continuous and reversible, so the offensive commitment shrinks as the quality falls and returns to full strength once fresh observations restore it, and a weak reading never stops a warning or avoids evacuation.

As an example, take a region where smoke has degraded the optical channel over one sector. The features that feed suppression feasibility lose confidence, the gate on that concept closes, and its activation falls back toward the decayed priority. The rule base still calls for a strong suppression effort from the fire threat level, but suppression feasibility no longer supports it, so the candidate is unbalanced and the quality drops below the gate. The graduated fail-safe attenuates the suppression effort and resource deployment toward the watchful baseline, while the public warning that the same cycle raised for the exposed settlement is issued at full strength. When the wind clears the sector two cycles later, confidence recovers, the quality rises above the gate, and the suppression order returns to full strength without any external action.

The **Global DSS** coordinates the regions and performs no inference of its own. It runs no features, concepts, or rules, and it only ranks the regions and shares the capacity between them. Each Local DSS sends up three small values, which are its operational priority, its resource request, and a short confidence summary, and nothing else leaves the region, so the channel stays narrow and easy to audit. In return the coordinator sends back the allocation, the normalized priority weights, and the acceptance thresholds.

The coordinator ranks the regions by operational priority, the concept at the top of the hierarchy that blends a region's fire threat level and asset exposure risk and fires no local rule. It marks the highest as the focus and gives each region an attention share $\sigma_i$. A region whose operational priority reaches a fixed fraction of the highest priority is attended and keeps the full share, with the attention threshold at 0.35 in this thesis; a region below that fraction is monitored, and its share decreases with its relative priority, as ( 67 ), where $p_i$ is the crisp value of the operational priority concept for region while $p_{max}$ is the largest of these values across all regions in the cycle. The floor on a monitored share is 0.50. $N_{Local_{DSS}}^{focus}$ is the number of local DSS which will be focused by Global DSS.

$$\sigma_i = \begin{cases} 1, & \text{if } p_i \geq \tau_{att} p_{\max} \\ 0.5 + 0.5 \dfrac{p_i}{p_{\max}}, & \text{otherwise} \end{cases} \qquad (67)$$

$$N_{Local_{DSS}}^{focus} = |\{i : p_i \geq \tau_{att} p_{\max}\}|$$

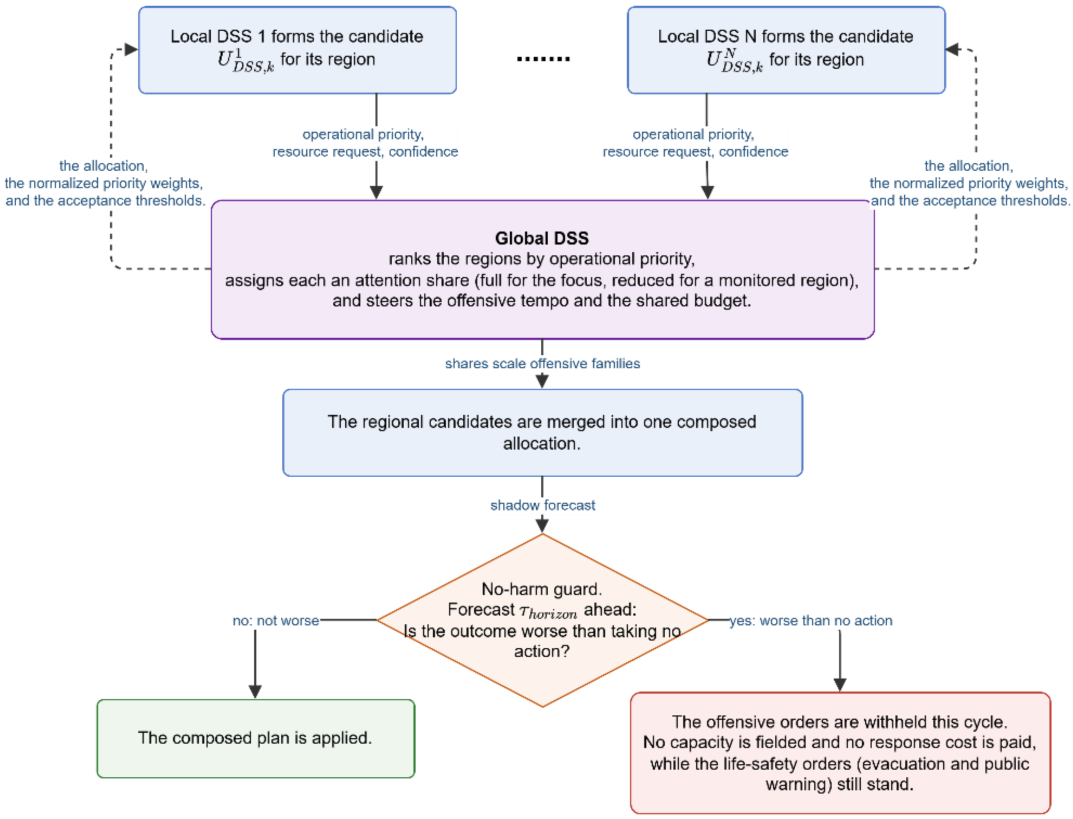


Figure 4.11 Global coordination and the no-harm fail-safe

The share acts on the offensive side only. It scales the suppression effort and resource deployment and puts more of the shared budget on the focus, while evacuation and public warning are left untouched. The share also sets how good an order must be to be accepted. An attended region keeps the normal quality threshold, but a monitored region must clear a higher one, so an offensive order in a low-priority region needs stronger decision quality before it uses shared capacity. The regional candidates are then merged into one plan, which the no-harm guard forecasts. When the plan is forecast worse than taking no action, its offensive orders are withheld for the cycle while the life-safety orders still stand. Figure 4.11 summarizes global coordination and the guard.

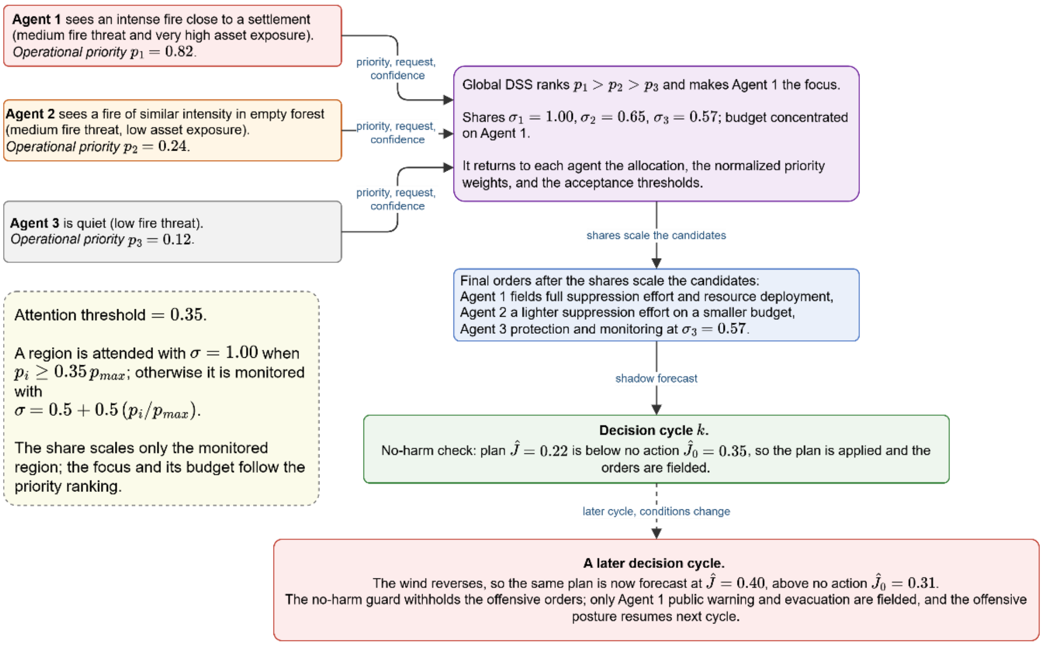


Figure 4.12 Example of local decisions, global coordination, and the fail-safe

Figure 4.12 follows one decision cycle with representative numbers. The priorities come from observations. Agent 1 sees an intense fire close to a settlement, so its asset exposure risk is very high and its operational priority is 0.82; Agent 2 sees a fire of similar intensity in empty forest, so its asset exposure risk is low and its priority is only 0.24; Agent 3 is quiet at 0.12. Each agent sends up its priority, its resource request, and a confidence summary.

The coordinator ranks the three, makes Agent 1 the focus, sets the attention shares to 1.00 for the focus Agent 1, 0.65 for the monitored Agent 2, and 0.57 for the monitored Agent 3, and returns to each agent the allocation, the normalized priority weights, and the acceptance thresholds. The budget concentrates on the highest priority, the final orders are a full suppression effort and resource deployment for Agent 1, a lighter suppression effort on a smaller budget for Agent 2, and protection with monitoring for Agent 3.

In this cycle the composed plan is forecast at 0.22 against 0.35 for taking no action, so it clears the no-harm check and the orders are fielded. In a later cycle the wind reverses, and the same plan is forecast at 0.40 against 0.31, so the no-harm guard withholds the offensive orders; only Agent 1's public warning and evacuation are fielded, and the offensive posture resumes once the forecast turns favorable again.

# CHAPTER 5

# SIMULATION AND PERFORMANCE RESULTS

This chapter evaluates the DisasterAware framework empirically. Its purpose is twofold: To demonstrate that the three contributions C1–C3 hold not only analytically but under simulated operations, and to establish that the simulation environment on which this demonstration rests is itself a faithful proxy of real fire behavior. The remainder of this chapter is organized as follows:

Section 5.1 derives the evaluation rubric: The eight criteria that the wildfire decision support literature asks an operational DSS to demonstrate, from predictive fidelity to scalability, against which the results of the whole chapter are evaluated.

Section 5.2 validates the simulation core: Four blind hindcasts of documented wildfires, driven by real terrain, fuel, and weather data, scored against the satellite record on spatial agreement and arrival time.

Section 5.3 states the experimental ground: The constraints and assumptions under which every experiment runs, and the simulation and decision parameters shared by all campaigns.

Section 5.4 quantifies the sensitivity of the outcome to parameter bias: With every parameter at its nominal value, one parameter at a time is displaced across its bias range, and the response of the physical cost identifies which deviations move the outcome, which deviations the loop tolerates, and why.

Section 5.5 reports the performance of the decision layer: The improvement ladder from no action to the full open decision space, the record of the adaptation stages and the generative products, and the timeliness and scalability of the loop itself.

Section 5.6 returns to the rubric of Section 5.1: The criterion by criterion compliance map, together with the criteria that are met and the reasons they remain open.

## 5.1 Evaluation Criteria for Wildfire Decision Support Systems

What a wildfire DSS must demonstrate has not been established by a single, agreed method. Published evaluations differ in what they measure, on which fires they measure it, and against which reference they score it, so results reported for different systems are rarely comparable, and no shared rubric has emerged that a new framework could simply adopt. Beneath this variety the literature divides into two loosely connected traditions: One asks whether the models inside the tool are right, and the other asks whether the tool changes the decision for the better.

The technical tradition evaluates the predictive models inside a tool and states that they should be assessed on their intended use, the spatial progression of real fires, before their results are used (Cruz, Alexander, & Wakimoto, 2003; Alexander & Cruz, 2013). The decision-support tradition evaluates the tool's effect on the decision itself (Martell, A review of recent forest and wildland fire management decision support systems research, 2015) (Martell, A review of recent forest and wildland fire management decision support systems research, 2015), and its most recent review reports a concerning finding. Most wildfire DSS evaluations so far have measured adoption and use rather than whether the tool improves the decision, which the review identifies as the main gap in the field (Rapp, et al., 2026).

An evaluation chapter must therefore address both traditions together. **It must validate the model core as required by the technical tradition, and it must measure the decision effect as required by the decision-support tradition.** Table 5.1 lists eight criteria, S1 to S8, with the measurement used for each in the literature. Section 5.2 validates the model core against the technical tradition, Sections 5.4 and 5.5 measure the decision effect, and Section 5.6 audits the evidence against these criteria.

Table 5.1 Evaluation criteria for wildfire DSS

| Criterion | Measurement in the literature |
|---|---|
| **Predictive fidelity:** The simulation core reproduces observed fire behavior from real inputs | Retrospective hindcasts against historical fires; spatial-overlap and arrival-time scores (Duff, et al., 2016; Fox-Hughes, Bridge, & Faggian, 2024; Filippi, Mallet, & Nader, 2014) |
| **Decision quality:** Recommended interventions to improve the outcome relative to a no-tool alternative | Outcome deltas against no-action and baseline policies on matched scenarios (Rapp, et al., 2026; Duff & Tolhurst, 2015). |
| **Timeliness:** Decision latency compatible with operational tempo | Decision-cycle wall time; detection-to-first-order time (Noonan-Wright, et al., 2011) |
| **Robustness under degraded observation:** Graceful behavior under sensor loss, noise, and communication failure | Performance versus outage and noise level; fail-safe engagement statistics (Comes, Hiete, Wijngaards, & Schultmann, 2011) |
| **Uncertainty handling:** Input uncertainty is represented and propagated to the decision | Ensemble and input-uncertainty propagation studies (Benali, et al., 2017; Cruz & Alexander, 2013) |
| **Adaptability:** The system extends its own knowledge when it does not cover the situation | Learning curves; coverage growth; rediscovery of withheld knowledge (Angelov & Filev, 2004) |
| **Explainability and auditability:** Every recommendation is traceable to its evidence | Trace completeness; interpretability as an adoption gate (Fillmore & Paveglio, 2023) |
| **Scalability:** Cost grows manageably with domain size and number of regions | Runtime and quality versus grid size and agent count (Pais, Carrasco, Moudio, & Shen, 2021). |

## 5.2 Validation of the Simulation Core

In DisasterAware, the simulation core is a close enough model of real fire behavior to support the decision-layer results. The test is a blind hindcast. The simulator is given only the real inputs of a documented fire, with no tuning on that fire. All spread parameters are frozen at the literature-calibrated values of Appendix A before any hindcast is run, and no parameter is adjusted on any of the four cases. It runs for the recorded duration, and its burned area is compared with the observed burned area.

The inputs come from independent public data. Terrain is taken from the Copernicus GLO-30 elevation model (DEM, 2026). Fuel is taken from the ESA WorldCover 2021 land-cover product and mapped to the internal fuel classes (2021, 2026). Weather is taken from the ERA5 hourly at the ignition point, and dead-fuel moisture is set by the equilibrium-moisture model (Open-Meteo, 2026). The ignition points and the reference burned area come from NASA FIRMS VIIRS 375 m active-fire detections (VIIRS, 2026). The reference mask is limited to the simulated time window and to the main detection cluster. This approach follows the widespread practice of testing fire models on independent, observed fires rather than on the data used to build them. The validation set has four documented fires in Turkey, chosen to cover different fuel, terrain, and wind conditions: Manavgat (Antalya, July 2021), Marmaris (Muğla, July 2021), Milas–Kemerköy (Muğla, August 2021), and Çanakkale–Kayadere (August 2023). Their case definitions are given in Table 5.2.

Table 5.2 Historical validation cases and their configuration

| **Case** | **Bounding box (°E, °N)** | **Start date** | **Duration (h)** |
|---|---|---|---|
| **Sc1: Manavgat (Antalya) 2021** | 31.30–31.80 E, 36.72–37.08 N | 2021-07-28 | 48 |
| **Sc2: Marmaris (Muğla) 2021** | 28.10–28.55 E, 36.75–37.05 N | 2021-07-29 | 48 |
| **Sc3: Milas (Muğla) 2021** | 27.70–28.35 E, 36.95–37.35 N | 2021-08-01 | 72 |
| **Sc4: Kayadere (Çanakkale) 2023** | 26.50–27.00 E, 39.90–40.20 N | 2023-08-22 | 48 |

The simulation core models free fire spread and does not represent suppression. A free run therefore overpredicts the final burned area, so area-overlap scores are not used as the primary measure, because they would conflate the spread error with the missing suppression and with the sparse satellite footprint. The core is instead evaluated on how the fire front propagates. Let $FR_{Sim}$ be the set of cells the model burns and $FR_{Re}$ the set of cells the satellite records as burned, and let $\partial FR_{Sim}$ and $\partial FR_{Re}$ be their fronts, the outer boundary cells of each set. Three criteria are used, each robust to suppression and to the length of the run: **Coverage**, **front position error**, and **arrival-time agreement**.

**Coverage** is represented as the probability of detection (POD). It is the share of the observed burn that the model reaches, the number of cells common to $FR_{Mod}$ and $FR_{Re}$ divided by the size of $FR_{Sat}$, ( 68 ). It stays informative even when the free run overshoots because extra simulated cells do not lower it.

$$POD = \frac{|FR_{Sim} \cap FR_{Re}|}{|FR_{Re}|} \tag{68}$$

The acceptance level of 0.70 on the probability of detection is taken from the level at which the wildfire-simulation literature reports good agreement between simulated and observed footprints. Parameterizing FARSITE with vegetation-derived fuel maps raised the Sorensen coefficient from 0.38 for standard inputs to 0.70 (Price, 2022), and calibrated cell-based growth models reach 0.7 to 0.9 while uncalibrated semi-empirical models typically fall between 0.5 and 0.7 (Pais, Carrasco, Martell, Weintraub, & Woodruff, 2021). **A probability of detection of at least 0.70 therefore marks the boundary of a reproduction of the observed extent that is as reliable as a well-behaved uncalibrated model. Since the probability of detection is insensitive to overprediction, this level is a necessary rather than a sufficient indication of agreement, and the front-position and arrival-time criteria carry the discriminating burden.**

**Front position error** measures how far the simulated front lies from the observed one. With $D(x, \mathcal{G}_{Set})$ the distance from a cell $x \in \mathcal{G}$ to the nearest cell of a set $\mathcal{G}_{set}$, as in ( 69 ), the front-to-front distances have a mean $\overline{d_{Front}}$, given in ( 70 ) and reported as **front mean**. Front position error expressed in meters is a standard output of FARSITE-style validation (Finney, 1998). In this study each distance is judged relative to the equivalent radius of the observed burn area ( 71 ), where $N_{burned}$ is the number of burned cells and $A_{cell}$ is the area of a cell. **The error of front position should be less than 0.3 (success greater than 0.7 similar to POD).**

$$D(x, \mathcal{G}_{Set}) = min_{y \in \mathcal{G}_{Set}} \|x - y\| \tag{69}$$

$$\overline{d_{Front}} = \frac{\Sigma_{x \in \partial FR_{Sim}} D(x, \partial FR_{Re}) + \Sigma_{y \in \partial FR_{Re}} D(y, \partial FR_{Sim})}{|\partial FR_{Sim}| + |\partial FR_{Re}|} \tag{70}$$

$$R = \sqrt{(|N_{burned}|\ A_{cell})/\pi}, \mathrm{TH}_{Front} = 1 - \frac{\overline{d_{Front}}}{R} \tag{71}$$

**Arrival-time agreement** assesses the rate of spread directly. Over the $N\_cells$ common to $FR_{Sim}$ and $FR_{Re}$, it compares the simulated arrival time at each cell with the first satellite detection time there. The **mean absolute time (MAE) difference** between them is $\tau_{MAE}$ ( 72 ). The arrival-time error is the mean absolute difference between the time at which the simulated front reaches each cell and the first satellite detection time of that cell, taken over the cells that both burned. It is the direct rate-of-spread test and is the least sensitive to suppression and to the length of the run. The error is judged relative to the observed fire duration $T$, the span of the satellite detections. **The error of MAE ( 73 ) is accepted when it is less than 0.3; the largest observed ratio is 0.29.** The observed duration may exceed the simulated window of Table 5.2, because the satellite record of a case continues after the simulated window closes; the agreement is computed over the cells the two records share.

$$\tau_{MAE} = \frac{1}{N_{cells}} \Sigma_{c \in FR_{Sim} \cap FR_{Re}} \left| t_c^{Sim} - t_c^{Re} \right|;\ N_{cells} = |FR_{Sim} \cap FR_{Re}| \tag{72}$$

$$\mathrm{TH}_{MAE} = 1 - \tau_{MAE}/T \tag{73}$$

Table 5.3 reports the agreement metrics per case, averaged over 50 stochastic replicates. Figure 5.1 shows the corresponding agreement maps, in which correctly predicted burn, overprediction, and missed burn are distinguished. It shows that the simulation core has an acceptable overall success rate over real scenarios. The overall success of a case is the arithmetic mean of the three normalized scores. It is a reporting convenience rather than a literature-based composite measure, which is why the three scores are also reported separately.

Table 5.3 Hindcast agreement metrics per historical case

| | Sc1 | Sc2 | Sc3 | Sc4 |
|---|---|---|---|---|
| $A_{cell}$ $(m^2)$ | 8100 | 8100 | 8100 | 8100 |
| $N_{burned}$ | 1288 | 3922 | 387 | 4915 |
| $R$ $(m)$ | 1823.03 | 3179.96 | 1000.19 | 3559.83 |
| $T$ $(hr)$ | 14.00 | 72.80 | 14.00 | 58.70 |
| $\overline{d_{Front}}$ (m) | 329.00 | 305.00 | 113.00 | 391.00 |
| $\tau_{MAE}$ (h) | 1.46 | 1.39 | 4.04 | 2.58 |
| $1-\frac{\overline{d_{Front}}}{R}$ | **82%** | **90%** | **89%** | **89%** |
| $1-\frac{\tau_{MAE}}{T}$ | **90%** | **98%** | **71%** | **96%** |
| $POD$ (–) | **92%** | **93%** | **97%** | **85%** |
| **Overall Success** | **88%** | **94%** | **86%** | **90%** |

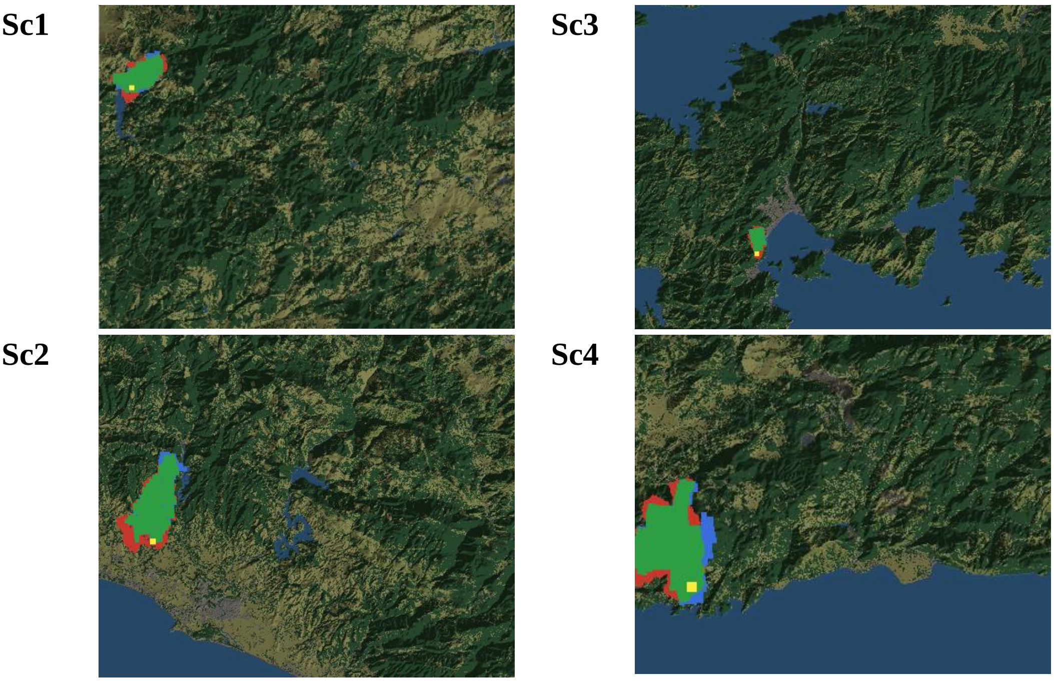


Figure 5.1 Agreement maps for the historical hindcasts (Green: Correctly predicted burn, Red: Simulated only, Blue: Observed only, Yellow: Ignition)

## 5.3 Simulation Constraints and Assumptions

All experiments run on the DisasterAware testbed, whose two cores are presented in Chapters 3 and 4 and are kept strictly separated by design: The simulation core advances the physical state of the fire, and the decision core reads that state only through the observation model and acts on it only through the intervention families.

Two scenario families are used. **Synthetic scenarios** are generated procedurally on grids of 200×200 cells at 90 m resolution with terrain presets (rolling hills, Mediterranean coast), seeded settlements and road networks, and capacity-limited resource bases; ignition points are chosen among burnable cells farthest from the bases, so that trivial fires do not mask parameter effects. **Historical scenarios** reconstruct documented fires on 90 m grids from real terrain, fuel, and weather data. Unless stated otherwise, the decision engine runs with the thresholds of Chapter 4, and the full seed rule base of Appendix E; deviations (thin seed profiles, adaptation off, perturbed observation) are the controlled variables of the individual experiments and are stated per experiment. Table 5.4 lists the parameters of all runs in one place, grouped by subsystem.

Table 5.4 Simulation and decision parameters of all experiments

| Parameter | Sym | Value / range | Unit |
|---|---|---|---|
| **Grid size (synthetic)** | $-$ | $200 \times 200$ | cells |
| **Cell size (synthetic / hindcast)** | $C_{Size}$ | 90 | m |
| **Simulation step** | $\Delta t$ | $2 - 30$ | min |
| **Decision cycle** | $-$ | $2 - 20\ (def: 12)$ | min |
| **Attention Threshold** | $\tau_{att}$ | 0.35 | — |
| **Satisficing bound** | $J_{TH}$ | 0.35 | — |
| **Quality gate** | $\eta$ | 0.60 | — |
| **Trial basis (shadow forecast)** | $-$ | $\geq 45$ | min |
| **Monte Carlo runs per condition** | $N_{MC}$ | 50 | runs |
| **Random seed policy** | $-$ | $1 - 50$ | — |
| **Hardware / wall time per run** | $-$ | i9-10900K, 32 GB RAM | — |

Modeling and decision layer assumptions, common at operational-scale fire simulation (Sullivan, Wildland surface fire spread modelling 1: Physical and quasi-physical models, 2009a) (Sullivan, Wildland surface fire spread modelling 1: Physical and quasi-physical models, 2009a), are listed in Table 5.5.

Table 5.5 Environment and decision layer assumptions

| |
|---|
| **Environment layer assumptions** |
| Precipitation is a single domain-uniform value. |
| Dead-fuel moisture follows the equilibrium-moisture model, not measured fuel sticks. |
| Fuel classes come from a land-cover product through a fixed mapping, which flattens within-class variability. |
| Ember transport and crown-fire transitions enter only through the spread terms. |
| Crews and aircraft are modeled as capacity fields, not individual units. |
| **Decision layer assumptions** |
| Decisions are taken on a fixed decision cycle rather than continuously. |
| Sensing and inter-region communication are reliable, except where an experiment perturbs them. |
| The cost priorities that weight J are fixed and known for a given run. |
| Any change proposed by the generative component takes effect only after passing the verification gates. |
| The no-harm guard treats the shadow forecast as the proxy for the no-action outcome over its horizon. |
| Evacuation is charged only through the response capacity it consumes; the societal cost of displacement lies outside the objective. |
| The population term counts exposure inside actively burning cells only; smoke and proximity exposure outside the fire front are not modeled. |
| Fire duration is priced only through the burned area and the accumulated exposure it produces, not as a separate cost term. |

## 5.4 Sensitivity Analysis: Response to Parameter Bias

This section quantifies how the outcome responds when the operational parameters of the closed loop deviate from their configured values. Every parameter of Table 5.6 has a nominal value, the value the framework actually runs with, and a bias is a controlled deviation from that nominal value. The question of the section is therefore not which parameter values are best, but how far the outcome moves when a known parameter is set wrong, and how much bias each parameter tolerates before the outcome degrades. This framing matters operationally: In a fielded system the parameters are set by configuration and doctrine, not re-tuned per incident, so the practical risk is not a suboptimal choice but a miscalibrated one.

The method applies one bias at a time (Morris, 1991; Saltelli, et al., 2008): A single parameter is displaced across its bias range while every other parameter holds its nominal value, and the response is recorded on the physical decision cost, the burned-area, asset, and population terms of the forecast. Displacing one parameter at a time keeps the attribution unambiguous, since any movement of the outcome belongs to the displaced parameter alone; interactions between simultaneous miscalibrations are outside the scope of this design. Each bias level is repeated over five independent worlds, and the figures show the mean with its 95% interval, because a single world cannot separate the effect of a bias from the noise of the map. The study comprises 880 runs in total. (Morris, 1991; Saltelli, et al., 2008)

A bias study can only show what its operating point allows. Where the fire is beaten under every setting, or lost under every setting, no bias can move the result, and the study would report a robustness that belongs to the situation rather than to the system. The two situational axes, the fire load and the resource level, are therefore swept first to locate an operating point at which the decision layer is neither winning nor losing outright, and the biases of the operational parameters are then applied around that point. The shape of each response curve is read in two ways: A flat stretch around the nominal value is the tolerance of the parameter, the miscalibration the loop absorbs, and the point where the curve turns is the margin beyond which the error becomes visible in the outcome. Table 5.6 lists the parameters, their nominal values, and their bias ranges.

Table 5.6 Sensitivity design: Parameters, nominal values, and bias ranges

| Parameter | Nominal value | Bias range | Definition |
|---|---|---|---|
| **Simultaneous ignitions** | 4 | 1-12 | Number of fires started at step 0, placed on the least reachable burnable cells. Sets the fire load. |
| **Resource level** | 0.25 | 0.10 - 1.00 | Scale on the availability of the pool the planner suggests for this map. It binds as a total budget across the regions. |
| **Local regions ($\boldsymbol{N}$)** | 4 | 1-16 | Number of sub areas the grid is partitioned into. Each is evaluated on its own and competes for the shared pool. |
| **Attention threshold ($\boldsymbol{\tau_{att}}$)** | 0.35 | 0.05 - 0.95 | Priority at which a region is attended rather than monitored. A monitored region gets a smaller share and a tighter gate. |
| **No-harm horizon** | 45 min | 45 - 90 min | Lookahead over which candidate orders and the no action baseline are compared before a decision is committed. |
| **Satisficing bound ($\boldsymbol{J_{TH}}$)** | 0.35 | 0.15-0.60 | Cost at which the orders in force are accepted. The effective bound is $min(J_{TH}, 0.95\, J_0)$, so it tightens when inaction is already cheap. |
| **Fail-safe quality gate ($\boldsymbol{\eta}$)** | 0.60 | 0.30-0.90 | Minimum decision quality for offensive orders at full strength. Below it they are scaled by Q/η, while evacuation and warning are never reduced. |

### 5.4.1 Simultaneous Ignitions vs Resource Level

Figure 5.2 depicts the physical decision cost over fire load and suppression pool, expressed as a share of the free burn of the same world. At a tenth of the pool, once there is more than one ignition, the cost stays between 77 and 83% of the free burn whatever else is set, so the fire is lost throughout. With a full or a three quarter pool and up to four ignitions it stays between 14 and 32%, so the fire is held throughout.


Figure 5.2 The physical cost over simultaneous ignitions and resource level

Only a few cells fall inside the decidable band, and the sweeps that follow are run at the one nearest the midpoint of the free burn, four simultaneous ignitions with a quarter of the pool. That setting retains 50% of the free burn cost on the mean and 28% in the least severe of the worlds.

### 5.4.2 Simultaneous Ignitions vs Observation Coverage

Figure 5.3 - Figure 5.6 report the physical cost of the decision layer over the fire load and the observation level, one figure per resource pool, and extend the resource sweep of Figure 5.2 by fixing the pool and varying the observation instead. Each cell

gives the cost as a share of the free burn of the same world, with the rows varying the deployed sensors and the columns the simultaneous ignitions.

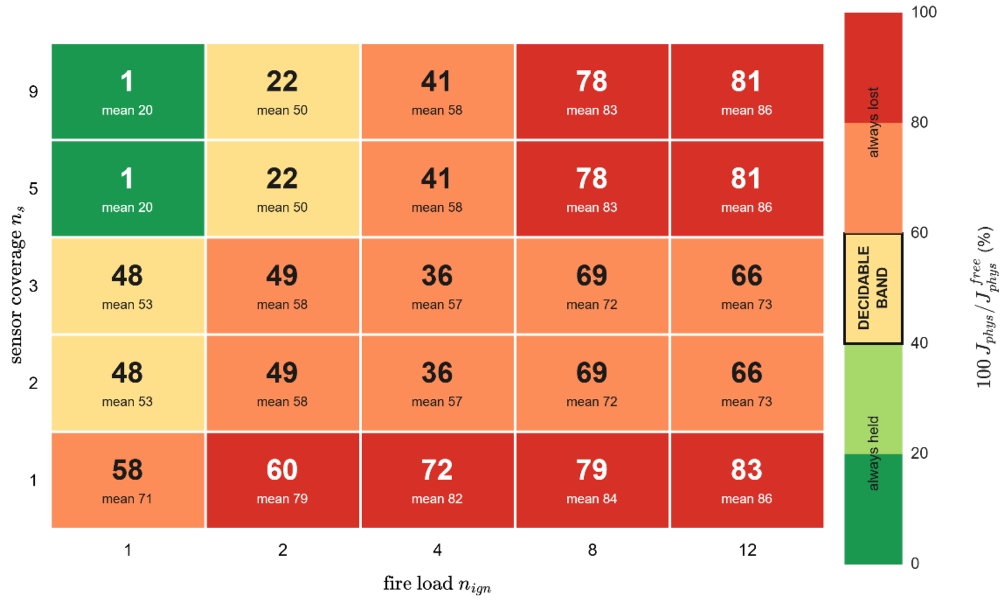


Figure 5.3 The physical cost over ignitions and coverage, pool is 0.25


Figure 5.4 The physical cost over ignitions and coverage, pool is 0.5

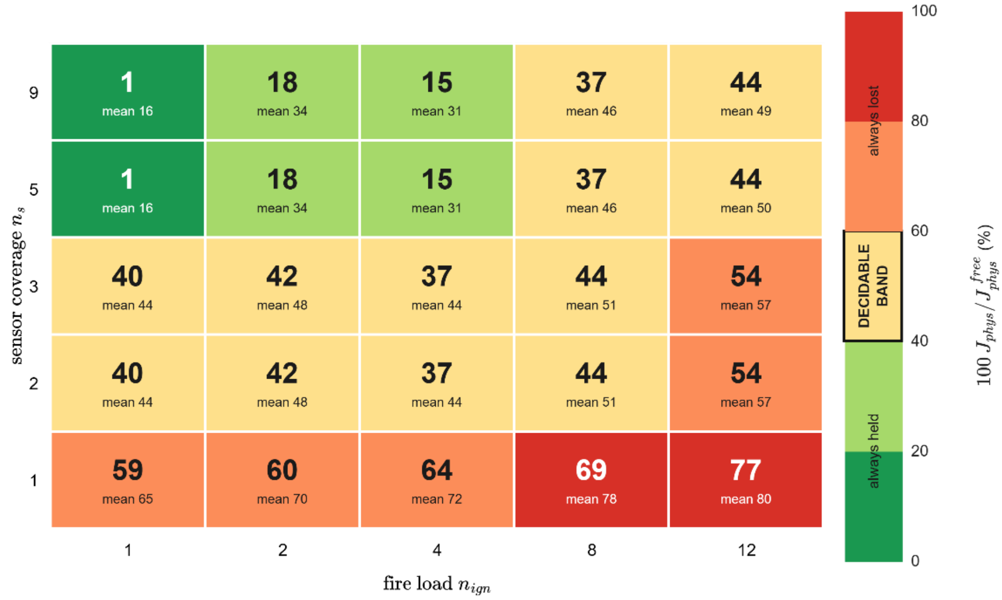


Figure 5.5 The physical cost over ignitions and coverage, pool is 0.75

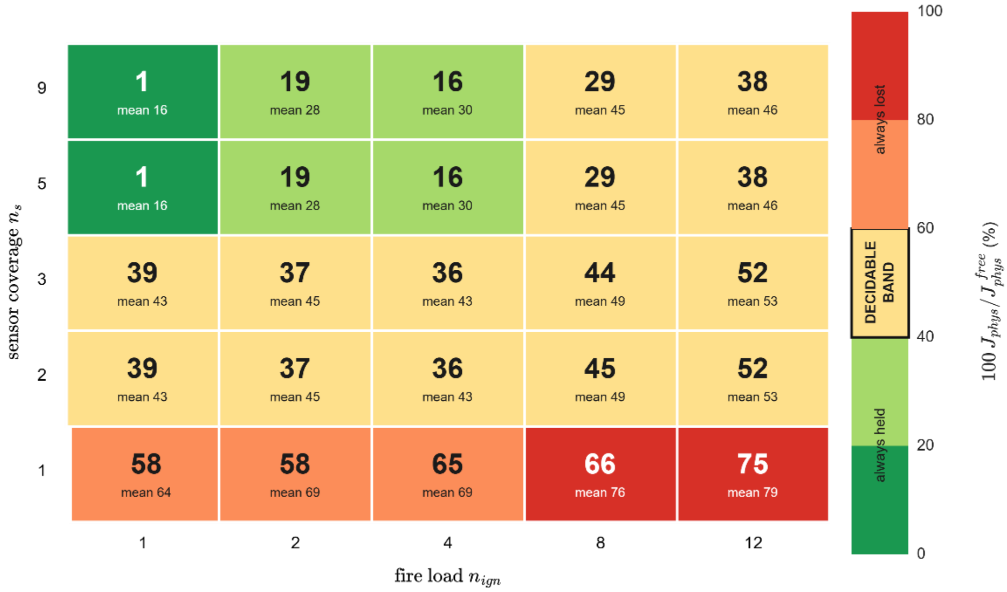


Figure 5.6 The physical cost over ignitions and coverage, pool is 1

Observation decides the outcome at a light fire load, where a single ignition falls from about 70% of the free burn with 1 sensor to about 20% with full coverage, and coverage saturates early because 5 and 9 sensors, and 2 and 3 sensors, give the same result. Resource decides the outcome at a heavy fire load, where 12 ignitions with full coverage drop from 86% at the quarter pool to about 46% at the full pool while the single-ignition cells barely move.

The two inputs are complementary rather than interchangeable: A blind map is not recovered by resource, and a scarce pool is not recovered by observation, so the decision layer changes the outcome only inside the decidable band, where both are sufficient for the fire at hand. As a bias reading, the outcome tolerates a loss of coverage down to five sensors with no measurable cost, and degrades smoothly rather than abruptly below that level.

### 5.4.3 Decision Cycle and Decision Horizon

The decision cycle is the interval at which the decision layer replans, so it sets how long a committed plan is allowed to age while the front continues to advance. For a DSS it is the difference between orders that describe the fire as it is and orders that describe the fire as it was, because a longer cycle does not change what the layer is able to decide, only how often it decides. The decision cycle is computed as ( 74 ).

$$\Delta \mathrm{t}_{\mathrm{decision}} = \max\left(\Delta t, \min\left(\frac{C_{size}}{2R_{\mathrm{spread}}}, \frac{\tau_{Horizon}}{5}\right)\right) \qquad (74)$$

The fire front advances no more than half a cell between two consecutive decisions $C_{size}/(2R_{\mathrm{spread}})$, so that orders still refer to the cells the fire actually occupies. The second requires that the plan be revised at least five times before the horizon over which it was judged expires $\tau_{Horizon}/5$, so that the no-harm guard can correct its own forecast before the commitment matures. The smaller of the two bounds is therefore selected, and the result is floored at the integration step, since the decision layer cannot act more often than the physics is advanced. Figure 5.7 and Figure 5.8 evaluate the two bounds over the rates of spread and the no harm horizons.

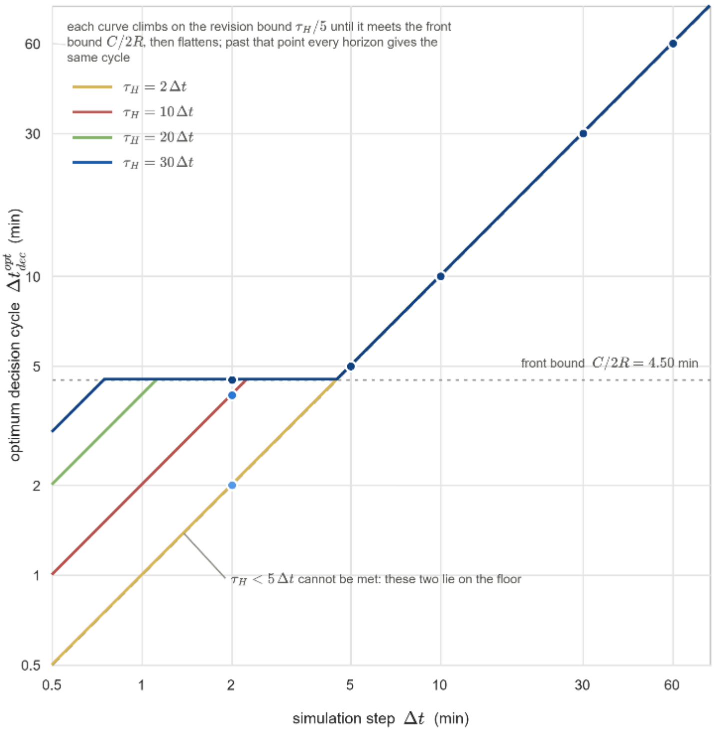


Figure 5.7 Optimum decision cycle for low ROS and no harm horizons

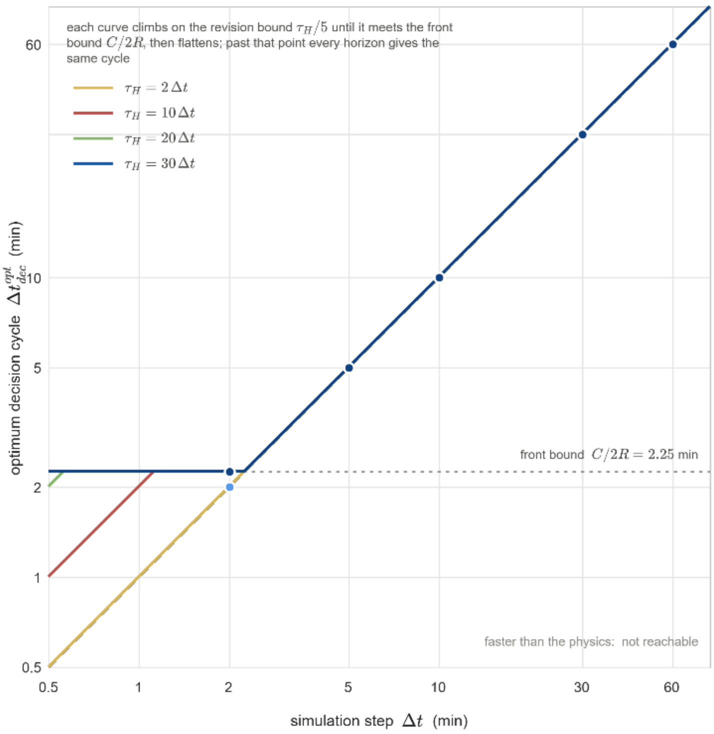


Figure 5.8 Optimum decision cycle for high ROS and no harm horizons

Figure 5.9 presents the paired gain over the decision cycle for the two fire types. That the basic fire shows the same ordering as the complex fire establishes that the benefit of a short cycle is not an artifact of a strained resource pool, although the five-world half width of 11.75 points at the best level leaves the size of the effect uncertain. In bias terms, the cycle derived in ( 74 ) is the nominal value, and a positive bias that lengthens the cycle degrades the paired gain monotonically for both fire types, so the cost of replanning late is graded rather than cliff-edged.

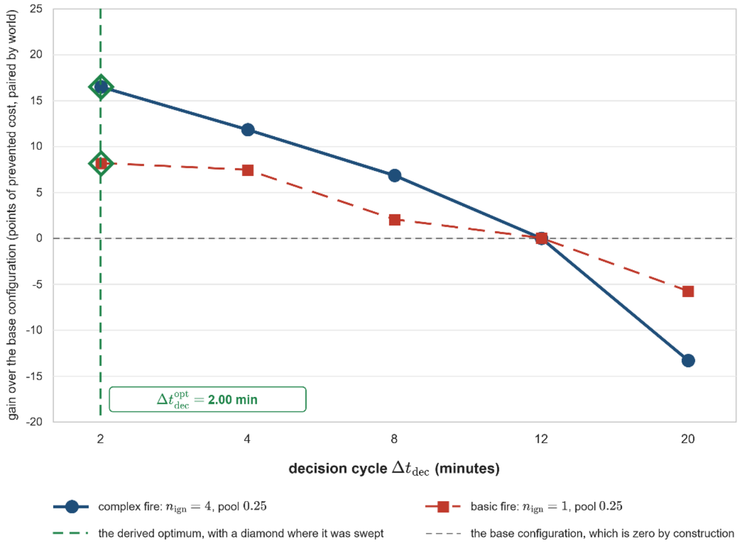


Figure 5.9 Sensitivity of the outcome to the decision cycle

### 5.4.4 Satisficing Bound and Quality Gate

Neither the satisficing bound nor the fail-safe quality gate acts on the outcome directly. Each governs a rate on the running incident. The bound decides how often the standing decision is judged insufficient. The gate decides how often an offensive order is reduced before it is fielded. Figure 5.10 and Figure 5.11 report the two rates over the whole unit interval.

In Figure 5.10, the bound sets how often the adaptation ladder is climbed. At 0.05 every decision cycle declares a cost deficit. At the configured 0.35 the share is 42.0%. Above 0.45 the curve is flat at 29.5%, because the relative margin against no action is then always the smaller of the two requirements and the ceiling no longer binds. A lower ceiling makes the standing decision harder to accept, so the system escalates more often, which is the direction the acceptance test asks for. Between the nominal ceiling of 0.35 and 0.45 the engagement rate falls by about 1.3 percentage points per 0.01 of upward bias, and below the nominal it rises steeply toward the saturated case at 0.05, so the acceptance machinery tolerates an upward bias of the ceiling and responds proportionally to a downward one.

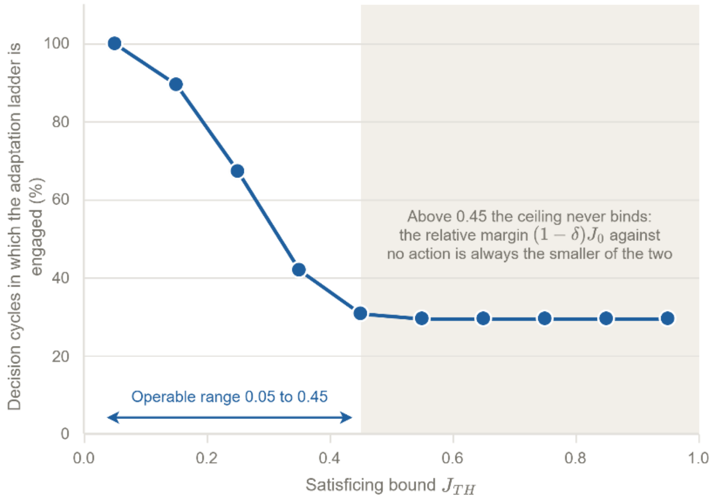


Figure 5.10 Effect of the satisficing bound on the rate at which the adaptation ladder is engaged

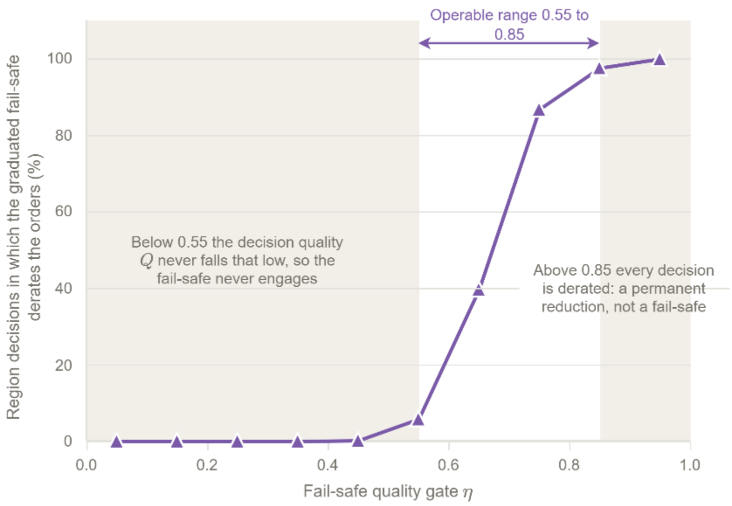


Figure 5.11 Effect of the fail-safe quality gate on the rate at offensive orders

Both settings are defined on the unit interval, yet each acts over a fraction of it. The reason is the same in the two cases. A threshold on a bounded quantity acts only where that quantity varies. The ceiling is compared against the forecast cost of taking no action, which spans 0.06 to 0.46 with a median of 0.23 at this operating point, so a ceiling above that span is never the binding requirement. The gate is compared against the decision quality, which stays between 0.55 and 0.85 for this rule base and this scenario. The range over which a threshold can be swept is therefore a property of the distribution it cuts, and not of the interval on which the parameter is defined. The scope of this analysis is set by the data behind it. The biases are applied one at a time around a single operating point, so interactions between simultaneous biases are not measured, and the biased quantities are the parameters of the decision loop; a bias of the plant itself, such as a mis-specified fuel map or a mis-forecast wind field, enters the evidence only through the degraded-observation scenario of Section 5.5 and would require dedicated runs to be separated. Within this scope the steep directions are the decision cycle and a resource pool below one half, and the tolerant directions are the observation coverage, the ceiling above its nominal value, and the quality gate inside the band that the decision quality actually occupies.

## 5.5 Performance of the Decision Layer

In the performance test, there are four configurations facing identical fires as listed in Table 5.7. $\boldsymbol{T_{F5}}$ against $\boldsymbol{T_0}$ prices the minimum decision layer itself; $\boldsymbol{T_{F5+Ev}}$ against $\boldsymbol{T_{F5}}$ prices evFIS based parametric adaptation; the full configuration $\boldsymbol{T_{DisasterAware}}$ against all three prices the open decision space.

Table 5.7 DSS test configurations

| |
|---|
| $\boldsymbol{T_0}$ issues no orders and fixes the cost of inaction. |
| $\boldsymbol{T_{F5}}$ runs the decision layer on the minimal seed base of five rules |
| $\boldsymbol{T_{F5+Ev}}$ adds the evFIS adaptation stages ① and ②, consequent tuning, and resolution growth. |
| $\boldsymbol{T_{DisasterAware}}$ runs the full staged adaptation, the open decision space in full. |

**Each configuration is evaluated as a Monte Carlo over the same fixed seed set in every scenario.** Every seed generates one scenario-conformant world, the map, the ignitions, and the weather drawn to the specification of the scenario, and all configurations replay that identical world, so the outcome differences come from the decision mechanism alone; $\mathrm{Test}_0$, run on the same history, additionally serves as the exact no-action counterfactual. Evaluation seeds are disjoint from every seed used during development and calibration. Episodes run until extinction or a defined interval (i.e., 12 hours). As the cost terms accumulate over time, cross-configuration cost comparisons are recorded at a common checkpoint (i.e., 6 hours), while suppression success and time to extinction are reported over the full episode. Comparisons are based on two costs. **Physical cost $\boldsymbol{J_{Phy}}$** is against the no-intervention reference and is using renormalized cost of assets, people, and burned area because $\mathrm{Test}_0$ pays no response cost by definition and a comparison on the total cost would reward inaction. The **total cost $\boldsymbol{J_{Tot}}$** is reported for other configurations. The costs ( 75 ) are based on ( 48 ).

$$
\begin{aligned}
J_{Phy} &= \frac{\sum_j w_j J_j}{\sum_j w_j}, \qquad j \in \{burn, asset, pop\} \\
J_{Tot} &= \frac{\sum_j w_j J_j}{\sum_j w_j}, \qquad j \in \{burn, asset, pop, resp, delay\}
\end{aligned}
\tag{75}
$$

### 5.5.1 Evaluation Scenarios

The tests run on five scenario families built from the landscape type and the resource sufficiency, and one degraded-observation space. The landscape axis separates the two operational faces of the problem, namely remote forest, where the burned-area term dominates and suppression and containment are focused, and the wildland-urban interface (WUI), where assets and population dominate and protection, evacuation, and warning are focused. The resource axis separates the comfortable regime, in which almost any allocation succeeds, from the capacity-limited regime, in which allocation quality is what separates the configurations; the realized resource sufficiency ratio is reported with the results. Weather is held at critical fire weather, dry fuels, and strong wind, in all scenarios, so that the scenario axes remain the only controlled difference. For each scenario, fifty distinct worlds are generated; Table 5.8 defines the five scenarios, and one representative world of each is illustrated in Figure 5.12, Figure 5.13 and Figure 5.14. The intervention and the layers are illustrated in Figure 5.15, respectively.

Table 5.8 Evaluation scenarios

| ID | Landscape | Resource* | Ignitions | What it assesses |
|---|---|---|---|---|
| **S1** | Remote Forest (≈2 small settlements) | Sufficient | 1 remote | Baseline suppression competence when capacity is not the constraint |
| **S2** | | Limited | 3 remote | Area protection under scarcity; allocation quality |
| **S3** | WUI (≈ 15 settlements, high assets) | Sufficient | 2 near the interface | Life/asset prioritization; evacuation and warning discipline |
| **S4** | | Limited | | Full operational stress; the demonstrative scenario |
| **S5** | as S4 | as S4 | as S4 | S4 under degraded observation |
| *Resource effectiveness exceeds 50% when resources are sufficient and drops below 50% when they are limited.* | | | | |

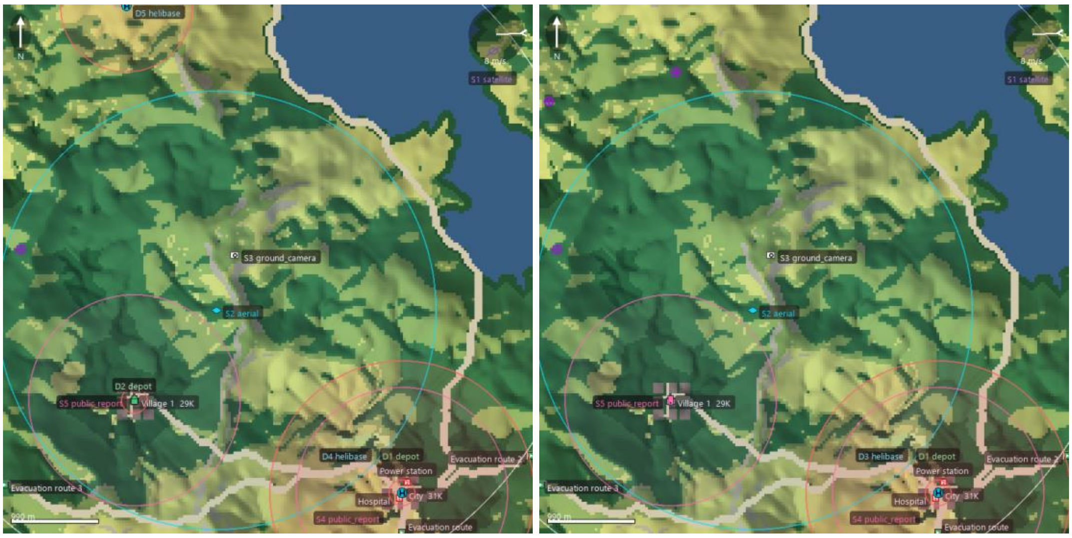


Figure 5.12 Representative world of S1 (left) and S2 (right)

Figure 5.13 Representative world of S3 (left) S4 (right)


Figure 5.14 Sample interventions (full set) on a representative world

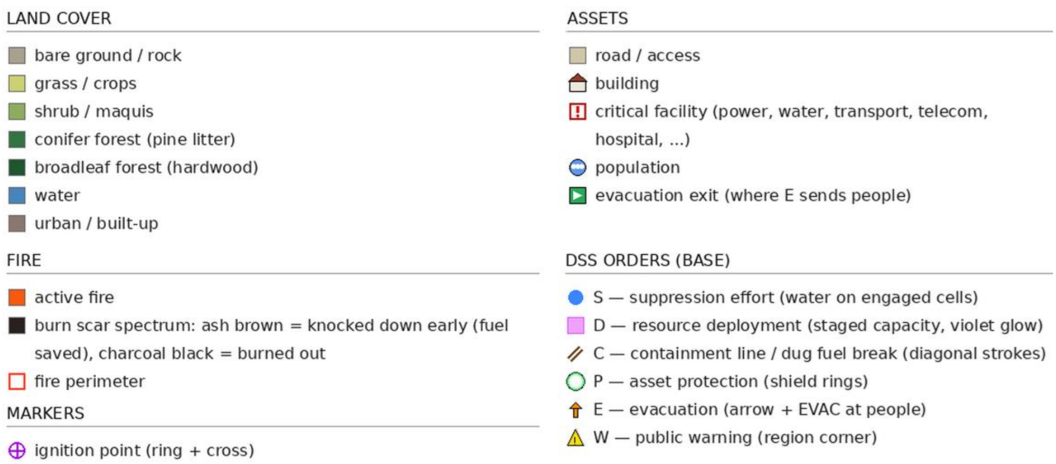


Figure 5.15 Legends for representative world

### 5.5.2 Improvement Ladder Performance

Table 5.9 and Table 5.10 report the physical outcome and the matching decision-cost terms per scenario at the six-hour horizon. The configurations form a ladder, and each configuration is expected to show a visible gain over the one below it.

$\boldsymbol{T_{F5}}$ lowers the mean burned area from 220 ha under free burn to 84 ha, a $\mathbf{62\%}$ reduction, and lowers the mean physical cost from 0.506 to 0.163.

$\boldsymbol{T_{F5+Ev}}$ adds a small but consistent gain over the static rules. It brings the mean burned area to 78 ha and the mean affected population from 609 to 511, and it lowers the burned area in four of the five scenarios. The mean total decision cost falls with it, from 0.185 to 0.169, and the saving comes from the delay term, which drops from 0.318 to 0.233: The tuned consequents commit the same resources sooner rather than committing more of them.

$\boldsymbol{T_{DisasterAware}}$ supplies the decisive second step: It halves the burned area again to 39 ha, an $\mathbf{82\%}$ reduction against free burn, lowers the mean affected population from about 511 to 300, and brings the mean total decision cost to 0.112 and the mean physical cost to 0.090, the lowest of the four configurations after its own response effort is charged. Two readings of this margin belong together. First, the against-no-action comparison is made on the physical cost, and the no-action total of 0.446, which carries no response and no delay term, is shown for reference only. Second, the full configuration also holds access to the verified action-list channels that no restricted configuration can order, so its margin combines the open decision mechanism with the wider verified action vocabulary; the ladder isolates each capability step, not the two factors inside this last step.

The average outcome, physical cost and total cost are shown in Figure 5.16, Figure 5.17 and Figure 5.18, respectively. The ± values of Table 5.9 are standard deviations over the Monte Carlo runs of the configuration under that scenario.

Scenario S5 repeats S4 under degraded observation and prices the confidence machinery directly. The mean burned area of the full configuration rises from 35.7 to 39.3 hectares, a loss of ten per cent, and the configuration remains the best of the four, so reduced observation degrades the response instead of breaking it.

Table 5.9 Physical outcome per scenario and configuration before the six-hour checkpoint per scenario

| | **Burned (ha)** | **Burned Forest (ha)** | **Affected Pop.** | **Evac. Pop.** | **Out (min) <360 min** |
|---|---|---|---|---|---|
| **S1** | | | | | |
| $\boldsymbol{T_0}$ | 243.3 ± 16.1 | 111.1 | 3117.0 | 0.0 | — |
| $\boldsymbol{T_{F5}}$ | 96.7 ± 53.6 | 41.0 | 253.0 | 0.0 | 63.0 |
| $\boldsymbol{T_{F5+Ev}}$ | 84.7 ± 46.2 | 36.0 | 206.0 | 166.0 | 81.0 |
| $\boldsymbol{T_{DisasterAware}}$ | 22.8 ± 24.5 | 10.4 | 54.0 | 296.0 | 81.0 |
| **S2** | | | | | |
| $\boldsymbol{T_0}$ | 242.2 ± 23.7 | 108.3 | 3318.0 | 0.0 | — |
| $\boldsymbol{T_{F5}}$ | 157.3 ± 49.3 | 67.5 | 727.0 | 0.0 | — |
| $\boldsymbol{T_{F5+Ev}}$ | 154.6 ± 39.9 | 67.1 | 633.0 | 0.0 | — |
| $\boldsymbol{T_{DisasterAware}}$ | 84.8 ± 37.6 | 35.5 | 314.0 | 0.0 | — |
| **S3** | | | | | |
| $\boldsymbol{T_0}$ | 178.9 ± 37.4 | 42.7 | 4868.0 | 0.0 | — |
| $\boldsymbol{T_{F5}}$ | 28.0 ± 13.9 | 3.0 | 261.0 | 0.0 | 204.0 |
| $\boldsymbol{T_{F5+Ev}}$ | 30.5 ± 12.0 | 6.6 | 207.0 | 0.0 | 168.0 |
| $\boldsymbol{T_{DisasterAware}}$ | 11.4 ± 7.4 | 5.5 | 57.0 | 0.0 | 246.0 |
| **S4** | | | | | |
| $\boldsymbol{T_0}$ | 218.1 ± 23.7 | 48.2 | 6202.0 | 0.0 | — |
| $\boldsymbol{T_{F5}}$ | 66.2 ± 25.8 | 6.9 | 889.0 | 0.0 | — |
| $\boldsymbol{T_{F5+Ev}}$ | 54.6 ± 18.9 | 8.9 | 739.0 | 0.0 | 187.0 |
| $\boldsymbol{T_{DisasterAware}}$ | 35.7 ± 17.0 | 6.5 | 501.0 | 276.0 | 204.0 |
| **S5** | | | | | |
| $\boldsymbol{T_0}$ | 218.1 ± 23.7 | 48.2 | 6202.0 | 0.0 | — |
| $\boldsymbol{T_{F5}}$ | 69.3 ± 22.7 | 8.1 | 916.0 | 0.0 | — |
| $\boldsymbol{T_{F5+Ev}}$ | 66.3 ± 19.7 | 9.7 | 771.0 | 0.0 | 209.0 |
| $\boldsymbol{T_{DisasterAware}}$ | 39.3 ± 10.0 | 5.6 | 573.0 | 0.0 | 209.0 |

Table 5.10 Decision-cost terms at the six-hour checkpoint per scenario and configuration

| Scenario | $J_{burn}$ | $J_{asset}$ | $J_{pop}$ | $J_{resp}$ | $J_{delay}$ | $J_{total}$ | $J_{phys}$ |
|---|---|---|---|---|---|---|---|
| **S1** | | | | | | | |
| $T_0$ | 0.64 | 0.804 | 0.107 | — | — | 0.456 | 0.517 |
| $T_{F5}$ | 0.337 | 0.093 | 0.002 | 0.445 | 0.417 | 0.177 | 0.144 |
| $T_{F5+Ev}$ | 0.317 | 0.070 | 0.002 | 0.427 | 0.386 | 0.162 | 0.13 |
| $T_{DisasterAware}$ | 0.123 | 0.028 | 0.001 | 0.207 | 0.499 | 0.086 | 0.051 |
| **S2** | | | | | | | |
| $T_0$ | 0.636 | 0.839 | 0.131 | — | — | 0.473 | 0.536 |
| $T_{F5}$ | 0.502 | 0.248 | 0.011 | 0.766 | 0.235 | 0.283 | 0.254 |
| $T_{F5+Ev}$ | 0.510 | 0.224 | 0.010 | 0.816 | 0.181 | 0.277 | 0.248 |
| $T_{DisasterAware}$ | 0.364 | 0.078 | 0.002 | 0.775 | 0.274 | 0.192 | 0.148 |
| **S3** | | | | | | | |
| $T_0$ | 0.565 | 0.639 | 0.089 | — | — | 0.380 | 0.431 |
| $T_{F5}$ | 0.163 | 0.056 | 0.001 | 0.082 | 0.339 | 0.090 | 0.073 |
| $T_{F5+Ev}$ | 0.179 | 0.054 | <0.001 | 0.085 | 0.213 | 0.086 | 0.078 |
| $T_{DisasterAware}$ | 0.078 | 0.017 | <0.001 | 0.066 | 0.223 | 0.045 | 0.032 |
| **S4** | | | | | | | |
| $T_0$ | 0.624 | 0.808 | 0.137 | — | — | 0.461 | 0.523 |
| $T_{F5}$ | 0.312 | 0.185 | 0.003 | 0.310 | 0.304 | 0.183 | 0.167 |
| $T_{F5+Ev}$ | 0.279 | 0.142 | 0.002 | 0.228 | 0.195 | 0.149 | 0.141 |
| $T_{DisasterAware}$ | 0.209 | 0.094 | 0.003 | 0.168 | 0.227 | 0.113 | 0.102 |
| **S5** | | | | | | | |
| $T_0$ | 0.624 | 0.808 | 0.137 | — | — | 0.461 | 0.523 |
| $T_{F5}$ | 0.330 | 0.195 | 0.003 | 0.311 | 0.293 | 0.191 | 0.176 |
| $T_{F5+Ev}$ | 0.323 | 0.156 | 0.002 | 0.286 | 0.190 | 0.169 | 0.160 |
| $T_{DisasterAware}$ | 0.229 | 0.115 | 0.002 | 0.199 | 0.207 | 0.125 | 0.115 |

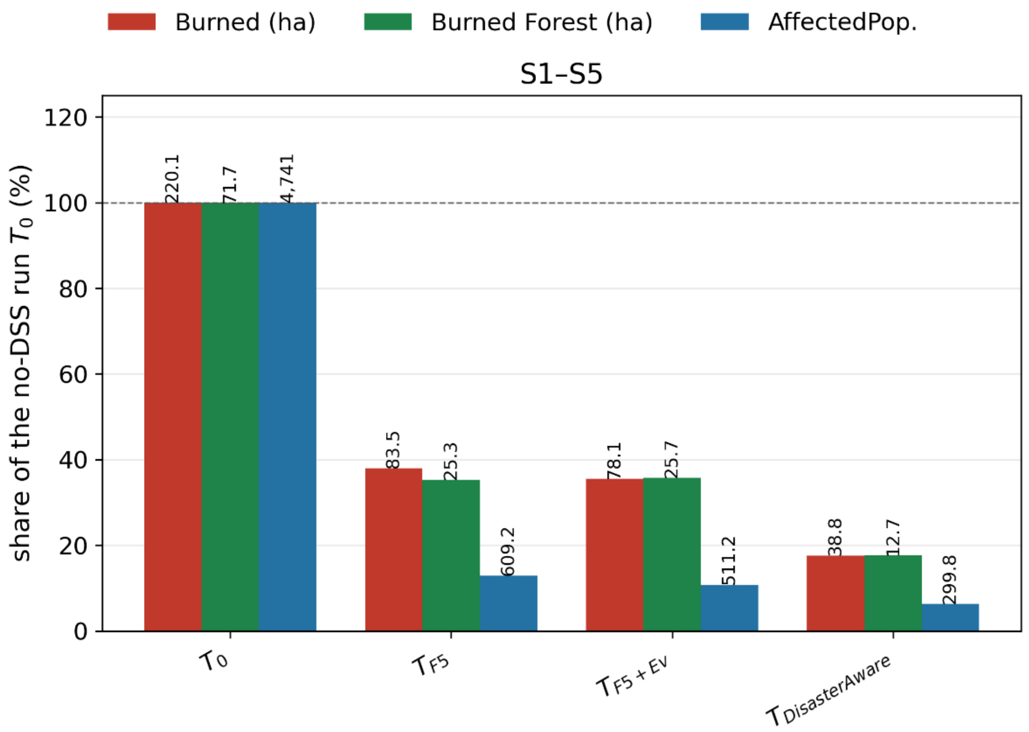


Figure 5.16 Average burned area, burned forest, and affected population of the four configurations, as shares of the no-DSS run, under the five scenarios

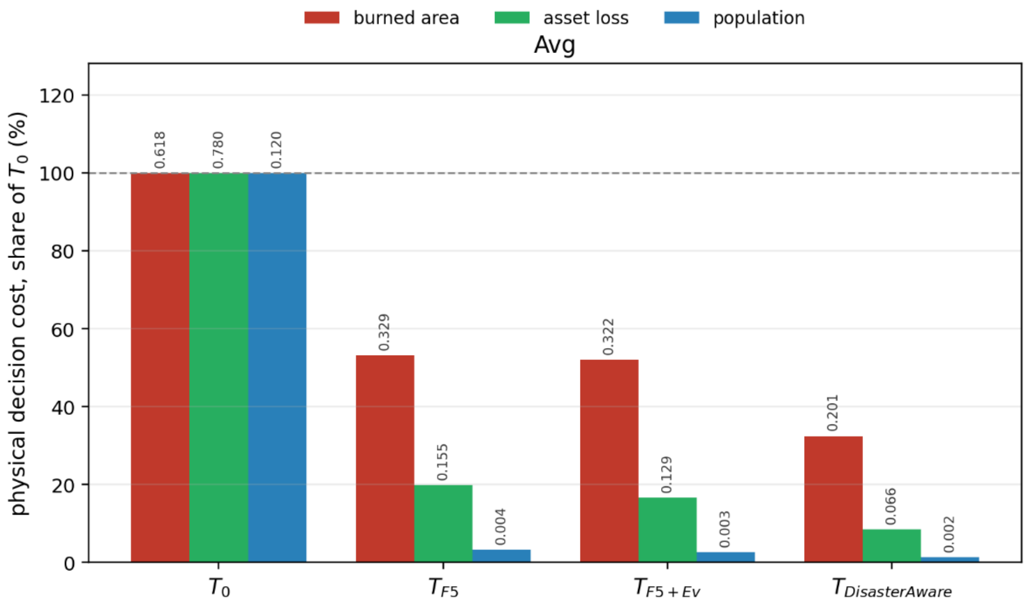


Figure 5.17 Average physical cost of the four configurations by component, as shares of the no-DSS run, under the five scenarios

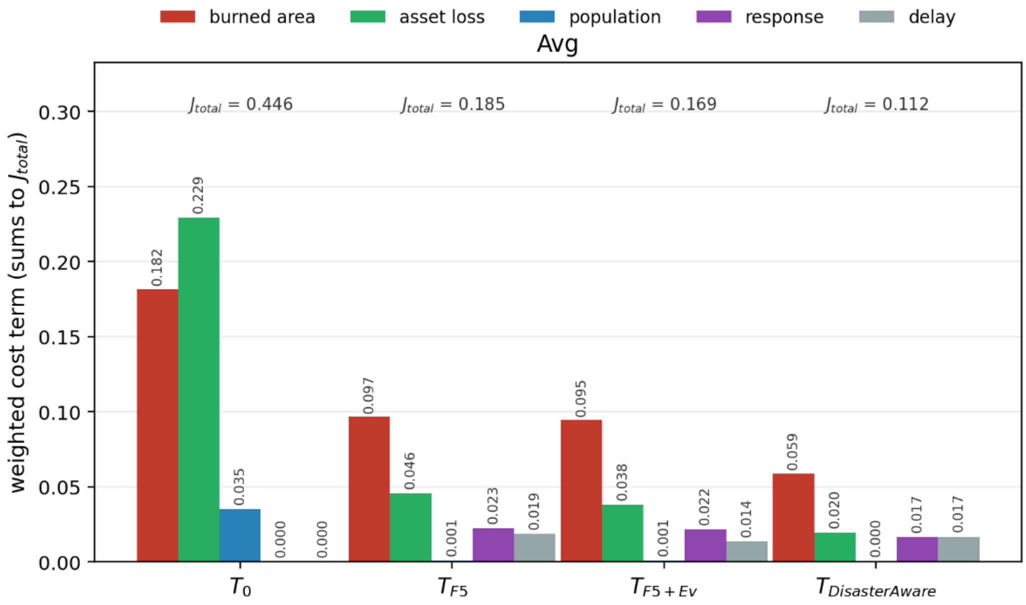


Figure 5.18 Average total cost of the four configurations by weighted component under the five scenarios

### 5.5.3 Open Decision Space Performance

Adaptation runs only on demand. It is triggered by a forecast cost above the satisficing bound, by a coverage gap while fire is alive, or by renewed spread under applied orders. The volume of trials therefore measures how often the standing decision fell short, and the acceptance rate measures how strictly the tests filtered the responses. The evaluation campaign comprises 160 recorded runs with 7 130 decision cycles, and 142 of these runs carry the adaptation analyses aggregated below, over 6 140 decision cycles. The record of the three adaptation stages over the analyzed runs is as follows.

Adaptation stage ①, the parametric layer, made 616 consequent-step trials over the analyzed runs and kept 129 of them (21%). The stage-① record is therefore a sequence of small, bounded corrections, and each of them passed its own forecast test at the moment of acceptance. Adaptation stage ②, the resolution layer, made 359 trials and kept 40 (11%); further candidate situations were filtered out before a trial was even attempted. The accepted trials instantiate new rules on antecedent cells that were empty while fire was alive. Adaptation stage ③, the generative layer, was invoked 232 times, and 47 proposals were admitted through the applicable verification gates, 36 as plain rules and 11 as packages that brought new vocabulary. Admissions are a flow, and the rule base is a stock: After deduplication and the strength-based pruning of the persistent store, 42 generated rules survive in the persistent store, and twenty representative rules are listed under their final naming in Appendix E. They carry the seven generated macro interventions and the one generated intermediate concept listed in Table 5.12.

The two evolving-fuzzy funnels appear on the left of Figure 5.19 and the generative cascade on its right. Low acceptance rates are the intended behavior of a gated design: Trials are cheap and rejected safely on the forecast, and the selectivity of the gates is what makes the admitted edits trustworthy.

Over the 160 recorded runs and 7 130 decision cycles the rule base produced 157 247 candidate interventions and 151 745 of them reached the field; 5 502 were withheld between the rule output and the order, the graduated fail-safe engaged in 5 338 region cycles, and the no-harm veto held the offensive channels in 113 cycles. The counts are collected in Table 5.11. The number of orders the physics could not honor after execution is not reported here, because the pool record that would carry it was added to the cycle log after these runs were made; the logs are silent on that quantity rather than clear of it.

Table 5.11 Constraint outcomes over 160 runs, 7 130 decision cycles

| Quantity | Value |
|---|---|
| **Candidate interventions** | 157 247 |
| **Fielded interventions** | 151 745 |
| **Withheld between rule output and order** | 5 502 |
| **Graduated fail-safe engagements (region cycles)** | 5 338 |
| **No-harm veto (cycles)** | 113 |
| **Post-execution actuator shortfall** | not logged |

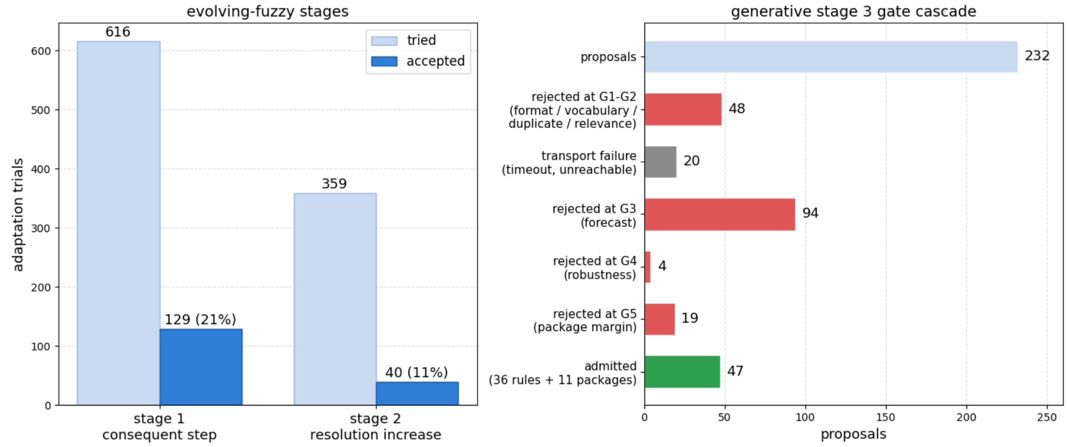


Figure 5.19 Left: EvFIS trials and acceptances at stage ① and stage ②. Right: The generative gates over proposals

#### 5.5.3.1 Performance of Adaptation Stages ① and ②

Every modification applied by the evFIS adaptation stages ① and ② is recorded. Each entry is a bounded, reversible edit that passed its own forecast test at the moment it was admitted; nothing in the record is a slow parameter drift. In total, the 129 accepted trials applied 167 consequent updates (103 escalations and 64 de-escalations), one accepted trial adjusting the consequents of up to two fired rules; the record further holds 5 membership-boundary shifts, 4 term insertions, and 40 rules instantiated on empty antecedent cells. There are five types of modification mechanisms.

**Consequent step (escalation):** This operation increases the intensity of selected consequents by 0.05. It is applied when the forecast cost remains above the satisficing bound and the associated rule continues to fire. This indicates that the rule recommends appropriate actions, but their current intensities are insufficient.

*G5: Downwind_retardant_shield 0.90 → 0.95; asset_protection 0.70 → 0.75; resource_deployment 0.70 → 0.75*

**Consequent step (de-escalation):** This operation decreases the intensity of selected consequents by 0.05. It is applied during the later phases of an incident when the response and delay components dominate the forecast cost. This indicates that the current actions are stronger than the remaining situation requires.

*G33: Suppression_effort 0.90 → 0.85; water_drafting 0.80 → 0.75; containment_line 0.70 → 0.65 (t = 90 min)*

**Membership-boundary shift:** This operation moves a shared boundary between two adjacent membership functions. As a result, the existing rules begin to fire earlier or later for the same input value. It is applied when the fire remains active but no rule fires above the coverage-gap threshold.

*suppression_feasibility: Shared edge moved inside the H–VH band (0.67–0.84)*

**Term insertion:** This operation adds a new term to a crowded part of a concept. It is applied when a coverage gap remains after membership-boundary adjustments, indicating that the existing partition does not provide sufficient resolution.

*suppression_feasibility: New term inserted at 0.74, splitting the H–VH band*

**Rule instantiation:** This operation creates a new rule for an antecedent combination that is not covered by any existing rule. It is applied when the fire remains active and the current input falls into an uncovered antecedent cell. The new rule is admitted because it fills this coverage gap and is non-inferior to the available alternatives.

*A1: IF suppression_feasibility VH ∧ asset_exposure_risk VH THEN asset_protection 0.56, resource_deployment 0.23*

#### 5.5.3.2 Performance of Adaptation Stage ③

Table 5.12 lists every vocabulary object of the generative stage ③ that survives in the persistent store after deduplication and pruning: Seven named macro interventions and one intermediate concept, each born at a stated moment of a specific incident under a logged cost deficit. Three of the base channels these products draw on, water drafting, aerial retardant drop, and tactical burn, belong to the action list that the physics validates although no seed rule orders them, and the situation summary offers a water-dependent channel only where the map carries a water body; a macro that draws on water drafting therefore composes a capability the seed doctrine ignored, not a new physical effect.

One product is renamed. The logged definition of the macro formerly called downwind_backburn is a containment line at full intensity with suppression effort behind it, and it carries no igniting channel and no igniting clause, so the name claimed a counter-fire the definition does not perform. It is listed as downwind_containment_shield. The macro that does carry fire is counterfire_strip, which composes the tactical burn with a containment line.

Table 5.12 The generative product ledger: The objects created by stage ③ and the action-list base channels they draw on, with their definitions

| Product | Definition |
|---|---|
| **Action-list base channel: *tactical burn*** | A firing crew ignites a strip ahead of the front, so a counter-fire consumes the fuel the head is running toward. |
| **Action-list base channel: *water drafting*** | Engines and aircraft draft from the nearest water body, raising sustained capacity near water; on a map without water it does nothing. |
| **Action-list base channel: Retardant drop** | Aircraft coat the fuel ahead of the head with long-term retardant, so the coated cells resist ignition even after they dry. |
| **Macro: downwind_containment_shield** | containment_line 1.0 + suppression_effort 0.7 |
| **Macro: Downwind_retardant_shield** | retardant_drop 1.0 + containment_line 0.8 |
| **Macro: Drafting_sustained_attack** | water_drafting 1.0 + suppression_effort 0.9 |
| **Macro: *head_knockdown*** | water_drafting 1.0 + suppression_effort 0.9 + retardant_drop 0.6 |
| **Macro: Wet_containment_line** | water_drafting 1.0 + containment_line 0.8 |
| **Macro: Counterfire_strip** | tactical_burn 1.0 + containment_line 0.6 |
| **Macro: Drafting_retardant_shuttle** | retardant_drop 1.0 + water_drafting 0.6 |
| **Concept: *interface_exposure*** | 0.5 asset_exposure_risk + 0.5 evacuation_pressure |

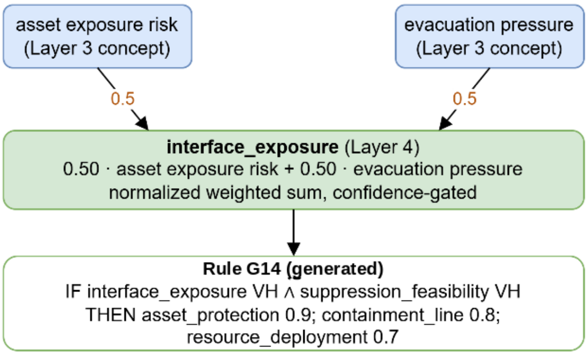


Figure 5.20 Generated intermediate concept and the admitted rule

One of these products is a new intermediate concept illustrated in Figure 5.20. The five seed concepts could not name the pressure that builds where exposed assets and evacuation demand meet, so the generative stage defined interface_exposure, a Layer-4 concept equal to the normalized sum of existing asset exposure risk and evacuation pressure in Figure 4.2. It enters the concept hierarchy like a native concept, so later rules may cite it. The concept and the admitted rule G14 that fires on it to protect the wildland-urban interface are shown below.

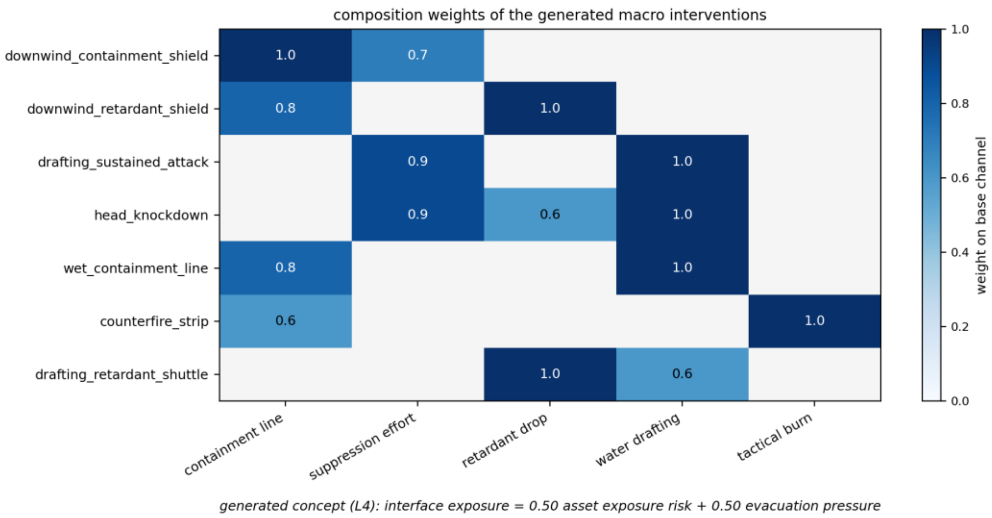


Figure 5.21 Composition of generated macro interventions over physical channels

The composition weights of the seven macros are shown in Figure 5.21. A weight is a per-channel intensity, so wet_containment_line applies full water drafting together with a partial containment line, and the weights are not required to sum to one. The macro names are assigned by the proposer and describe intent rather than composition; downwind_containment_shield, for example, contains no igniting effect and acts through containment and suppression.

Table 5.14 lists five of the admitted generative rules, which consume generated macro interventions, in full: Every antecedent is a decision concept with a linguistic term, and every consequent is an intervention with an intensity. Rule G14 shows the composed case, a generated rule that fires on the generated concept interface_exposure to protect the wildland-urban interface, so a concept the system invented and a rule that uses it enter through the same gates. Twenty representative admitted rules are listed in Appendix E.

Table 5.13 One decision, end to end

| Step | What the record holds |
|---|---|
| **Observation** | fire intensity 0.97 (confidence 0.97), ignition proximity 1.00 (0.97), asset exposure 1.00 (1.00), fuel load 0.70 (0.12), suppression availability 1.00 (1.00) |
| **Concept gate** | fire threat level 0.12, asset exposure risk 0.97, suppression feasibility 1.00, intervention urgency 0.12, evacuation pressure 0.97 |
| **Concept activation, effective** | fire threat level 0.36, asset exposure risk 0.85, suppression feasibility 0.79, intervention urgency 0.38, evacuation pressure 0.62 |
| **Rules fired, firing strength** | G5 0.50, R20 0.19, G2 0.19, G4 0.19, G1 0.17, G3 0.17, G6 0.17, G7 0.17, G8 0.17, G9 0.17, R26 0.11, R7 0.07 |
| **Candidate orders** | water drafting 0.84, containment line 0.83, retardant drop 0.82, suppression effort 0.80, asset protection 0.75, resource deployment 0.70, evacuation 0.66 |
| **Satisficing test** | forecast cost 0.0490 against a no-action forecast of 0.2568 and a bound of 0.2500, accepted |
| **Quality gate** | decision quality 0.776 against a gate of 0.60, no derating |
| **Fielded order** | the candidate as written (water drafting 0.84, containment line 0.83, retardant drop 0.82, suppression effort 0.80, asset protection 0.75, resource deployment 0.70, evacuation 0.66), at an attention share of 1.00, the region being attended |

The rule base and the gates are also observable one decision at a time. Table 5.13 reproduces a single decision of a recorded incident from end to end, five minutes after ignition, for the region the coordinator was attending. The row order is the order the layer works in, so each line is produced by the line above it: The sensors deliver ten readings with a confidence apiece, of which the row lists the five that feed the concepts of the fired rules while the complete vector is stored in the cycle record; the confidences gate the concepts, the gated concepts fire the rules, the fired rules compose the candidate orders, and the two acceptance tests decide whether those orders are fielded as written.

Table 5.14 Five of the admitted generative rules consuming generated macro interventions and the generated concept

| Rule | IF | THEN |
|---|---|---|
| **G4** | suppression_feasibility VH ∧ intervention_urgency VH ∧ fire_threat_level M | downwind_containment_shield 0.9; retardant_drop 0.8; asset_protection 0.6 |
| **G6** | fire_threat_level ≥ H ∧ suppression_feasibility VH | head_knockdown 0.95; resource_deployment 0.8 |
| **G7** | suppression_feasibility VH ∧ fire_threat_level ≥ H ∧ evacuation_pressure VH | wet_containment_line 0.9; resource_deployment 0.8 |
| **G8** | suppression_feasibility VH ∧ fire_threat_level H | counterfire_strip 0.9; resource_deployment 0.8 |
| **G14** | interface_exposure VH ∧ suppression_feasibility VH | asset_protection 0.9; containment_line 0.8; resource_deployment 0.7 |

Figure 5.22 depicts one recorded incident from ignition to containment. The upper panel shows the cumulative burned area, which rises and then flattens as the fire is held, together with the count of actively burning cells, which peaks near the middle

of the incident and is then driven back down to near zero. The lower panel marks the adaptation events by stage, with accepted trials as triangles and rejected trials as crosses. The accepted trials cluster in the early minutes, before the active front collapses, while later proposals are mostly rejected because little is left to improve.

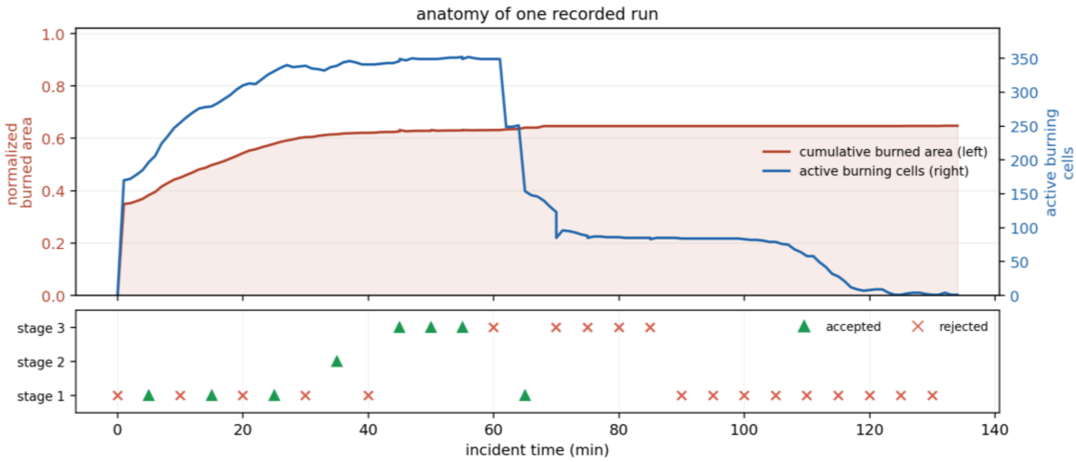


Figure 5.22 One recorded incident from ignition to containment

#### 5.5.3.3 Discovered Clause Actuators

The products of Table 5.12 are compositions of orders the doctrine already carries. A second campaign tests whether the stage can also create an intervention that carries its own geometry, that is, an order stating where on the map to act and how, in terms no base family expresses. It is reported separately because it runs the gate chain that ends in G5b. Five seeds were run in two arms over 205 decision cycles on the same worlds and under the same gates, the arms differing only in whether the geometry diagnosis of Section 4.5 is raised in the situation the proposer receives. The stage was entered 35 times and admitted 19 proposals.

The proposer is a generative model, with one call per proposal and the standing mission brief of Appendix F. Table 5.15 and Figure 5.23 report what was proposed and what survived, by kind of object.

Table 5.15 What stage ③ proposed and what the gates admitted, by object kind

| Object kind | Proposed | Admitted | Where the refusals fell |
|---|---|---|---|
| **Plain rule** | 13 | 6 | G3: 6, G4 A/B: 1 |
| **Intermediate concept** | 7 | 5 | G5: 2 |
| **Macro intervention** | 1 | 1 | None |
| **Clause actuator** | 14 | 7 | G5: 7 |
| **Total** | 35 | 19 | G3: 6, G4 A/B: 1, G5: 9 |

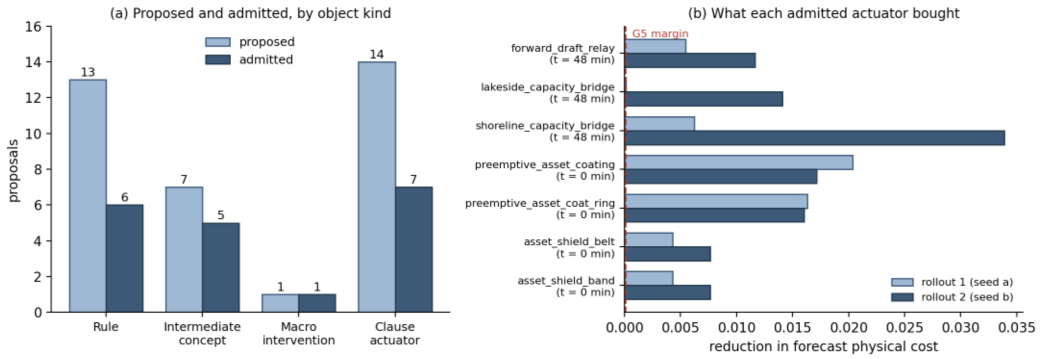


Figure 5.23 Generative products: (a) proposed and admitted by object kind, (b) the forecast cost each admitted actuator removed on the two reseeded rollouts

Seven of the fourteen proposed clause actuators were admitted, and they are listed in Table 5.16 with the moment of the incident at which each was written. Two families appear and they are separated in time. The actuators written in the first cycle coat the assets and the populated cells before the fire reaches them, which is a preventive tactic with no counterpart in the seed doctrine, since the base asset protection family raises a defensive posture but lays no retardant. The actuators written at minute 48 draft from the water body into the burning cells and the head and evacuate the population behind them, which is the substitution described in Section 4.5.3: On an incident whose pool is already spent, drafting raises the yield of crews that are committed instead of asking for crews that are not.

Table 5.16 The admitted clause actuators and their definitions

| Actuator | Written at | Clauses |
|---|---|---|
| **asset_shield_band** | 0 min | coat on assets [0, 6] at 1.0; coat on populated [0, 6] at 1.0; draft on assets [0, 8] at 1.0 |
| **asset_shield_belt** | 0 min | coat on assets [0, 4] at 1.0; coat on populated [0, 5] at 1.0; draft on at_fire [0, 6] at 0.8 |
| **preemptive_asset_coat_ring** | 0 min | coat on assets [0, 4] at 1.0; coat on populated [0, 4] at 1.0; coat on head [1, 6] at 1.0 |
| **preemptive_asset_coating** | 0 min | coat on assets [0, 4] at 1.0; coat on populated [0, 3] at 1.0 |
| **forward_draft_relay** | 48 min | draft on at_fire [0, 3] at 1.0; draft on ring [0, 8] at 1.0; evacuate on populated [0, 15] at 1.0 |
| **lakeside_capacity_bridge** | 48 min | draft on at_fire [0, 3] at 1.0; draft on head [0, 8] at 1.0; evacuate on populated [0, 6] at 1.0 |
| **shoreline_capacity_bridge** | 48 min | draft on at_fire [0, 8] at 1.0; draft on ring [0, 15] at 1.0; evacuate on populated [0, 15] at 1.0 |

The gain reported in Figure 5.23 belongs to the actuator and not to its company. Every admitted actuator was the only consequent of the rule that ordered it, so the two rollouts of G5 compare a situation in which the new object acts against one in which nothing acts in its place. G5b, which repeats the comparison with the new object struck out and the ordinary orders of its rule left standing, refused no proposals in this campaign. That is reported as it stands rather than as a property of the gate:

The situation given to the proposer also states that a new object must stand on its own, so no package arrived padded and the gate was never put to work.

One admission sits at the margin rather than comfortably above it. In Figure 5.23 (b) the first rollout of lakeside_capacity_bridge removed a forecast cost of the order of the margin itself, while its second rollout removed 0.014. The admission is regular, since G5 requires both rollouts to clear the margin and both did, but the case shows the limit of an AND-condition with a small margin: An object can enter on a gain that one future barely registers. The later history of the same actuator, which a separately admitted rule ordered again at minute 114, is the evidence that the gain was real rather than a fluke of the first seed.

The refusals carry as much information as the admissions. Of the seven refused actuators, six returned a forecast identical to the one without them on the first rollout, which means the tactic did not touch anything the forecast records, and one raised the cost. An identical forecast has three causes in this framework, all of them geometric: The sector resolved to no cells at the moment the actuator was written, the coating fell below the extinction moisture of the fuel it was laid on, or the effect drew on a pool with nothing left to give. Two of the sectors, assets and populated, resolve without reference to the written range, so two actuators that differ only in their ranges compile to the same order and are measured identically, which is why the two actuators of the first row pair in Table 5.16 removed identical forecast costs.

Two of the admitted objects show the vocabulary being used rather than merely being grown. The actuator forward_draft_relay is ordered by a rule whose antecedent is capacity_starved_head_pressure, an intermediate concept the same campaign created, so one generated object is invoked through another. The actuator lakeside_capacity_bridge, written at minute 48, is ordered again at minute 114 of the same incident by a separately admitted rule that was not part of the package which created it. This is the property the open decision space claims: An object admitted through the gates enters the vocabulary on the same footing as a doctrinal one and is available to whatever the rest of the incident writes.

Both arms produced admitted actuators, four of seven in the arm that raises the geometry diagnosis and three of seven in the arm that suppresses it. The diagnosis is therefore not what makes the open space productive; it is a statement about the incident that changes what the proposer attends to, and the gates decide the rest.

### 5.5.4 Timeliness and Scalability of the Decision Layer

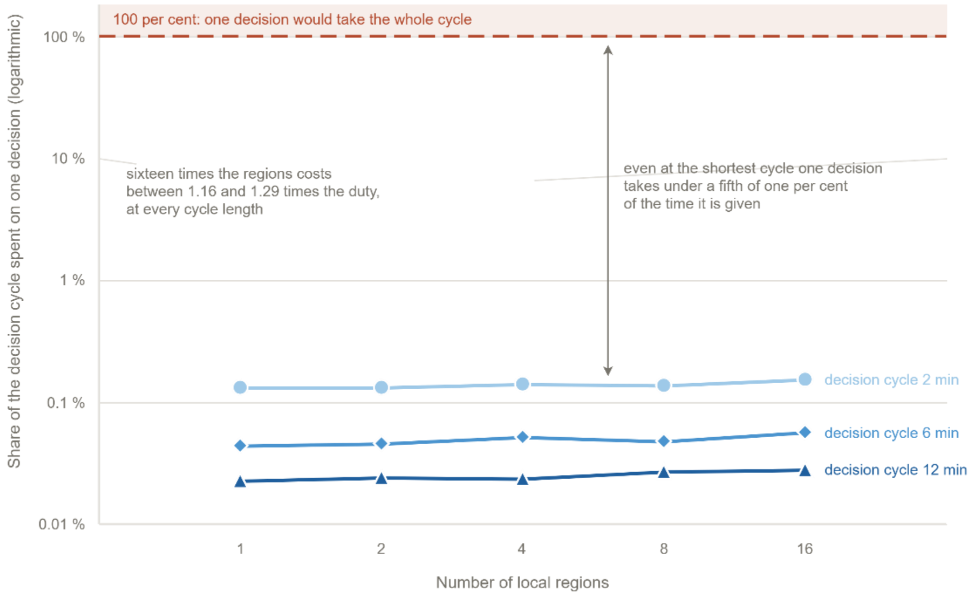


Figure 5.24 The cost of the decision layer against the number of local agents

Scalability asks how the cost grows as the problem grows. One decision covers the whole loop: The features and their confidences, the concept gates, the rule base, the global coordination, the two shadow forecasts of the acceptance test, and any adaptation stage that runs. The cost is reported as a share of the decision cycle because the cycle length is set by the deployment and not by the framework.

Figure 5.24 gives that share for one to sixteen local regions at cycles of two, six and twelve minutes. Every point lies between 0.02 and 0.16%, and 16 times the agents cost only 1.16 to 1.29 times the share. Each agent reasons only over its own region, so more regions split the same work instead of multiplying it. Two further sweeps locate the real driver: Six times the forecast horizon raised the median decision by 1.4, and four times the grid raised it by 6.9. The number of agents can therefore be raised by an order of magnitude at almost no cost, while the size of the domain remains the binding driver of computational load.

Timeliness asks whether the layer decides fast enough for the tempo of the incident. Table 5.17 reports the wall time of one decision over ten worlds. The median is 274 milliseconds, the ninetieth percentile 909 milliseconds and the slowest decision 1.28 seconds, against a decision cycle of twelve minutes. The spread comes from the adaptation stages, which run only when the acceptance test fails and each add a shadow forecast. In all ten worlds the first offensive order is fielded two minutes after ignition, at the first decision. Even the slowest decision uses less than two thousandths of the time available, and no step of the loop delays the first response.

Table 5.17 Decision latency of the closed loop

| Quantity | Value |
|---|---|
| **Decision latency, median** | 274 ms |
| **Decision latency, ninetieth percentile** | 909 ms |
| **Decision latency, slowest observed** | 1 277 ms |
| **Decision cycle** | 12 min (720 000 ms) |
| **Median latency as a share of the cycle** | 0.04% |
| **Ignition to first fielded offensive order** | 2 min, in all ten worlds |

## 5.6 Compliance with the Evaluation Criteria and Chapter Summary

Section 5.1 turns the literature into eight criteria. This section returns to them and states, for each, where the evidence sits and what it supports. Table 5.18 gives the map. Four criteria are met in full. Four are met on the axis this chapter measured and left open on another, and these are stated here rather than in Chapter 6, because a rubric that reports only its successes carries no information.

Table 5.18 Compliance with the evaluation criteria

| Criterion | Evidence in this chapter |
|---|---|
| **Predictive fidelity** | Section 5.2, Table 5.3 and Figure 5.1. Four blind hindcasts driven by real terrain, fuel, and weather; overall agreement between 86 and 94% on the front-propagation criteria and probability of detection between 85 and 97%; the final burned area is overpredicted by design in the suppression-free core. |
| **Decision quality** | Section 5.5.2, Table 5.9, Table 5.10, Figure 5.16 to Figure 5.18. Against no action the full configuration lowers the mean burned area from 220 to 39 ha (82%) and the mean physical cost from 0.506 to 0.090; the full configuration attains the lowest cost in every scenario, and the intermediate rung improves the mean but not every scenario. No baseline policy beyond no action is tested. |
| **Timeliness** | Section 5.5.4, Table 5.17. Median decision 274 milliseconds against a twelve-minute cycle, ninetieth percentile 909 milliseconds; the first offensive order is fielded two minutes after ignition in all ten worlds. |
| **Robustness under degraded observation** | Section 5.5.1 and Section 5.5.2, scenario S5 of Table 5.8 and Table 5.9, gate behavior in Figure 5.11. Under degraded observation the mean burned area rises from 35.7 to 39.3 ha (10%), and the configuration remains the best of the four. The criterion is met for observation loss; sensor noise and communication failure are not tested. |
| **Uncertainty handling** | Section 4.3; Section 5.2 and Section 5.5.2 for the spread over replicates and repeated worlds. Every feature carries a confidence that gates its concept; ensemble propagation of weather and fuel uncertainty was not run. |

Table 5.18 Compliance with the evaluation criteria (cont'd)

| | |
|---|---|
| **Adaptability** | Section 5.5.3, Table 5.12, Table 5.14, Figure 5.19 to Figure 5.21; the clause-actuator campaign in Table 5.15, Table 5.16 and Figure 5.23. Stage ① kept 129 of 616 trials and stage ② 40 of 359; stage ③ admitted seven macro interventions and one intermediate concept, each with a logged birth moment. |
| **Explainability and auditability** | Section 5.5.3, Table 5.12, Table 5.14, Figure 5.22. Every admitted edit, rejected proposal, and generated object is logged with the deficit that triggered it, the gate that judged it, and the rule that consumes it. Trace completeness is demonstrated; interpretability to the operator has not yet been evaluated. |
| **Scalability** | Section 5.5.4, Figure 5.24. One decision costs 0.02 to 0.16% of the decision cycle, and sixteen times the agents cost 1.16 to 1.29 times that share; the cost follows the number of cells, not the number of agents. |

The chapter set out to show that the three contributions hold under simulated operations and that the simulator they rest on is a faithful proxy of real fire behavior. The hindcasts settle the second point. The improvement ladder settles the first for the decision layer, since the concept driven rule base beats no action, the evolving stages ① and ② beat the static base, and the open decision space beats both. The sensitivity study locates the operating point at which those comparisons carry information and identifies the settings the loop is actually sensitive to. Two items remain open and are named in the table. Uncertainty is represented and carried into the decision but is not propagated as an ensemble, and scalability is demonstrated against the number of agents but not bounded against the size of the domain. Both are taken up in Chapter 6.

The evidence of this chapter maps onto the three sub-claims of Section 1.2 as follows. H1 rests on the improvement ladder and the scaling study: The full open configuration outperforms the static base in every scenario family, and one decision costs less than 0.2 per cent of the cycle as the number of local agents grows from one to sixteen. The scaling study partitions a fixed domain ever more finely, so it measures the overhead of distribution rather than growth of the domain itself; the domain-size driver is reported separately in Section 5.5.4. H2 rests on the adaptation ledger and the sensitivity study: Every fielded order carries its concept, rule, and confidence trace, every admitted or rejected edit is logged with its reason, and the response of the loop to its operational parameters is quantified. H3 rests on the end-to-end campaign: Across the five scenario families and the degraded-observation variant, every fielded intervention passed the applicable pre-execution gates, with the counts reported in the constraint outcome ledger, and the first offensive order was fielded at the first decision cycle.

The clause-actuator campaign of Section 5.5.3 adds the vocabulary level to that evidence line. The open configuration did not only choose better among the orders it was given; it composed seven interventions that the seed doctrine does not contain, each one admitted only after lowering the forecast physical cost on two independently reseeded futures, and one of them was later ordered by a rule that had no part in creating it.

Table 5.19 Evaluation of the research hypotheses

| | **Evidence base** | **Acceptance criterion** | **Observed result** |
|---|---|---|---|
| **H1** | The improvement ladder of Section 5.5.2 and the scaling study of Section 5.5.4. | The open configurations outperform the static rule base in every scenario family, at a decision latency compatible with the decision cycle. | The open decision space gives the lowest cost in all five scenario families, and one decision costs less than 0.2 per cent of the cycle from one to sixteen agents. |

Table 5.19 Evaluation of the research hypotheses (cont'd)

| | Evidence base | Acceptance criterion | Observed result |
|---|---|---|---|
| **H2** | The adaptation ledger, the decision trace of Table 5.13, and the sensitivity study of Section 5.4. | Every fielded order carries its concept, rule, and confidence trace, and the response of the closed loop to its operational parameters is quantified. | The trace and the ledgers are complete over the recorded runs, and the sensitivity of the outcome to the decision cycle, the satisficing bound, and the quality gate is reported. |
| **H3** | The end-to-end campaign of Section 5.5 with the degraded-observation variant and the constraint ledger of Table 5.11. | Every fielded intervention passes the applicable pre-execution gates, and the first offensive order arrives within the first decision cycle. | All 151 745 fielded orders passed the applicable pre-execution gates, 5 502 candidates were withheld, and the first offensive order was fielded at the first decision cycle. |

# CHAPTER 6

# CONCLUSION AND FUTURE WORKS

This chapter closes the thesis. The preceding chapters state three contributions, build the framework that embodies them, and test that framework against a rubric fixed in advance. What remains is to assess the results as a whole. Section 6.1 revisits the contributions and sets out, for each, what the evidence supports and where it stops. Section 0 turns the points where it stops into the two studies that should follow this work. The assessment follows the chain the thesis was built on: The gaps identified in Chapter 1, the hypotheses raised against them, the contributions that carry those hypotheses, and the performance results of Chapter 5 that support each one.

## 6.1 Conclusions

This thesis proposes DisasterAware, a distributed decision support system for wildfire response. The system reads ten observation features from heterogeneous sensing, and each feature carries confidence. Five decision concepts translate these features into operational language, and a sparse Takagi-Sugeno rule-base written over the concepts produces intervention intensities. A satisficing acceptance test and a graduated fail-safe decide whether a candidate decision may be fielded, and a global coordinator that performs no inference of its own ranks the local agents and shares the resource pool between them. When the standing decision falls short, a three stage ladder adapts the rule base: First its parameters (stage ①), then its resolution (stage ②), and finally its vocabulary (stage ③), where a generative stage proposes new rules, macro interventions and concepts that must pass the same gates before they act. The whole framework runs against a simulation core that is kept strictly separate from the decision core, so that each side can be validated on its own.

The argument runs from three structural gaps to three tested hypotheses. Chapter 1 identifies the gaps: G1, closed decision spaces in centralized architectures; G2, unrepresented epistemic uncertainty and limited explainability; and G3, simulation-decision coupling without an end-to-end operational workflow. Against them the thesis raises three hypotheses: H1, that a hierarchically distributed, concept-abstracted architecture generates adaptive interventions over an open decision space at a latency compatible with the decision cycle; H2, that concept-driven reasoning with confidence-gated uncertainty encoding yields intrinsically traceable decisions without post-hoc attribution; and H3, that an end-to-end workflow over strictly separated cores fields only interventions that pass every applicable pre-execution gate. Each hypothesis is carried by one contribution, each contribution answers one gap, and Chapter 5 judges all three against criteria fixed in advance; Table 5.19 records the verdicts, and the paragraphs below state, for each contribution, how the gap is closed and which results support the closure.

The first contribution, the open decision space, answers G1 and carries H1, and it is delivered. A closed catalogue over five concepts with five terms each needs 3 125 rules, which no expert can author or audit. The framework instead starts from five seeded rules and adds only what the field demands: 40 rules instantiated on uncovered situations, seven generated macro interventions and one generated concept over the recorded evaluation runs. A separate campaign on the clause grammar added seven admitted clause actuators, orders that carry their own map geometry, together with four further intermediate concepts and one further macro intervention, all admitted through the same gate chain. Every addition passes a forecast test before it is admitted, and the additions matter for the outcome, since the full open decision space halves the burned area of the static rule base. The distributed side of the contribution holds as well, because sixteen times the number of agents raises the cost of a decision by a factor of at most 1.29, so the hierarchy opens the number of agents at almost no price.

The second contribution, uncertainty handling with built-in explanation, answers G2 and carries H2, and it is delivered. Every fielded order traces to the concepts that demand it, the rules that fire, and the confidence behind each feature, without any post hoc attribution method. Every edit the adaptation ladder admits is logged with the deficit that triggers it, every rejected proposal with the gate and the reason that rejects it, and every generated object with its definition, its birth moment and the rule that consumes it. The sensitivity study completes the contribution: It states which settings move the outcome and which do not, and it explains the flat cases from the mechanism rather than leaving them as unexplained robustness. Under degraded observation the confidence gates do what they are designed to do, since the full configuration loses ten per cent of its performance and remains the best of the four tested.

The third contribution, the end-to-end demonstration over strictly separated cores, answers G3 and carries H3, and it is delivered. The simulation core is validated first and independently: Four blind hindcasts driven by real terrain, fuel and weather data agree with the satellite record between 86 and 94 per cent on the front-propagation criteria; the final burned area is overpredicted by design, since the core runs without suppression while the recorded fires were fought. On that validated ground the decision core runs end to end, from ignition through observation, reasoning, coordination, and adaptation to containment. Against no action the seeded rule base lowers the mean burned area from 220 to 84 hectares, the evolving stages ① and ② take it to 78, and the full framework to 39, a reduction of 82 per cent, while the mean physical cost falls from 0.506 to 0.090. The full configuration attains the lowest cost in every scenario, although the intermediate rung does not improve every scenario, and the loop that produces it spends 274 milliseconds at the median against a decision cycle of twelve minutes.

Taken together, the three contributions are not only implemented but shown to work, each against the criterion set for it in advance. The framework opens the decision space and pays almost nothing for its hierarchy, explains every order it gives from its own record, and improves the outcome of every scenario it is evaluated on, on a simulator that is itself validated against real fires. The hypothesis of this thesis is therefore supported by the evidence, and the limits that remain, stated openly in this chapter, mark the path of the future work rather than a gap in the argument.

Three boundaries of this evidence deserve a direct statement. The first boundary concerns the comparison base. Every quantitative comparison in Chapter 5 is internal. The configurations of the ladder differ only in the capabilities they enable, and the paired runs share their worlds, so the measured differences isolate the contribution of each capability. No external system appears in these comparisons. Deployed tools such as WFDSS support the documentation and evaluation of decisions but do not generate interventions over an open decision space, so a like-for-like comparison target does not exist in the reviewed literature. The internal ladder with its paired with-DSS and without-DSS design is therefore the appropriate form of the comparison for the claims of this thesis.

The second boundary concerns the validation base of the simulator. The hindcast study covers four historical fires, and the agreement of 86 to 94 per cent is reported over stochastic replicates of these cases. The simulation core is designed as a faithful proxy for the decision layer rather than as a contribution to fire science, and testability was preferred over maximal fidelity by design. A broader historical case base would strengthen the proxy claim and is a natural extension of the validation.

The third boundary concerns the generative stage. The proposer is a large language model, so the individual proposals are not reproducible in a bitwise sense. The campaign is therefore documented with the pinned model, the transport, the standing prompt, and the seed design, which makes the procedure repeatable, and the admission decisions do not rest on the model at any point. Every proposal is judged by deterministic gates that end in reseeded forecast comparisons on the simulator, and the verdict and the margin of every gate are logged. A different proposer would produce different candidates, but nothing enters the rule base on the authority of the model.

## 6.2 Future Works

The evaluation of this thesis is entirely in a simulated setting, and this is both its strength and its boundary. Simulation makes the comparisons controlled, repeatable and safe, but it also means that the two most important questions that remain are questions about the world outside the simulator. The first is whether the framework serves the people for whom it is built. The second is whether its architecture carries beyond the hazard it was built on. These two directions define the field side of the future work.

The first direction is validation with operators in the field. The framework has been demonstrated end to end against a simulator and against the satellite record of four historical fires, but it has not yet faced an incident commander. The traceability record is designed to be read, so the first study should place that record in front of experienced fire officers and ask whether they read it, whether they trust it, and whether the reasons it gives change the decisions they make. A natural form for this study is a tabletop exercise, in which commanders manage a simulated incident once with the framework and once without it, so that the effect of the tool can be separated from the skill of the operator. Such an exercise would also evaluate the parts of the design that a simulator cannot: Whether the concepts speak the language of the field, whether the pace of the interface matches the pace of command, and whether an operator accepts an order that a machine has adapted during the incident. What survives this contact belongs to the framework; what does not survive it marks the next design cycle.

The second direction is transfer to other hazards. Nothing in the decision layer is specific to fire. The concepts, the confidence gates, the satisficing acceptance, and the graduated fail-safe are defined over an abstract state and an abstract set of interventions, and the simulation core is coupled to them only through a narrow interface. Floods, industrial plumes and search and rescue pose the same kind of problem, because each is a process that spreads over a landscape, is observed incompletely, and must be answered with limited resources under time pressure. A transfer study would keep the decision layer unchanged, replace the simulation core with a model of the new hazard, and rewrite only the feature definitions and the seed rules. The cost of that rewriting is itself the result of the study, since it measures how much of the framework is architecture and how much is wildfire. A successful transfer would show that the contributions of this thesis describe a general way of building decision support under uncertainty; a failed one would show precisely where the generality ends.

Two further studies remain inside the simulator, and both take up a boundary stated in Chapter 5. The first study concerns uncertainty. Chapter 5 represents epistemic uncertainty through confidence gating and reports the spread of every outcome over stochastic replicates, but it does not propagate weather and fuel uncertainty as an ensemble. An ensemble study would run the closed loop over a set of equally plausible forecasts and report the decisions that remain stable across the set, which would turn the present per-replicate spread into a statement about forecast uncertainty.

The second study concerns scale. The scaling result of Chapter 5 bounds the cost of the hierarchy against the number of local agents. The complementary study would grow the domain itself, holding the number of agents fixed while the cell count and the incident size increase, and would state where the decision cycle stops being sufficient.

These four studies order themselves naturally. The operator study validates the interface the framework already exposes, the transfer study tests the generality of its architecture, and the ensemble and scale studies widen the envelope of the evidence. None of them requires a change to the framework itself; each extends the evidence around it.

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# APPENDICES

## A. Simulation Mathematics for Evolution of Burning Status

This appendix provides the formulations underlying the simulation in Chapter 3. The rate of spread $R_{spread,k}(x, y)$ is the forward propagation of wildfire at grid cell (x, y) and time step $k$. It is computed internally within the hybrid fire spread model using meteorological forcing, terrain properties, fuel characteristics, and fuel moisture. The spread formulation is not intended to reproduce physical wildfire propagation in metric units. It provides a grid-consistent, DSS-oriented abstraction inspired by semi-empirical wildfire behavior models.

The formulation adopted in this work is a semi-empirical, Rothermel-like mapping designed to preserve structural interpretability (i.e., the output $R_{spread,k}$ is expressed in normalized cells-per-step rather than in metric units) while remaining computationally stable and easily calibratable for DSS simulation purposes (Rothermel, 1972). The rate of spread $R_{spread,k}(x, y)$ is calculated by $f_{ros}$ ( 29 ). Each multiplicative component of the rate-of-spread function $f_{ros}$ corresponds directly to a physically interpretable effect and is tied to an already-defined state or input variable ( 76 ). All terms are evaluated at $(x, y)$ and time $k$.

$$R_{spread,k}\ (x, y) = r_{base}\ \left(F_{type}(x, y)\right) \times g_{moist}(x, y) \times g_{wind}(x, y) \times g_{slope}(x, y) \times\ g_{aspect}(x, y) \tag{76}$$

Fuel-Type Base Spread Rate $r_{base}$ $(\cdot)$ defines the baseline spread potential of each fuel class under nominal environmental conditions. It is calibrated in units of cell per simulation step. Moisture damping factor $g_{moist}$ is ( 77 ) where $F_{moist,k}(x, y)$ is surface fuel moisture fraction and $m_{ext}\left(F_{type}\right)$ is extinction moisture threshold for the given fuel type

$$g_{moist}(x, y) = \max\left(0{,}1 - \frac{F_{moist,k}(x, y)}{m_{ext}\left(F_{type}(x, y)\right)}\right) \tag{77}$$

If fuel moisture equals the extinction threshold, spread is suppressed ( 78 ). The damping structure is selected for stability and interpretability; it prevents negative spread rates and ensures bounded behavior.

$$F_{moist,k} \geq m_{ext} \Rightarrow g_{moist} = 0 \qquad (78)$$

Wind amplification factor $g_{wind}$ is ( 79 ) where $W_{ws,k}(x,y)$ is the wind speed, $a_w(F_{type})$ is a wind sensitivity coefficient and $w_0$ is a scaling constant.. This factor models the acceleration of fire spread due to wind speed. The hyperbolic tangent ensures that spread increases with wind speed but does not grow unbounded for extreme winds. The use of $\tanh(\cdot)$ guarantees numerical stability. The wind scaling constant $w_0$ controls the saturation behavior of wind amplification. A value of 10 m/s is selected to reflect moderate-to-strong wind regimes while preserving numerical stability.

$$g_{wind}(x,y) = 1 + a_w(F_{type}(x,y)) \cdot tanh\left(\frac{W_{ws,k}(x,y)}{w_0}\right) \qquad (79)$$

Slope acceleration term $g_{slope}$ is ( 80 ) where $G_{slope}(x,y)$ is terrain slope (radians) and $a_s(F_{type})$ is slope sensitivity. Fire spreads faster on steep terrain due to flame preheating of fuels on the upslope side. The factor is isotropic within a cell: it accelerates propagation on steep cells regardless of the local spread direction. The tangent function reflects nonlinear growth with increasing slope angle.

$$g_{slope}(x,y) = \min\left(3, 1 + a_s\left(F_{type}(x,y)\right) \cdot \tan\left(G_{slope}(x,y)\right)\right) \qquad (80)$$

Aspect-wind orientation term $g_{aspect}$ is ( 81 ) where $G_{aspect}(x,y)$ is terrain orientation angle, $W_{wd,k}(x,y)$ is wind direction field, $a_{asp}(F_{type})$ is aspect coefficient. This term captures interaction between terrain orientation and wind direction. If wind direction aligns with terrain orientation, $\cos(\cdot) \approx 1$ and spread increases. If wind opposes terrain orientation, $\cos(\cdot) < 0$ and spread decreases.

$$g_{aspect}(x,y) = 1 + a_{asp}\left(F_{type}(x,y)\right) \cdot \cos\left(G_{aspect}(x,y) - W_{wd,k}(x,y)\right) \qquad (81)$$

Different fuel types exhibit distinct intrinsic combustion and propagation characteristics due to variations in fuel structure, packing ratio, moisture retention capacity, and heat-release behavior. These differences are parameterized in Table

A.1 through a fuel-dependent base spread rate and sensitivity coefficients that modulate environmental amplification effects. The base spread parameter $r_{base}$ is expressed in units of grid cells per simulation step and therefore must be interpreted relative to the spatial grid resolution $\Delta x$ and the discrete simulation time step $\Delta t$.

In contrast, the extinction moisture threshold $m_{ext}$ is a dimensionless mass fraction, representing the critical fuel moisture level above which sustained combustion is suppressed. The environmental coefficients $a_w$, $a_s$, and $a_{asp}$ are dimensionless parameters, governing the relative amplification of spread rate due to wind speed, terrain slope, and wind-aspect alignment, respectively. Together, these parameters provide a resolution-consistent, yet physically interpretable abstraction of fuel-dependent wildfire spread dynamics suitable for discrete-time grid-based simulation.

Table A.1 Spread rate parameters for each fuel type

| $\boldsymbol{Fuel\,Type}$ | $\boldsymbol{r_{base}}$ | $\boldsymbol{m_{ext}}$ | $\boldsymbol{a_w}$ | $\boldsymbol{a_s}$ | $\boldsymbol{a_{asp}}$ |
|---|---|---|---|---|---|
| **Grass** | 2.00 | 0.15 | 25.0 | 2.0 | 0.30 |
| **Shrub** | 0.80 | 0.20 | 15.0 | 2.0 | 0.25 |
| **Pine litter** | 0.80 | 0.25 | 6.0 | 2.2 | 0.20 |
| **Hardwood** | 0.30 | 0.30 | 4.0 | 1.8 | 0.15 |

The parameter values of Table A.1 are calibrated against the standard fire-behavior literature rather than chosen ad hoc. At the reference discretization of 90 m cells and 30 min steps, one cell per step corresponds to 3 m/min, so the base rates carry a direct metric interpretation. When a scenario uses a different discretization, the base rates are rescaled by the factor $(90\frac{m}{\Delta x}) \cdot (\frac{\Delta t}{30}\ min)$, so the metric spread rates are preserved. With the wind amplification factor, the calibrated values reproduce the benchmark head-fire rates of the standard fuel models: Grass reaching tens of m/min, shrub 5-10 m/min, long-needle litter 2-4 m/min and compact hardwood litter below 1 m/min. The slope acceleration term is additionally saturated at three times the base rate since the tangent term otherwise diverges on the near-vertical cells of real digital elevation models.

## B. Simulation Mathematics for Fuel Mass Evolution

This section provides detailed mathematical formulation and explanation for both the fractional burn rate calculation and the suppression of fuel reduction maps.

**Fractional burn rate calculation**

The discrete combustion consumption ratio $F_{burn,k}(x,y)$ represents the fraction of available fuel mass consumed at grid cell $(x,y)$ during a single simulation time step $\Delta t$ under active burning conditions. Unlike fuel load $F_{load,k}$, which is a dynamic state variable, $F_{burn,k}$ is not stored as part of the state vector. Instead, it is computed algebraically at each simulation step as a bounded function of fuel type and surface fuel moisture ( 82 ) where $b_{base}(F_{type}) \in (0,1)$ is the baseline combustion coefficient associated with the fuel class, $F_{moist,k}(x,y) \in [0,1]$ is the surface fuel moisture fraction, $\mathrm{sat}_{[0,1]}(\cdot)$ is the saturation operator ensuring boundedness. The term $(1-F_{moist,k})$: A linear moisture damping factor, reducing combustion efficiency as moisture increases..

$$F_{burn,k}(x,y) = \mathrm{sat}_{[0,1]}\left(b_{base}\left(F_{type}(x,y)\right) \cdot \left(1 - F_{moist,k}(x,y)\right)\right) \qquad (82)$$

The coefficient $b_{base}$ is expressed as a *fraction of fuel consumed per simulation time step* and therefore depends on the chosen time resolution $\Delta t$. It is not a universal physical constant but a calibrated simulation parameter. Different fuel classes exhibit distinct combustion rates due to structural density, packing ratio, and heat-release characteristics. These differences are abstracted through the fuel-dependent baseline coefficient $b_{base}$. Surface fuel moisture inhibits combustion efficiency. When $F_{moist,k} \to 1$, the burn fraction approaches zero. When $F_{moist,k} \to 0$, combustion proceeds at the baseline rate.

The baseline combustion coefficients used in this study are summarized in Table B.1. Values are selected to produce realistic burn residence times under nominal dry conditions for the adopted grid resolution and simulation time step.

Table B.1 Baseline combustion coefficients per fuel type

| Fuel Type | $b_{base}$(fraction per step) | Typical Burn Duration (qualitative) |
|---|---|---|
| **Grass** | 0.85 | Short, fast consumption |
| **Shrub** | 0.35 | Moderate consumption |
| **Pine litter** | 0.15 | Sustained surface burn |
| **Hardwood** | 0.08 | Slow combustion |

These coefficients are calibrated such that, in dry conditions and absent suppression, fuel mass decreases significantly over several time steps without causing numerical instability.

**Suppression of fuel reduction map**

The suppression function $f_{supp}(\cdot)$ produces a **step-normalized fuel reduction** applied at grid cell $(x, y)$ during simulation step $k$. Unlike continuous-time rate formulations, the present model operates in discrete simulation steps. Therefore, the suppression term directly represents the amount of fuel removed within a single time step, expressed in normalized load units and bounded by the remaining load. It acts as an absolute reduction, unlike the combustion term, which removes a fraction of the remaining load through $F_{burn,k}$. The suppression-induced fuel reduction term is defined as ( 83 ) where $\alpha_s \in [0,1]$ is global suppression scaling coefficient, $\eta_{cap,k}$ is operational capa city factor, $\eta_{avail,k}$ is operational availability factor, $\eta_{reach,k}(x, y)$ is reachability factor, $\eta_{eff,k}(x, y)$ is local suppression effectiveness factor. This multiplicative structure guarantees bounded and interpretable mitigation dynamics.

$$F_{red,k}(x, y) = \alpha_s \cdot \eta_{cap,k} \cdot \eta_{avail,k} \cdot \eta_{reach,k}(x, y) \cdot \eta_{eff,k}(x, y) \qquad (83)$$

Global scaling term $\alpha_s \in [0,1]$ is a global suppression scaling coefficient representing overall doctrine or scenario-level mitigation strength per simulation step. In practical terms, $\alpha_s$ absorbs the modeling choice of “how strong one step of suppression can be” under nominal conditions.

Suppression cannot exceed the operational capability that is actually present. This is captured by separating *capacity magnitude* $\eta_{cap,k}$ ( 84 ) where from *availability gating* $\eta_{avail,k}$ ( 85 ). $R_{cap,k}$ is the available suppression capacity (e.g., water volume, manpower equivalent, sortie capability) at step $k$ and $R_{cap,max}$ is the maximum reference capacity used for normalization

$$\eta_{cap,k} = \frac{R_{cap,k}}{R_{cap,max}} \tag{84}$$

Thus $\eta_{cap,k} \in [0,1]$ expresses the fraction of maximal capability currently available. $R_{avail,k} \in \{0,1\}$ is availability of the suppression resources

$$\eta_{avail,k} = R_{avail,k} \tag{85}$$

This factor encodes a strict necessary condition. If resources are not deployable, suppression impact is zero regardless of other terms.

Even if resources exist, mitigation effectiveness must decay with increasing travel time and poor accessibility. This is represented by a smooth reachability term ( 86 ) where $R_{time,k}(x,y)$ is the estimated travel time to reach $(x, y)$ at step $k$, $G_{access}(x,y) \in [0,1]$ is the static accessibility index of the terrain and $\beta_t > 0$ is the decay rate with travel delay

$$\eta_{reach,k}(x,y) = \exp\left(-\beta_t\, R_{time,k}(x,y)\right) \cdot G_{access}(x,y) \tag{86}$$

The exponential structure ensures boundedness and numerical stability. Operationally, it implements the principle that interventions arriving late cannot mitigate near-term fire growth at the same step, and inaccessible terrain reduces the usable effectiveness of resources.

Suppression success is not independent of local fire severity. High intensity fires typically resist suppression, reducing the effective mitigation that can be achieved within a single step. This is modeled by ( 87 ). $R_{eff,k}(x,y) \in [0,1]$ is the nominal suppression efficiency at $(x, y)$, $I_k(x,y) \in [0,1]$ is the local fire intensity proxy and $\gamma_I \geq 0$ is intensity resistance.

$$\eta_{eff,k}(x,y) = \frac{R_{eff,k}(x,y)}{1 + \gamma_I \, I_k(x,y)} \quad (87)$$

As $I_k$ increases, the denominator increases and $\eta_{eff,k}$ decreases. This captures the intuitive and operationally observed phenomenon that extreme fire behavior reduces the marginal impact of suppression. Suppression effectiveness is computed using the current-state intensity $I_k(x,y)$, ensuring that mitigation decisions are based on observable fire conditions at time step $k$, prior to state transition.

The final multiplicative formulation of $f_{supp}$ in ( 32 ) is explicitly written as ( 88 ). Each factor enters multiplicatively, so a vanishing capacity, availability, or reachability annuls the mitigation, high intensity attenuates it, and the product remains bounded for discrete-time simulation.

$$\begin{aligned} F_{red,k}^{raw}(x,y) = \alpha_s \cdot \left(\frac{R_{cap,k}}{R_{cap,max}}\right) \cdot R_{avail,k} \cdot \exp\left(-\beta_t R_{time,k}(x,y)\right) \\ \cdot G_{access}(x,y) \cdot \frac{R_{eff,k}(x,y)}{1 + \gamma_I I_k(x,y)} \end{aligned} \quad (88)$$

The calibration parameters summarized in Table B.2 regulate the strength, spatial decay, and operational sensitivity of the suppression mapping. These parameters are not arbitrary numerical coefficients; rather, they encode interpretable characteristics of wildfire response systems under discrete-time grid simulation.

To preserve physical realism, suppression-driven fuel reduction cannot exceed available fuel mass $0 \leq F_{red,k}(x,y) \leq F_{load,k}(x,y)$. Implementation uses the saturation rule $F_{red,k}(x,y) = \min\{F_{load,k}(x,y), F_{red,k}^{raw}(x,y)\}$. This prevents negative fuel states and ensures compatibility with the fuel evolution equation.

The global suppression gain $\alpha_s$ represents the maximum achievable fuel reduction fraction within a single simulation step and is therefore dimensionless (fraction per step). The travel-time decay parameter $\beta_t$ has units of inverse minutes; the reference value 0.03 per minute. It governs the exponential attenuation of suppression effectiveness as arrival time increases. The intensity resistance coefficient $\gamma_I$ is dimensionless and models the nonlinear reduction in suppression efficiency under increasing fire intensity.

Calibration may be performed using historical wildfire response data, operational doctrine constraints, or structured sensitivity analysis across scenario ensembles.

Table B.2 Suppression mapping calibration parameters

| **Parameter** | **Typical Range** | **Effect When Increased** |
|---|---|---|
| **Global suppression gain** $\boldsymbol{\alpha_s}$ | 0.01 – 0.30 | Stronger overall mitigation per simulation step |
| **Travel-time decay coefficient** $\boldsymbol{\beta_t}$ | 0.01 – 0.10 (per minute) | Faster decay of suppression effectiveness with increasing delay |
| **Intensity resistance factor** $\boldsymbol{\gamma_I}$ | 0.5 – 5.0 | High-intensity fires become progressively harder to suppress |

## C. Simulation Mathematics for Fire Intensity Evolution

The fire intensity proxy $I_k(x, y)$ is a bounded indicator representing local combustion strength. It does not model flame temperature or heat flux directly but provides a normalized scalar quantity suitable for suppression resistance modeling; risk prioritization; decision-layer feature extraction; and visualization. To ensure numerical stability, scale invariance, and resolution consistency, intensity is constructed using normalized inputs ( 89 ). $F_{load,\max}$ is the maximum fuel load, $W_{ws,\max}$ is a reference maximum wind speed (e.g., 20 m/s), $G_{slope,\max}$ is the maximum slope angle allowed in the terrain model.

$$0 \leq \tilde{F}, \widetilde{W}, \tilde{S} \leq 1$$

$$\tilde{F}_k(x, y) = \frac{F_{load,k}(x,y)}{F_{load,\max}}, \widetilde{W}_k(x, y) = \frac{W_{ws,k}(x,y)}{W_{ws,\max}}, \tilde{S}(x, y) = \frac{\tan\left(G_{slope}(x,y)\right)}{\tan\left(G_{slope,\max}\right)} \qquad (89)$$

The local fire intensity proxy is defined as ( 90 ). The intensity proxy is normalized and bounded within the interval [0,1] in order to ensure numerical stability and consistent interpretation across grid cells. $\beta > 0$ is the global intensity gain parameter, $\gamma_W \in [0,1]$ is the wind weighting coefficient, $\gamma_S \in [0,1]$ is the slope weighting coefficient. $B_{k+1}$ ensures that non-burning cells have zero intensity.

$$I_{k+1}(x, y) = B_{k+1}(x, y) \cdot \tanh\left(\beta\left(\tilde{F}_k(x, y) + \gamma_W \widetilde{W}_k(x, y) + \gamma_S \tilde{S}(x, y)\right)\right) \qquad (90)$$

The normalized structure ensures that fuel load remains the dominant contributor; wind and slope function as amplifying modifiers; intensity remains bounded in [0,1]. The fire intensity evolution parameters are listed in Table C.1.

Table C.1 Fire intensity evolution parameters

| | Range | Value | Interpretation |
|---|---|---|---|
| $\boldsymbol{\beta}$ | $1 \leq \beta \leq 3$ | 2.0 | Moderate global intensity amplification |
| $\boldsymbol{\gamma_W}$ | $0 \leq \gamma_W \leq 0.7$ | 0.5 | Wind moderately increases intensity |
| $\boldsymbol{\gamma_S}$ | $0 \leq \gamma_S \leq 0.5$ | 0.3 | Slope has secondary influence |

## D. Complete Concept Mapping Specification

This appendix states every numerical constant of the concept layer involving the wiring of the hierarchy ( 41 ); the aggregation rule ( 42 ); the gating rule ( 43 ). The term partitions of all features and concepts are the five-term trapezoidal partitions of the background chapter, applied to the unit interval. The saturated end terms carry full membership at the boundaries of the unit interval.

Table D.1 Level 1 to 4 aggregation weights over the previous levels

| **Concept** | **Contributor set** | **Weights $\boldsymbol{\omega}$** |
|---|---|---|
| $\boldsymbol{c_1^{(1)}}$ **fire severity** | $z_1$ fire intensity, $z_2$ spread potential | 0.60, 0.40 |
| $\boldsymbol{c_2^{(1)}}$ **spread hazard** | $z_2$ spread potential, $z_3$ weather severity, $z_4$ ignition proximity | 0.45, 0.35, 0.20 |
| $\boldsymbol{c_3^{(1)}}$ **fuel hazard** | $z_3$ weather severity, $z_5$ fuel load | 0.40, 0.60 |
| $\boldsymbol{c_4^{(1)}}$ **asset value** | $z_6$ asset exposure | 1.00 |
| $\boldsymbol{c_5^{(1)}}$ **crew reachability** | $z_7$ resource accessibility, $z_8$ access and road status | 0.55, 0.45 |
| $\boldsymbol{c_6^{(1)}}$ **logistics support** | $z_9$ suppression availability | 1.00 |
| $\boldsymbol{c_1^{(2)}}$ **fire threat level** | $c_1^{(1)}$, $c_2^{(1)}$, $c_3^{(1)}$ | 0.50, 0.30, 0.20 |
| $\boldsymbol{c_2^{(2)}}$ **asset exposure risk** | $c_2^{(1)}$, $c_4^{(1)}$ | 0.40, 0.60 |
| $\boldsymbol{c_3^{(2)}}$ **suppression feasibility** | $c_5^{(1)}$, $c_6^{(1)}$ | 0.50, 0.50 |
| $\boldsymbol{c_1^{(3)}}$ **intervention urgency** | $c_1^{(2)}$, $c_2^{(2)}$, $z_{10}$ temporal urgency | 0.40, 0.35, 0.25 |
| $\boldsymbol{c_2^{(3)}}$ **evacuation pressure** | $c_2^{(2)}$, $1 - z_8$ (inverted access and road status), $z_{10}$ temporal urgency | 0.45, 0.25, 0.30 |
| $\boldsymbol{c_1^{(4)}}$ **operational priority** | $c_1^{(2)}$, $c_2^{(2)}$ | 0.55, 0.45 |

Table D.2 Parameters of the concept layer

| Constant | Value | Role |
|---|---|---|
| **$\rho$ persistence decay** | 0.90 | Geometric forgetting rate of the prior in ( 43 ) |
| **$\alpha_{min}$ coverage floor** | 0.05 | Lower bound of the aggregate firing strength guaranteed by the overlapping partitions |

Table D.3 Trapezoidal parameters

| Term | Trapezoidal edge values (a, b, c, d) |
|---|---|
| **very low** | (0.00, 0.00, 0.15, 0.30) |
| **low** | (0.15, 0.30, 0.35, 0.50) |
| **medium** | (0.35, 0.50, 0.55, 0.70) |
| **high** | (0.55, 0.70, 0.75, 0.90) |
| **very high** | (0.75, 0.90, 1.00, 1.00) |

The membership side of equation ( 42 ) is computed with vectors, exactly as in the code. Each feature value is fuzzified into a five-term membership vector over the partition of Table D.3, and the aggregation in ( 42 ) combines these vectors term by term. A concept activation is therefore a five-term membership vector, not a single number. Rule antecedents use this vector directly, because the membership of a linguistic term is just the matching component of the vector. A single scalar is needed only in three places: The priority sent to the coordinator, the quality measure, and reporting. In those cases the vector is reduced to its expected value over the term centers. The worked examples in the decision chapter follow the highest component of each vector. Since every component passes through the same weighted sums, this component is a faithful sample of the full computation. There is one exception in the wiring. Evacuation pressure uses the complement of access and road status, as given in the wiring equation. With Table D.3 and these conventions, the membership side of ( 42 ) is fully defined. Per-pair term tables are not required, and adaptation can later refine the shared partition by splitting terms where the field needs it.

## E. Representative Decision Rules

The evaluation runs the decision layer on the minimal profile. Table E.1 includes only the strongest seed of that family per intervention family. There are five rules: R1 for asset protection, R2 for evacuation and public warning, R3 for containment line, R4 for resource deployment, and R5 for suppression effort. These five rules are the irreducible core of the doctrine. Every intervention family remains reachable, coverage of the term space is deliberately surrendered, and everything else must be recovered by the staged adaptation of the decision layer.

Each rule follows the canonical form of ( 44 ). The antecedent is a conjunction over the gated activations of the decision concepts, namely ***fire threat level***, ***asset exposure risk***, ***suppression feasibility***, ***intervention urgency***, *and* ***evacuation pressure.*** Each concept is described by one of the five terms very low, low, medium, high, and very high; a concept absent from the antecedent is a do-not-care condition. The consequent assigns normalized intensities in the unit interval to one or more of the six intervention types, namely ***suppression effort***, ***resource deployment***, ***containment line***, ***asset protection***, ***evacuation***, *and* ***public warnin*****g**.

Each intensity reaches the simulation only through the resource input layer, and the channel differs by family: Suppression effort, containment line and asset protection place capacity with different spatial patterns, namely on the active front, on a band ahead of it and around the protected assets; resource deployment raises availability and shortens response time; and evacuation and public warning are population-side orders that move no suppression resource and act on the cost alone. Rule effects are of two kinds. A raise adds one linguistic term, that is 0.25 on the normalized intensity scale, whereas a cap bounds the intensity from above at the value the rule states. Both are blended by the firing strength of the rule, so a weakly fired effect shifts the intensity only slightly. Rule identifiers are stable across the thesis, so a rule cited in the evaluation chapter can be looked up here.

Table E.1 Representative decision rules by operational regime

| ID | Antecedent<br>(IF, over gated decision concepts) | Consequent<br>(THEN, intervention intensities) |
|---|---|---|
| **R1** | asset exposure risk is very high AND suppression feasibility is low | asset protection 0.9, containment line 0.6 |
| **Evacuation and warning regime** | | |
| **R2** | evacuation pressure is very high AND suppression feasibility is very low | evacuation 1.0, public warning 1.0, resource deployment 0.7 for evacuation support (offensive intensities withdrawn) |
| **Feasibility-limited regime** | | |
| **R3** | fire threat level is very high AND suppression feasibility is very low | containment line 0.9, evacuation 0.6, public warning 0.8 |
| **Life-safety boundary** | | |
| **R4** | intervention urgency is very high | resource deployment 1.0 (priority preemption across regions) |
| **Backbone coverage rules: Fire threat level (primary: Suppression effort)** | | |
| **R5** | fire threat level is very high | suppression effort 1.0, resource deployment 0.8 |

Every rule in the operational base carries one of three provenance classes, recorded in its explanation trail. Expert-seeded rules constitute the initial base and encode doctrine; evFIS-tuned rules are expert-seeded rules whose membership shoulders or consequent coefficients have been modified by the first adaptation operator, with the modification history retained; generative rules entered the base through the verification gates and the admission of ( 64 ), and remain permanently marked as machine-proposed. The evaluation chapter reports, per scenario, the size of the instantiated active base and the share of each provenance class, which quantifies how far the open decision space is actually exercised.

**Rules and interventions produced by the open decision space.** The evaluation seeds only the five rules of the minimal profile; each rule defined in Table E.2 was written or changed by the adaptation stages during a run and admitted through the verification gates of Section 4.5. Table E.2 lists only a subset of the generated rules.

Table E.2 The generative rules admitted over the recorded runs

| ID | Antecedent (IF, over gated decision concepts) | Consequent (THEN, intervention intensities) |
|---|---|---|
| **G1** | fire threat level H AND suppression feasibility VH | suppression effort 0.9; retardant drop 0.8; resource deployment 0.7 |
| **G2** | suppression feasibility VH AND intervention urgency VH | suppression effort 0.9; retardant drop 0.8; resource deployment 0.7 |
| **G3** | suppression feasibility VH AND fire threat level H AND intervention urgency VH | suppression effort 0.95; retardant drop 0.85; resource deployment 0.9 |
| **G4** | suppression feasibility VH AND intervention urgency VH AND fire threat level M | downwind_containment_shield 0.9; retardant drop 0.8; asset protection 0.6 |
| **G5** | asset exposure risk VH AND suppression feasibility VH | downwind retardant shield 0.9; asset protection 0.7; resource deployment 0.7 |
| **G6** | fire_threat_level ≥ H AND suppression_feasibility VH | head_knockdown 0.95; resource_deployment 0.8 |
| **G7** | suppression feasibility VH AND fire threat level ≥H AND evacuation pressure VH | wet containment line 0.9; resource deployment 0.8 |
| **G8** | suppression_feasibility H ∧ fire_threat_level H | counterfire_strip 0.9; resource_deployment 0.8 |
| **G9** | suppression feasibility VH AND asset exposure risk H AND evacuation pressure H | suppression effort 0.9; water drafting 0.8; resource deployment 0.7 |
| **G10** | asset exposure risk VH AND suppression feasibility VH AND intervention urgency VL | asset protection 0.9; containment line 0.8; retardant drop 0.7 |
| **G11** | suppression feasibility VH AND asset exposure risk VH AND fire threat level VL | suppression effort 0.9; water drafting 0.8; resource deployment 0.8 |
| **G12** | suppression feasibility VH AND asset exposure risk VH AND evacuation pressure VH | suppression effort 0.9; water drafting 0.8; retardant drop 0.7 |
| **G13** | suppression feasibility VH AND asset exposure risk VH AND intervention urgency VH | suppression effort 0.95; water drafting 0.8; retardant drop 0.7 |

Table E.2 The generative rules admitted over the recorded runs (cont'd)

| ID | Antecedent (IF, over gated decision concepts) | Consequent (THEN, intervention intensities) |
|---|---|---|
| **G14** | interface exposure VH AND suppression feasibility VH | asset protection 0.90; containment line 0.80; resource deployment 0.70 |
| **G15** | fire threat level H AND asset exposure risk VH AND suppression feasibility VH | suppression effort 0.9; retardant drop 0.8; asset protection 0.7 |
| **G16** | asset exposure risk VH AND suppression feasibility VH AND evacuation pressure VL | containment line 0.8; retardant drop 0.7; resource deployment 0.7 |
| **G17** | fire threat level H AND suppression feasibility L | containment line 0.9; retardant drop 0.8 |
| **G18** | suppression feasibility VH AND fire threat level VL AND intervention urgency VL | suppression effort 0.9; water drafting 0.8; containment line 0.7 |
| **G19** | intervention urgency H AND suppression feasibility L AND asset exposure risk H | retardant drop 0.9; containment line 0.8; water drafting 0.7 |
| **G20** | asset exposure risk H AND evacuation pressure VH AND suppression feasibility L | water drafting 0.9; suppression effort 0.8; retardant drop 0.7 |

*Term set: VL = very low, L = low, M = medium, H = high, VH = very high; "≥ X" denotes a premise satisfied at level X or above. Intervention intensities are on [0, 1] and are reproduced verbatim from the source list. No premise set occurs twice within a block, and no undefined linguistic label remains.*

Objects created by the open decision space. A generated rule is written in the vocabulary in force at the time, and part of that vocabulary is itself generated. Table E.3 lists representative objects admitted by the campaign of Section 5.5.3, with the definition recorded in the lineage store; a rule of the previous table that cites a name not listed here refers to an admitted object whose definition is recorded in the same store. A concept is a normalized weight vector over existing inputs, a macro is a weighted composition of base families, and a clause actuator is a list of verified effects with the sector and range each acts on.

Table E.3 Vocabulary objects admitted by the generative stage, with their recorded definitions (Examples)

| Kind | Name | Recorded definition |
|---|---|---|
| **Clause actuator** | asset_shield_band | coat on assets [0, 6] at 1.0; coat on populated [0, 6] at 1.0; draft on assets [0, 8] at 1.0 |
| **Clause actuator** | asset_shield_belt | coat on assets [0, 4] at 1.0; coat on populated [0, 5] at 1.0; draft on at_fire [0, 6] at 0.8 |
| **Clause actuator** | forward_draft_relay | draft on at_fire [0, 3] at 1.0; draft on ring [0, 8] at 1.0; evacuate on populated [0, 15] at 1.0 |
| **Clause actuator** | lakeside_capacity_bridge | draft on at_fire [0, 3] at 1.0; draft on head [0, 8] at 1.0; evacuate on populated [0, 6] at 1.0 |
| **Clause actuator** | preemptive_asset_coat_ring | coat on assets [0, 4] at 1.0; coat on populated [0, 4] at 1.0; coat on head [1, 6] at 1.0 |
| **Clause actuator** | preemptive_asset_coating | coat on assets [0, 4] at 1.0; coat on populated [0, 3] at 1.0 |
| **Clause actuator** | shoreline_capacity_bridge | draft on at_fire [0, 8] at 1.0; draft on ring [0, 15] at 1.0; evacuate on populated [0, 15] at 1.0 |
| **Intermediate concept** | capacity_starved_head_pressure | fire_threat_level 0.50, intervention_urgency 0.30, suppression_feasibility 0.20 |
| **Intermediate concept** | head_run_pressure | fire_threat_level 0.40, intervention_urgency 0.30, evacuation_pressure 0.20, suppression_feasibility 0.10 |
| **Intermediate concept** | village_interface_pressure | intervention_urgency 0.40, evacuation_pressure 0.30, fire_threat_level 0.30 |
| **Macro intervention** | waterborne_relay_and_shelter | water_drafting 1.00, evacuation 0.60, public_warning 0.50 |

## F. Generative AI Prompt Specifications

A single interface couples the framework to the LLM: The stage-③ generative proposer, which runs inside the live decision cycle. The interface carries no free-form instructions. It is a fixed content: A fixed system instruction, a deterministic situation context assembled by the engine, and a machine-readable reply whose grammar is closed over the framework's own vocabulary, so that anything outside the admissible decision space is unparseable. The model identifier and the transport are pinned per campaign and recorded in every run log.

**The stage-③ proposer: Inputs and admissible outputs**

At each invocation the proposer receives a compact, auditable summary of the current situation (i.e., the decision-concept activations, the observation confidence, the recent decision-cost trajectory, the rule-base statistics, and the location where residual cost concentrates) together with the complete current vocabulary: All decision concepts including learned intermediate concepts, all interventions including learned macros, the five fuzzy terms, and an actuator library annotated with map availability (a water-dependent tactic is declared unavailable on a map without a water body, so it cannot be ordered by construction).

The reply is one candidate rule: At most three antecedents, each pairing a decision concept with a fuzzy term, and at most three consequents, each pairing an intervention with an intensity in [0,1]. Optionally, the proposal may be a package that grows the vocabulary by exactly one new object, which the accompanying rule is required to use.

Three package forms exist. A new intermediate concept is a named, weighted composition of at most four existing features or concepts, with the weights normalized on installation. A new macro intervention is a named, weighted bundle of at most three base channels, the weights interpreted as relative proportions of the bundle. A new clause actuator is a genuinely new tactic rather than a blend: An ordered set of clauses, each applying one verified physical effect (i.e., wetting, clearing, igniting, coating, evacuating, priming, or drafting) to a named sector of the fire (head, flank, rear, ring, at-fire, assets, populated) over a radial cell band with an amount. The instruction requires the tactic to be drawn from wildfire practice and named honestly; a backburn, for example, is the igniting effect applied to a ring sector ahead of the head, a maneuver no weighted blend of the base channels can express.

What remains fixed is the physical basis itself: The observation features, the decision axes, and the seven verified physical effects are the surface the simulation core can actually execute, so the generative stage is combinatorially open but physically closed; in other words, new observables and new tactics over the existing physics, never new physics. Every proposal, packaged or not, then passes the verification gates G1–G5b of Section 4.54.5 before any order can be influenced by it.

**Reply discipline and transport**

The proposer runs under a system instruction that admits exactly one minified JSON object and nothing else, with no tools enabled and a bounded wall-clock budget (ninety seconds by default): A slow call becomes a skipped stage recorded in the log, never a stalled decision cycle. Obvious notation slips, such as a numeric threshold written where the grammar wants a linguistic term, are translated to the nearest term, and every such repair is stamped into the resulting rule's provenance note so the audit trail states what the model actually wrote. When the transport or the model is unavailable, the stage reports itself unavailable and the staged adaptation simply never escalates beyond the evolving-fuzzy stages.

**System instruction (abstract)**

You are a rule proposer for a wildfire response framework. You will be given a snapshot summary of the current fire situation and the framework's complete vocabulary. Your task is to propose a single candidate rule that reduces residual decision cost. Constraints are as follows:

- Use only the decision concepts, interventions, fuzzy terms, and map-available actuators given to you.
- At most three antecedents and three consequents; intensities lie in [0,1].
- You may optionally extend the vocabulary by exactly one new object (intermediate concept, macro intervention, or clause actuator); if you do, the proposed rule must use it.
- A new clause actuator may compose only the verified physical effects, must be drawn from real wildfire practice, and must be named honestly.
- Do not invent new observables, new physical effects, or any object not on the supplied lists.
- Output: One minified JSON object on a single line. No explanation, no rationale, no code fences, no other text.

**<u>Example mission brief (verbatim, one recorded run)</u>**

The mission brief below is an example of the standing prompt that is built once per incident and sent verbatim on every generative call of that run; it is also written to the run log for audit. It gives the model the map affordances (including the water body), the six seeded doctrine families, the actuator library that no seed rule orders, and the basis of seven verified physical effects from which a new clause actuator may be composed. The concrete values are those of one recorded run.

```
=== MISSION BRIEF (standing; read once, applies to every decision of this
incident) ===
MAP: 100 x 70 cells at 30 m; land cover: Non fuel 7%, grass 20%, shrub 19%,
pine litter 40%, hardwood 9%, water 3%, urban 2%.
- water: 206 cells of lake/river/sea, centred near (52,17); engines and aircraft
can draft/refill there, so sustained capacity near the water is real.
- assets on the map (protect by value and life):
  population 'Town residents' at (2,1), ~12600 persons, value 1.0
  building 'Town centre' at (2,1), value 1.0
  critical 'Hospital' at (5,0), value 1.0
  critical 'Power plant' at (7,4), value 0.9
  critical 'Water treatment' at (0,10), value 0.9
  critical 'Government office' at (0,1), value 0.8
  critical 'School' at (0,0), value 0.8
  population 'Village 1 residents' at (8,69), ~3150 persons, value 1.0
```

```
  building 'Village 1 centre' at (8,69), value 0.7
  critical 'Fire station (Village 1)' at (6,69), value 0.7
  evac_route 'Evacuation route' at (2,0), value 1.0
- resources: Staged capacity 535 (rcap sum), map-mean travel time 20 min, air
support: Yes.
- protection priority weights (WHERE effort goes): Critical 0.40, population
0.25, building 0.20, evacuation 0.15
- loss weights of the decision cost J (WHAT counts as a bad outcome): Burn
0.29, asset 0.29, population 0.29, response 0.06, delay 0.06. Trials and cross-
run comparisons are judged on the PHYSICAL part (burn + asset + population).
DOCTRINE FAMILIES (seed rules draw only on these): Suppression_effort
(suppression effort), resource_deployment (resource deployment),
containment_line (containment line), asset_protection (asset protection),
evacuation (evacuation), public_warning (public warning)
ACTUATOR LIBRARY (physics available beyond the doctrine; NO seed rule orders
them, discovering a use is your job):
  tactical_burn: A firing crew ignites a strip between the containment band and
the front so a counter-fire consumes the fuel the head fire is running toward.
Real fire: In strong wind or near assets it can backfire, and the forecast
gates will reject a reckless order.
  water_drafting: Engines and helicopters refill from the nearest lake, river,
or sea, raising sustained capacity near water. On a map without water it does
nothing.
  retardant_drop: Aircraft coat the fuel ahead of the head fire with long-term
retardant or soil, so the coated cells resist ignition even after the water in
them dries. Aerial delivery: It does not need road access, but it is a finite,
expensive pass and only the head sector is coated.
YOU MAY ALSO DEFINE A NEW ACTUATOR as data (a package with "clauses"): Each
clause = one verified effect (wet, clear, ignite, coat, evacuate, prime, draft)
on a sector (head, flank, rear, ring, at_fire, assets, populated) at a cell
range [rin, rout] from the front with an amount. That is a genuinely new tactic
(WHERE and HOW); a mere re-weighting of channels is the tuning stage's job and
fails the novelty gates.
WHEN TO INVENT WHAT (think like an incident commander):
  - a plain rule: The situation is expressible in the current concepts and an
existing action answers it.
  - a new CONCEPT: The same kind of situation keeps recurring and the five
decision concepts cannot name it (for example ember pressure on a settlement
edge, urban interface stress). Concepts change the architecture: They enter
Layer 3 and later rules can cite them.
  - a composite intervention: Two channels must function AS ONE with a fixed
ratio (hardening a town: Protection + line).
  - a clause actuator: The tactic needs its own geometry (a flank firing
operation, a deep pre-wetted band). Use only what THIS map supports: No water
means no drafting and no aerial drops; no population nearby means evacuate and
prime are wasted orders.
STANDING RULES OF THE GAME: Every proposal must FIRE in the situation it is
proposed for (use current dominant terms, or '>=' for a rising threat); orders
must be strong enough to move a 45-minute physical forecast; every candidate
is judged by simulation gates (form, vocabulary, relevance, availability, two
reseeded A/B forecasts, and a growth margin for new vocabulary); a rejected
proposal comes back to you once with the failing gate named, so fix exactly
that. Always return ONLY the JSON.
=== END OF MISSION BRIEF ===
```

# CURRICULUM VITAE

## PERSONAL INFORMATION

Surname, Name: AKMAN, Çağlar

## EDUCATION

| Degree | Institution | Year of Graduation |
|---|---|---|
| MS | METU Electrical and Electronics Engineering | 2017 |
| BS | METU Electrical and Electronics Engineering | 2008 |
| High School | Yusuf Ziya Öner Fen Lisesi, Antalya | 2002 |

## WORK EXPERIENCE

| Year | Place | Enrollment |
|---|---|---|
| 2014- Present | HAVELSAN | Modelling and Sensor Technologies Team Leader |
| 2013-2014 | AYESAS | Senior Software Engineer |
| 2008-2013 | KAREL | Senior Design Engineer |
| 2007-2007 | TUBITAK | Part Time Engineer |

## FOREIGN LANGUAGES

Advanced English, Beginner German

## PUBLICATIONS

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2. Akman, Ç. (Co-inventor), & HAVELSAN (Assignee). (2018, December 31). Hareket algılama, analiz ve takip için bir sistem (Patent No. TR 2018/21331). Türk Patent ve Marka Kurumu.
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## HOBBIES

Playing piano, playing chess, playing table tennis, and swimming